Physica bk8

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1 On GlossySnake

A LaTeX Proof-of-Concept of the GlossySnake project. By the nature of the machine-glossing, the inaccurate and malformatted glossing is expected.

2 Physica bk8

PHYSICA BK.8

PHYSICS Book 8

Κεφάλαιο 1

Chapter 1

Πότερον γέγονέ ποτε κίνησις οὐκ οὖσα πρότερον, καὶ φθείρεται

Whether has occurred ever a motion not being previously, and is destroyed

πάλιν οὕτως ὥστε κινεῖσθαι μηδέν, ἢ οὔτ' ἐγένετο οὔτε φθείρεται,

again thus so that to be moved nothing, or neither has become nor is destroyed,

ἀλλ' ἀεὶ ἦν καὶ ἀεὶ ἔσται, καὶ τοῦτ' ἀθάνατον καὶ ἄπαυστον

but always was and always will be, and this immortal and unceasing

ὑπάρχει τοῖς οὖσιν, οἷον ζωή τις οὖσα τοῖς φύσει συνεστῶσι

exists for those who are, like life some being for those by nature that coexist

πᾶσιν;

with all?

εἶναι μὲν οὖν κίνησιν πάντες φασὶν οἱ περὶ φύσεώς τι

to be indeed therefore motion all say the about nature something

λέγοντες διὰ τὸ κοσμοποιεῖν καὶ περὶ γενέσεως καὶ φθορᾶς εἶναι

speaking through the world-making and about generation and decay to be

τὴν θεωρίαν πᾶσαν αὐτοῖς, ἣν ἀδύνατον ὑπάρχειν μὴ κινήσεως

the theory all to them, which impossible to exist not of motion

οὔσης· ἀλλ' ὅσοι μὲν ἀπείρους τε κόσμους εἶναί φασιν,

being. but as many as indeed infinite and worlds to be say,

καὶ τοὺς μὲν γίγνεσθαι τοὺς δὲ φθείρεσθαι τῶν κόσμων,

and the indeed to come into being the but to perish of the worlds,

ἀεί φασιν εἶναι κίνησιν (ἀναγκαῖον γὰρ τὰς γενέσεις καὶ τὰς

always they say to be motion (necessary for the generations and the

φθορὰς εἶναι μετὰ κινήσεως αὐτῶν)· ὅσοι δ' ἕνα <ἢ ἀεὶ>

decays to be with motion of them). As many as indeed one or always>

ἢ μὴ ἀεί, καὶ περὶ τῆς κινήσεως ὑποτίθενται κατὰ

or not always, and concerning the motion are hypothesized according to

λόγον. εἰ δὴ ἐνδέχεταί ποτε μηδὲν κινεῖσθαι, διχῶς ἀνάγκη

reason. if indeed it may be at some time nothing to be moved, otherwise necessity

τοῦτο συμβαίνειν· ἢ γὰρ ὡς Ἀναξαγόρας λέγει (φησὶν γὰρ

this to happen; or for as Anaxagoras says (he says for

ἐκεῖνος, ὁμοῦ πάντων ὄντων καὶ ἠρεμούντων τὸν ἄπειρον χρόνον,

that one, together of all beings and those at rest the infinite time,

κίνησιν ἐμποιῆσαι τὸν νοῦν καὶ διακρῖναι), ἢ ὡς Ἐμπεδοκλῆς

to create motion to create the mind and to differentiate), or as Empedocles

ἐν μέρει κινεῖσθαι καὶ πάλιν ἠρεμεῖν, κινεῖσθαι μὲν ὅταν

in part to be moved and again to be at rest, to be moved on the one hand whenever

ἡ φιλία ἐκ πολλῶν ποιῇ τὸ ἓν ἢ τὸ νεῖκος πολλὰ ἐξ

the love out of many may produce the one or the strife many from

ἑνός, ἠρεμεῖν δ' ἐν τοῖς μεταξὺ χρόνοις, λέγων οὕτως ᾗ

one, to be at rest but in the between times, saying thus in which

μὲν ἓν ἐκ πλεόνων μεμάθηκε φύεσθαι, ἠδὲ

on the one hand one out of more has been accustomed to come into being, and

πάλιν διαφύντος ἑνὸς πλέον' ἐκτελέθουσιν, [251a] τῇ

again when one is separating from one more' they will grow, [251a] to the

μὲν γίγνονταί τε καὶ οὔ σφισιν ἔμπεδος αἰών· ᾗ δὲ

on the one hand they become and and not to them stable eternity; in which and

τάδ' ἀλλάσσοντα διαμπερὲς οὐδαμὰ λήγει, ταύτῃ δ' αἰὲν ἔασιν

these changing continuously nowhere ends, in this and always are

ἀκίνητοι κατὰ κύκλον. τὸ γὰρ "ᾗ δὲ τάδ' ἀλλάσσοντα"

unmoving in accordance with circle. the for "in which and these changing"

ἐνθένδε ἐκεῖσε λέγειν αὐτὸν ὑποληπτέον. σκεπτέον δὴ περὶ

from here to there to speak it must be assumed. one must consider indeed about

τούτων πῶς ἔχει·

these matters how it is;

πρὸ ἔργου γὰρ οὐ μόνον πρὸς τὴν περὶ φύσεως θεωρίαν

in front of work for not only towards the concerning nature theory

ἰδεῖν τὴν ἀλήθειαν, ἀλλὰ καὶ πρὸς τὴν μέθοδον τὴν περὶ τῆς

to see the truth, but and towards the method the concerning the

ἀρχῆς τῆς πρώτης. ἀρξώμεθα δὲ πρῶτον ἐκ τῶν διωρισμένων

beginning the first. let us begin and first from the defined

ἡμῖν ἐν τοῖς φυσικοῖς πρότερον. φαμὲν δὴ τὴν κίνησιν εἶναι

to us in the natural beforehand. we say indeed the motion to be

ἐνέργειαν τοῦ κινητοῦ ᾗ κινητόν. ἀναγκαῖον ἄρα ὑπάρχειν

an activity of the mobile by which mobile. necessary therefore to exist

τὰ πράγματα τὰ δυνάμενα κινεῖσθαι καθ' ἑκάστην κίνησιν.

the things the capable of being moved according to each motion.

καὶ χωρὶς δὲ τοῦ τῆς κινήσεως ὁρισμοῦ, πᾶς ἂν ὁμολογήσειεν

and without but the the motion definition, every if he would agree

ἀναγκαῖον εἶναι κινεῖσθαι τὸ δυνατὸν κινεῖσθαι καθ' ἑκάστην

necessary to be to be moved the possible to be moved according to each

κίνησιν, οἷον ἀλλοιοῦσθαι μὲν τὸ ἀλλοιωτόν, φέρεσθαι δὲ τὸ

motion, such as to be changed indeed the changeable, to be carried but the

κατὰ τόπον μεταβλητόν, ὥστε δεῖ πρότερον καυστὸν

according to place changeable, so that it is necessary beforehand fiery

εἶναι πρὶν κάεσθαι καὶ καυστικὸν πρὶν κάειν. οὐκοῦν καὶ ταῦτα

to be before being burned and caustic before burning. therefore and these

ἀναγκαῖον ἢ γενέσθαι ποτὲ οὐκ ὄντα ἢ ἀΐδια εἶναι. εἰ μὲν

necessary or to become at some time not existing or eternal to be. if indeed

τοίνυν ἐγένετο τῶν κινητῶν ἕκαστον, ἀναγκαῖον πρότερον τῆς

then it came to be of the movers each, it is necessary beforehand of the

ληφθείσης ἄλλην γενέσθαι μεταβολὴν καὶ κίνησιν, καθ'

having been taken another to become change and motion, according to

ἣν ἐγένετο τὸ δυνατὸν κινηθῆναι ἢ κινῆσαι· εἰ δ' ὄντα

which it came to be the possible to be moved or to move; if but existing

προϋπῆρχεν ἀεὶ κινήσεως μὴ οὔσης, ἄλογον μὲν φαίνεται

had previously existed always of motion not being, unreasonable indeed it seems

καὶ αὐτόθεν ἐπιστήσασιν, οὐ μὴν ἀλλὰ μᾶλλον ἔτι προάγουσι

and from itself to have knowledge, not however but rather still they lead forward

τοῦτο συμβαίνειν ἀναγκαῖον. εἰ γὰρ τῶν μὲν κινητῶν ὄντων

this to happen necessary. if for of the indeed movers of beings

τῶν δὲ κινητικῶν ὁτὲ μὲν ἔσται τι πρῶτον

the but of the moving sometimes on one hand there will be something first

κινοῦν, τὸ δὲ κινούμενον, ὁτὲ δ' οὐθέν, ἀλλ' ἠρεμεῖ,

causing to move, the but being moved, sometimes and nothing, but is at rest,

ἀναγκαῖον τοῦτο μεταβάλ λειν πρότερον· ἦν γάρ τι

necessary this to change to leave earlier. It was for something

αἴτιον τῆς ἠρεμίας· ἡ γὰρ ἠρέμησις στέρησις κινήσεως.

a cause of the rest; the for rest is a deprivation of movement.

ὥστε πρὸ τῆς πρώτης μεταβολῆς ἔσται μεταβολὴ προτέρα.

Therefore before the first change there will be a change prior.

τὰ μὲν γὰρ κινεῖ μοναχῶς, τὰ δὲ καὶ τὰς ἐναντίας κινήσεις,

the on one hand for moves alone, the but also the opposing movements,

οἷον τὸ μὲν πῦρ θερμαίνει, ψύχει δ' οὔ, ἡ δ' ἐπιστήμη δοκεῖ

such as the indeed fire warms, cools and not, the and knowledge seems

τῶν ἐναντίων εἶναι μία. φαίνεται μὲν οὖν κἀκεῖ τι

of the opposites to be one. It appears on one hand therefore and there something

εἶναι ὁμοιότροπον· τὸ γὰρ ψυχρὸν θερμαίνει στραφέν πως καὶ

to be in a similar manner; the for cold warms having turned somehow and

ἀπελθόν, ὥσπερ καὶ ἁμαρτάνει ἑκὼν ὁ ἐπιστήμων, ὅταν

having departed, just as and sins willingly the knowledgeable person, whenever

ἀνάπαλιν χρήσηται τῇ ἐπι [251b] στήμῃ. ἀλλ' οὖν ὅσα γε

again is used to the upon [251b] stick. But therefore as much as indeed

δυνατὰ ποιεῖν καὶ πάσχειν ἢ κινεῖν, τὰ δὲ κινεῖσθαι, οὐ πάντως

possible to do and to suffer or to move, the yet to be moved, not at all

δυνατά ἐστιν, ἀλλ' ὡδὶ ἔχοντα καὶ πλησιάζοντα ἀλλήλοις.

possible is, but in this way having and drawing near to one another.

ὥσθ' ὅταν πλησιάσῃ, κινεῖ, τὸ δὲ κινεῖται, καὶ ὅταν ὑπάρξῃ

So that whenever it draws near, it moves, the and is moved, and whenever it exists

ὡς ἦν τὸ μὲν κινητικὸν τὸ δὲ κινητόν. εἰ τοίνυν μὴ ἀεὶ

as it was the one moving the and the movable. If therefore not always

ἐκινεῖτο, δῆλον ὡς οὐχ οὕτως εἶχον ὡς ἦν δυνάμενα τὸ
were moved, it is clear that not thus they had as it was being able the
μὲν κινεῖσθαι τὸ δὲ κινεῖν, ἀλλ' ἔδει μεταβάλλειν θάτερον
one to be moved the and to move, but it was necessary to change the other
αὐτῶν· ἀνάγκη γὰρ ἐν τοῖς πρός τι τοῦτο συμβαίνειν, οἷον
of them; necessity for in the towards something this to happen, as if
εἰ μὴ ὂν διπλάσιον νῦν διπλάσιον, μεταβάλλειν, εἰ μὴ ἀμφότερα,
if not being double now double, to change, if not both,
θάτερον. ἔσται ἄρα τις προτέρα μεταβολὴ τῆς πρώτης.
the other. it will be therefore someone previous change of the first.
πρὸς δὲ τούτοις τὸ πρότερον καὶ ὕστερον πῶς ἔσται χρόνου
towards but these the previous and later how will it be of time
μὴ ὄντος; ἢ χρόνος μὴ οὔσης κινήσεως; εἰ δή ἐστιν ὁ χρόνος
not being? or time not being of movement? if indeed it is the time
κινήσεως ἀριθμὸς ἢ κίνησίς τις, εἴπερ ἀεὶ χρόνος ἔστιν,
of movement a number or movement something, if indeed always time is,
ἀνάγκη καὶ κίνησιν ἀΐδιον εἶναι. ἀλλὰ μὴν περί γε χρόνου
necessity and movement eternal to be. but indeed about at least time
ἔξω ἑνὸς ὁμονοητικῶς ἔχοντες φαίνονται πάντες· ἀγένητον γὰρ
outside of one in agreement having they seem all; ungenerated for
εἶναι λέγουσιν. καὶ διὰ τούτου Δημόκριτός γε δείκνυσιν ὡς
to be they say. and through this Democritus indeed shows that
ἀδύνατον ἅπαντα γεγονέναι· τὸν γὰρ χρόνον ἀγένητον εἶναι.
impossible all to have come into being. the for time ungenerated to be.
Πλάτων δὲ γεννᾷ μόνος· ἅμα μὲν γὰρ αὐτὸν τῷ οὐρανῷ
Plato but generates alone: at once on one hand for it with the universe
[γεγονέναι], τὸν δ' οὐρανὸν γεγονέναι φησίν. εἰ
[to have come into being], the and universe to have come into being he says. if

οὖν ἀδύνατόν ἐστιν καὶ εἶναι καὶ νοῆσαι χρόνον ἄνευ τοῦ

therefore impossible it is and to be and to understand time without the

νῦν, τὸ δὲ νῦν ἐστι μεσότης τις, καὶ ἀρχὴν καὶ τελευτὴν ἔχον

now, the but now is a middle some, and a beginning and end having

ἅμα, ἀρχὴν μὲν τοῦ ἐσομένου χρόνου, τελευτὴν δὲ τοῦ

together, a beginning on one hand of the forthcoming time, an end but of the

παρελθόντος, ἀνάγκη ἀεὶ εἶναι χρόνον. τὸ γὰρ ἔσχατον τοῦ

past, necessity always to be time. the for last of the

τελευταίου ληφθέντος χρόνου ἔν τινι τῶν νῦν ἔσται (οὐδὲν γὰρ

last taken time in some of the now will be nothing for

ἔστι λαβεῖν ἐν τῷ χρόνῳ παρὰ τὸ νῦν), ὥστ' ἐπεί ἐστιν ἀρχή τε

is to take in the time besides the now), so that since it is beginning and

καὶ τελευτὴ τὸ νῦν, ἀνάγκη αὐτοῦ ἐπ' ἀμφότερα εἶναι ἀεὶ χρόνον.

and end the now, necessity of it upon both to be always time.

ἀλλὰ μὴν εἴ γε χρόνον, φανερὸν ὅτι ἀνάγκη εἶναι καὶ κίνησιν,

but indeed if indeed time, evident that necessity to be and motion,

εἴπερ ὁ χρόνος πάθος τι κινήσεως.

if indeed the time is an experience something of motion.

ὁ δ' αὐτὸς λόγος καὶ περὶ τοῦ ἄφθαρτον εἶναι τὴν κίνησιν·

the but he word and about the imperishable to be the motion;

καθάπερ γὰρ ἐπὶ τοῦ γενέσθαι κίνησιν συνέβαινεν προτέραν εἶναί

just as for upon the becoming motion it happened earlier to be

τινα μεταβολὴν τῆς πρώ της, οὕτως ἐνταῦθα ὑστέραν τῆς

some change of the first (one), thus here later of the

τελευταίας· οὐ γὰρ ἅμα παύεται κινούμενον καὶ κινητὸν

last; not for at the same time ceases being moved and movable

ὄν, οἷον καιόμενον καὶ καυστὸν ὄν (ἐνδέχεται γὰρ καυστὸν
being, like burning and combustible being (it is possible for combustible
εἶναι μὴ καιόμενον), οὐδὲ [252a] κινητικὸν καὶ κινοῦν. καὶ τὸ
to be not burning), nor [252a] movable and moving. And the
φθαρτικὸν δὴ δεήσει φθαρῆναι ὅταν φθείρῃ· καὶ τὸ τούτου
corruptible indeed will need to be corrupted when it corrupts; and the of this
φθαρτικὸν πάλιν ὕστερον· καὶ γὰρ ἡ φθορὰ μεταβολή τίς ἐστιν.
corruptible again later; and for the corruption change which is.
εἰ δὴ ταῦτ' ἀδύνατα, δῆλον ὡς ἔστιν ἀΐδιος κίνησις, ἀλλ'
If indeed these things are impossible, it is clear that is eternal motion, but
οὐχ ὁτὲ μὲν ἦν ὁτὲ δ' οὔ· καὶ γὰρ ἔοικε τὸ οὕτω
not sometimes on the one hand was sometimes but not; and for seems the thus
λέγειν πλάσματι μᾶλλον.
to say to creation more.

ὁμοίως δὲ καὶ τὸ λέγειν ὅτι πέφυκεν οὕτως καὶ ταύτην δεῖ νομίζειν
similarly but and the saying that has become thus and this must be considered
εἶναι ἀρχήν, ὅπερ ἔοικεν Ἐμπεδοκλῆς ἂν εἰπεῖν, ὡς τὸ
to be a principle, which seems Empedocles to have said, as the
κρατεῖν καὶ κινεῖν ἐν μέρει τὴν φιλίαν καὶ τὸ νεῖκος ὑπάρχει τοῖς
ruling and moving in part the friendship and the strife exists in the
πράγμασιν ἐξ ἀνάγκης, ἠρεμεῖν δὲ τὸν μεταξὺ χρόνον. τάχα
things out of necessity, to be at rest but the intervening time. perhaps
δὲ καὶ οἱ μίαν ἀρχὴν ποιοῦντες, ὥσπερ Ἀναξαγόρας, οὕτως ἂν
but also the one principle making, as Anaxagoras, thus would
εἴποιεν. ἀλλὰ μὴν οὐδέν γε ἄτακτον τῶν φύσει καὶ κατὰ
say. but indeed, nothing at all disorderly of the nature and according to

φύσιν· ἡ γὰρ φύσις αἰτία πᾶσιν τάξεως. τὸ δ' ἄπειρον

nature. the for nature is the cause of all order. the and infinite

πρὸς τὸ ἄπειρον οὐδένα λόγον ἔχει· τάξις δὲ πᾶσα λόγος. τὸ

towards the infinite no reason has; order but all word. the

δ' ἄπειρον χρόνον ἠρεμεῖν, εἶτα κινηθῆναί ποτε, τούτου δὲ

and infinite time to rest, then to be moved at some time, of this but

μηδεμίαν εἶναι διαφοράν, ὅτι νῦν μᾶλλον ἢ πρότερον, μηδ' αὖ

no to be difference, that now rather than previously, nor again

τινὰ τάξιν ἔχειν, οὐκέτι φύσεως ἔργον. ἢ γὰρ ἁπλῶς ἔχει τὸ

any order to have, no longer nature's work. Or for simply has the

φύσει, καὶ οὐχ ὁτὲ μὲν οὕτως ὁτὲ δ' ἄλλως, οἷον τὸ πῦρ

nature, and not sometimes indeed thus sometimes but otherwise, such as the fire

ἄνω φύσει φέρεται καὶ οὐχ ὁτὲ μὲν ὁτὲ δ' οὔ· ἢ λόγον

upwards by nature is carried and not sometimes indeed sometimes but not. Or reason

ἔχει τὸ μὴ ἁπλοῦν. διόπερ βέλτιον ὡς Ἐμπεδοκλῆς, κἂν

has the not simple. Therefore it is better as Empedocles, and even if

εἴ τις ἕτερος εἴρηκεν οὕτως ἔχειν, ἐν μέ ρει τὸ πᾶν ἠρεμεῖν

someone says another has stated thus to have, in me flows the all to rest

καὶ κινεῖσθαι πάλιν· τάξιν γὰρ ἤδη τιν' ἔχει τὸ τοιοῦτον. ἀλλὰ

and to be moved again. order for already some has the such. but

καὶ τοῦτο δεῖ τὸν λέγοντα μὴ φάναι μόνον, ἀλλὰ καὶ τὴν

also this is necessary the speaker not to say only, but also and the

αἰτίαν αὐτοῦ λέγειν, καὶ μὴ τίθεσθαι μηδὲν μηδ' ἀξιοῦν ἀξίωμ'

cause of it to state, and not to be posited anything nor to demand a claim,

ἄλογον, ἀλλ' ἢ ἐπαγωγὴν ἢ ἀπόδειξιν φέρειν· αὐτὰ μὲν

unreasonable, but either induction or demonstration to bring forth. These indeed

γὰρ οὐκ αἴτια τὰ ὑποτεθέντα, οὐδὲ τοῦτ' ἦν τὸ φιλότητι ἢ νείκει

for not causes the that are supposed, nor this was the friendship or quarrel

εἶναι, ἀλλὰ τῆς μὲν τὸ συνάγειν, τοῦ δὲ τὸ διακρίνειν. εἰ δὲ

to be, but of the indeed the gathering, of the but the distinguishing. if but

προσοριεῖται τὸ ἐν μέρει, λεκτέον ἐφ' ὧν οὕτως, ὥσπερ

is defined the in part, it must be said about those in this way, just as

ὅτι ἔστιν τι ὃ συνάγει τοὺς ἀνθρώπους, ἡ φιλία, καὶ

that there is something which gathers the people, the friendship, and

φεύγουσιν οἱ ἐχθροὶ ἀλλήλους· τοῦτο γὰρ ὑποτίθεται καὶ ἐν

flee the enemies each other. this for is suggested and in

τῷ ὅλῳ εἶναι· φαίνεται γὰρ ἐπί τινων οὕτως. τὸ δὲ καὶ δι'

the whole to be. it appears for upon some in this way. the but also through

ἴσων χρόνων δεῖται λόγου τινός. ὅλως δὲ τὸ νομίζειν

equal times it is necessary of a reason of some. generally but the to think

ἀρχὴν εἶναι ταύτην ἱκανήν, εἴ τι αἰεὶ ἢ ἔστιν οὕτως ἢ

beginning to be this sufficient, if anything always or is in this way or

γίγνεται, οὐκ ὀρθῶς ἔχει ὑπολαβεῖν, ἐφ' ὃ Δημόκριτος ἀνάγει

happens, not rightly it holds to assume, on which Democritus brings up

τὰς περὶ φύσεως αἰτίας, ὡς οὕτω καὶ τὸ πρότερον ἐγίγνετο·

the concerning nature causes, as thus and the former happened.

τοῦ δὲ ἀεὶ οὐκ [252b] ἀξιοῖ ἀρχὴν ζητεῖν, λέγων ἐπί τινων

of the but always not [252b] he considers beginning to seek, saying upon some

ὀρθῶς, ὅτι δ' ἐπὶ πάντων, οὐκ ὀρθῶς. καὶ γὰρ τὸ τρίγωνον ἔχει

rightly, that but upon all, not rightly. and for the triangle has

δυσὶν ὀρθαῖς ἀεὶ τὰς γωνίας ἴσας, ἀλλ' ὅμως ἐστίν τι

two right always the angles equal, but nevertheless it is something

τῆς ἀϊδιότητος ταύτης ἕτερον αἴτιον· τῶν μέντοι ἀρχῶν οὐκ

of the eternity this another cause. of the however principles not

ἔστιν ἕτερον αἴτιον ἀϊδίων οὐσῶν. ὅτι μὲν οὖν οὐδεὶς ἦν

is another cause eternal beings. that indeed therefore no one was

χρόνος οὐδ' ἔσται ὅτε κίνη σις οὐκ ἦν ἢ οὐκ ἔσται, εἰρήσθω

time nor will be when motion there not was or not will be, let it be said

τοσαῦτα.

thus much.

Κεφάλαιο 2

Chapter 2

Τὰ δὲ ἐναντία τούτοις οὐ χαλεπὸν λύειν. δόξειε δ' ἂν ἐκ

The however opposite to these not difficult to resolve. it would seem but if from

τῶν τοιῶνδε σκοποῦσιν ἐνδέχεσθαι μάλιστα κίνησιν εἶναί ποτε

the such considerations to be possible especially motion to be at any time

μὴ οὖσαν ὅλως,

not being altogether,

πρῶτον μὲν ὅτι οὐδεμία ἀΐδιος μεταβολή· μεταβολὴ γὰρ ἅπασα

first indeed that no eternal change; change for all

πέφυκεν ἔκ τινος εἴς τι, ὥστε ἀνάγκη πάσης μεταβολῆς

has arisen from something into something, so that necessity of all change

εἶναι πέρας τὰ ἐναντία ἐν οἷς γίγνεται, εἰς ἄπειρον δὲ κινεῖσθαι

to be a limit the opposites in which come into being, into the infinite but to be moved

μηδέν.

nothing.

ἔτι ὁρῶμεν ὅτι δυνατὸν κινηθῆναι μήτε κινούμενον μήτ' ἔχον

still we see that it is possible to be moved neither something moved nor having

ἐν ἑαυτῷ μηδεμίαν κίνησιν, οἷον ἐπὶ τῶν ἀψύχων, ὧν οὔτε

in itself no motion, such as in the inanimates, of which neither

μέρος οὐδὲν οὔτε τὸ ὅλον κινούμενον ἀλλ' ἠρεμοῦν κινεῖταί

part nothing nor the whole moved but at rest it is being moved

ποτε· προσῆκεν δὲ ἢ ἀεὶ κινεῖσθαι ἢ μηδέποτε, εἴπερ μὴ
at some time; it is relevant but or always to be moved or never, if indeed not
γίγνεται οὐκ οὖσα.
it happens not being.

πολὺ δὲ μάλιστα τὸ τοιοῦτον ἐπὶ τῶν ἐμψύχων εἶναι φανερόν·
much and especially the such upon the living beings to be evident.
οὐδεμιᾶς γὰρ ἐν ἡμῖν ἐνούσης κινήσεως ἐνίοτε, ἀλλ' ἡσυχάζοντες
of no for in us being of movement, sometimes, but remaining quiet
ὅμως κινούμεθά ποτε, καὶ ἐγγίγνεται ἐν ἡμῖν ἐξ ἡμῶν αὐτῶν
nonetheless we are moved at times, and it comes to be in us from us themselves
ἀρχὴ κινήσεως, κἂν μηθὲν ἔξωθεν κινήσῃ. τοῦτο γὰρ ἐπὶ
a beginning of movement, even if nothing from outside moves. this for upon
τῶν ἀψύχων οὐχ ὁρῶμεν ὁμοίως, ἀλλ' ἀεὶ κινεῖ τι αὐτὰ
the non-living beings not do we see similarly, but always moves something them
τῶν ἔξωθεν ἕτερον· τὸ δὲ ζῷον αὐτό φαμεν ἑαυτὸ κινεῖν. ὥστ'
of the outside other. the and animal itself we say itself to move. so
εἴπερ ἠρεμεῖ ποτὲ πάμπαν, ἐν ἀκινήτῳ κίνησις ἂν γίγνοιτο
if ever it is at rest at times completely, in immobile movement would occur
ἐξ αὐτοῦ καὶ οὐκ ἔξωθεν. εἰ δ' ἐν ζῴῳ τοῦτο δυνατὸν
from itself and not from outside. if but in an animal this possible
γενέσθαι, τί κωλύει τὸ αὐτὸ συμβῆναι καὶ κατὰ τὸ πᾶν; εἰ
to happen, what hinders the same to happen and according to the whole? if
γὰρ ἐν μικρῷ κόσμῳ γίγνεται, καὶ ἐν μεγάλῳ· καὶ εἰ ἐν τῷ
indeed in small world it comes to be, and in large. and if in the
κόσμῳ, κἂν τῷ ἀπείρῳ, εἴπερ ἐνδέχεται κινεῖσθαι τὸ ἄπειρον καὶ
world, and in the infinite, if indeed it is possible to be moved the infinite and

ἠρεμεῖν ὅλον.

to be at rest altogether.

τούτων δὴ τὸ μὲν πρῶτον λεχθέν, τὸ μὴ τὴν αὐτὴν ἀεὶ καὶ

of these indeed the indeed first said, the not the same always and

μίαν τῷ ἀριθμῷ εἶναι τὴν κίνησιν τὴν εἰς τὰ ἀντικείμενα, ὀρθῶς

one to the number to be the motion the into the objects, correctly

λέγεται. τοῦτο μὲν γὰρ ἴσως ἀναγκαῖον, εἴπερ μὴ ἀεὶ μίαν

is said. this indeed for perhaps necessary, if indeed not always one

καὶ τὴν αὐτὴν εἶναι δυνατὸν τὴν τοῦ αὐτοῦ καὶ ἑνὸς κίνησιν·

and the same to be possible the of the same and of one motion.

λέγω δ' οἷον πότερον τῆς μιᾶς χορδῆς εἷς καὶ ὁ αὐτὸς φθόγγος,

I say but as if whether of the one string, one and the same sound,

ἢ ἀεὶ ἕτερος, ὁμοίως ἐχούσης καὶ κινουμένης. ἀλλ' ὅμως

or always another, similarly having and moving. but nonetheless

ὁποτέρως ποτ' ἔχει, οὐδὲν κωλύει τὴν αὐ [253a] τὴν εἶναί

whichever at any time has, nothing prevents the same [253a] the to be

τινα τῷ συνεχῆ εἶναι καὶ ἀΐδιον· δῆλον δ' ἔσται μᾶλλον ἐκ

someone to the continuous to be and eternal. evident but it will be rather from

τῶν ὕστερον.

the later.

τὸ δὲ κινεῖσθαι μὴ κινούμενον οὐδὲν ἄτοπον, ἂν ὁτὲ μὲν

the and to be moved not being moved nothing strange, if at any time on the one hand

ᾖ τὸ κινῆσον ἔξωθεν, ὁτὲ δὲ μή. τοῦτο μέντοι πῶς ἂν

it is the moving from outside, at one time but not. this however how if

εἴη, ζητητέον, λέγω δὲ ὥστε τὸ αὐτὸ ὑπὸ τοῦ αὐτοῦ

it may be, it must be sought, I say but so that the same by the same

κινητικοῦ ὄντος ὁτὲ μὲν κινεῖσθαι ὁτὲ δὲ μή·

moving being at one time on the one hand to be moved at one time but not.

οὐδὲν γὰρ ἄλλ' ἀπορεῖ ὁ τοῦτο λέγων ἢ διὰ τί οὐκ ἀεὶ τὰ

nothing for else it lacks the this saying or through what not always the

μὲν ἠρεμεῖ τῶν ὄντων τὰ δὲ κινεῖται.

on one hand are at rest of the beings the but are moved.

μάλιστα δ' ἂν δόξειεν τὸ τρίτον ἔχειν ἀπορίαν, ὡς ἐγγιγνομένης

indeed but if it should seem the third to have a difficulty, as arising

οὐκ ἐνούσης πρότερον κινήσεως, τὸ συμβαῖνον ἐπὶ τῶν ἐμψύχων·

not existing beforehand of motion, the occurring in the living beings.

ἠρεμοῦν γὰρ πρότερον μετὰ ταῦτα βαδίζει, κινήσαντος

(it) being at rest for previously along with these it walks, (having) been set in motion

τῶν ἔξωθεν οὐδενός, ὡς δοκεῖ. τοῦτο δ' ἐστὶ ψεῦδος. ὁρῶμεν

of the external nothing, as it seems. this but is a falsehood. we see

γὰρ ἀεί τι κινούμενον ἐν τῷ ζῴῳ τῶν συμφύτων· τούτου

for always something moving in the animal of the compounds. of this

δὲ τῆς κινήσεως οὐκ αὐτὸ τὸ ζῷον αἴτιον, ἀλλὰ τὸ

but of the motion not it itself the animal (is) the cause, but the

περιέχον ἴσως. αὐτὸ δέ φαμεν αὑτὸ κινεῖν οὐ πᾶσαν κίνησιν,

containing perhaps. itself and we say itself to move not all movement,

ἀλλὰ τὴν κατὰ τόπον. οὐδὲν οὖν κωλύει, μᾶλλον δ' ἴσως

but the according to place. nothing then hinders, rather but perhaps

ἀναγκαῖον, ἐν μὲν τῷ σώματι πολλὰς ἐγγίγνεσθαι κινήσεις

necessary, in on the one hand the body many to come into being movements

ὑπὸ τοῦ περιέχοντος, τούτων δ' ἐνίας τὴν διάνοιαν ἢ τὴν ὄρεξιν

by the encompassing, of these but some the intellect or the desire

κινεῖν, ἐκείνην δὲ τὸ ὅλον ἤδη ζῷον κινεῖν, οἷον συμβαίνει

to move, that but the whole already living being to move, as it happens

περὶ τοὺς ὕπνους· αἰσθητικῆς μὲν γὰρ οὐδεμιᾶς ἐνούσης

concerning the sleeps. of sensation on the one hand for of no existing

κινήσεως, ἐνούσης μέντοι τινός, ἐγείρεται τὰ ζῷα πάλιν. ἀλλὰ

motion, existing however of a certain is raised the animals again. But

γὰρ φανερὸν ἔσται καὶ περὶ τούτων ἐκ τῶν ἑπομένων.

for it will be evident it will be also concerning these from the following.

Κεφάλαιο 3

Chapter 3

Ἀρχὴ δὲ τῆς σκέψεως ἥπερ καὶ περὶ τῆς λεχθείσης ἀπορίας,

Beginning but of the thought which and regarding of the stated perplexity,

διὰ τί ποτε ἔνια τῶν ὄντων ὁτὲ μὲν κινεῖται

through what at some time some of the beings sometimes on one hand moves

ὁτὲ δὲ ἠρεμεῖ πάλιν. ἀνάγκη δὴ ἤτοι πάντα

sometimes on the other hand is at rest again. necessity indeed either all

ἠρεμεῖν ἀεί, ἢ πάντα ἀεὶ κινεῖσθαι, ἢ τὰ μὲν κινεῖσθαι

to be at rest always, or all always to be moved, or the on one hand to be moved

τὰ δ' ἠρεμεῖν, καὶ πάλιν τούτων ἤτοι τὰ μὲν κινούμενα

the but to be at rest, and again of these either the on one hand moving

κινεῖσθαι ἀεὶ τὰ δ' ἠρεμοῦντα ἠρεμεῖν, ἢ πάντα πεφυκέναι

to be moved always the but at rest, to be at rest, or all to have arisen

ὁμοίως κινεῖσθαι καὶ ἠρεμεῖν, ἢ τὸ λοιπὸν ἔτι καὶ τρίτον.

similarly to be moved and to be at rest, or the remaining still and third.

ἐνδέχεται γὰρ τὰ μὲν ἀεὶ τῶν ὄντων ἀκίνητα εἶναι,

it may happen for the on one hand always of the beings to be unmoved, to be,

τὰ δ' ἀεὶ κινούμενα, τὰ δ' ἀμφοτέρων μεταλαμβάνειν· ὅπερ

the but always moving, the but of both to partake; which

ἡμῖν λεκτέον ἐστίν· τοῦτο γὰρ ἔχει λύσιν τε πάντων τῶν

to us must be said it is; this for has a solution and of all the

ἀπορουμένων, καὶ τέλος ἡμῖν ταύτης τῆς πραγματείας ἐστίν.

ones in difficulty, and finally to us of this the treatise it is.

τὸ μὲν οὖν πάντ' ἠρεμεῖν, καὶ τούτου ζητεῖν λόγον

the on the one hand therefore all to be at rest, and of this to seek a reason

ἀφέντας τὴν αἴσθησιν, ἀρρωστία τίς ἐστιν διανοίας, καὶ περὶ

having set aside the perception, illness what it is of thought, and about

ὅλου τινὸς ἀλλ' οὐ περὶ μέρους ἀμφισβήτησις· οὐδὲ μόνον

the whole something else not about a part dispute. nor only

πρὸς τὸν φυσικόν, ἀλλὰ πρὸς πάσας τὰς [253b]

with respect to the natural, but with respect to all the [253b]

ἐπιστήμας ὡς εἰπεῖν καὶ πάσας τὰς δόξας διὰ τὸ κινήσει χρῆσθαι

sciences as to say and all the opinions through the movement to be used

πάσας. ἔτι δ' αἱ περὶ τῶν ἀρχῶν ἐνστάσεις, ὥσπερ ἐν

all. Furthermore and the concerning the principles objections, just as in

τοῖς περὶ τὰ μαθήματα λόγοις οὐδέν εἰσιν πρὸς τὸν

the concerning the subjects discussions nothing exist with respect to the

μαθηματικόν, ὁμοίως δὲ καὶ ἐπὶ τῶν ἄλλων, οὕτως οὐδὲ περὶ

mathematical, similarly but also concerning the others, thus nor about

τοῦ νῦν ῥηθέντος πρὸς τὸν φυσικόν· ὑπόθεσις γὰρ ὅτι ἡ

the now said with respect to the natural. hypothesis for that the

φύσις ἀρχὴ τῆς κινήσεως.

nature principle of the movement.

σχεδὸν δὲ καὶ τὸ φάναι κινεῖσθαι πάντα ψεῦδος μέν, ἧττον
nearly but and the to say to be moved everything is a lie on one hand, less
δὲ τούτου παρὰ τὴν μέθοδον· ἐτέθη μὲν γὰρ ἡ φύσις
but of this compared to the method; has been placed indeed for the nature
ἐν τοῖς φυσικοῖς ἀρχή, καθάπερ κινήσεως, καὶ ἠρεμίας, ὅμως
in the physical principles, as of motion, and rest, however
δὲ φυσικὸν ἡ κίνησις· καί φασί τινες κινεῖσθαι τῶν ὄντων
but the natural the motion; and they say some to be moved of the beings
οὐ τὰ μὲν τὰ δ' οὔ, ἀλλὰ πάντα καὶ ἀεί, ἀλλὰ λανθάνειν
not the indeed the but not, but everything and always, but to be concealed
τοῦτο τὴν ἡμετέραν αἴσθησιν· πρὸς οὓς καίπερ οὐ διορίζοντας
this the our perception; towards whom although not distinguishing
ποίαν κίνησιν λέγουσιν, ἢ πάσας, οὐ χαλεπὸν ἀπαντῆσαι. οὔτε
what kind of motion they say, or all, not difficult to encounter. Neither
γὰρ αὐξάνεσθαι οὔτε φθίνειν οἷόν τε συνεχῶς, ἀλλ' ἔστι
for to increase nor to diminish is it possible and continuously, but there is
καὶ τὸ μέσον. ἔστι δ' ὅμοιος ὁ λόγος τῷ περὶ τοῦ τὸν
also the middle. it is and similar the discourse to the concerning of the the
σταλαγμὸν κατατρίβειν καὶ τὰ ἐκφυόμενα τοὺς λίθους διαιρεῖν· οὐ
drop to wear down and the that emerge the stones to divide; not
γὰρ εἰ τοσόνδε ἐξέωσεν ἢ ἀφεῖλεν ὁ σταλαγμός, καὶ τὸ ἥμισυ ἐν
for if so much made to flow or removed the drop, and the half in
ἡμίσει χρόνῳ πρότερον· ἀλλ' ὥσπερ ἡ νεωλκία, καὶ οἱ σταλαγμοὶ
half time earlier; but just as the tugging, and the drops
οἱ τοσοιδὶ τοσονδὶ κινοῦσιν, τὸ δὲ μέρος αὐτῶν ἐν οὐδενὶ χρόνῳ
the so many so much move, the but part of them in no time

τοσοῦτον. διαιρεῖται μὲν οὖν τὸ ἀφαιρεθὲν εἰς πλείω, ἀλλ'

so much. it is divided indeed thus the taken away into more, but

οὐδὲν αὐτῶν ἐκινήθη χωρίς, ἀλλ' ἅμα. φανερὸν οὖν ὡς

nothing of them was moved separately, but together. it is evident thus that

οὐκ ἀναγκαῖον ἀεί τι ἀπιέναι, ὅτι διαιρεῖται ἡ φθίσις

not necessary always something to go away, because it is divided the decay

εἰς ἄπειρα, ἀλλ' ὅλον ποτὲ ἀπιέναι. ὁμοίως δὲ καὶ ἐπ' ἀλλοιώσεως

into infinite, but all once to go away. likewise and also upon transformation

ὁποιασοῦν· οὐ γὰρ εἰ μεριστὸν εἰς ἄπειρα τὸ ἀλλοιούμενον, διὰ

of any kind; not for if divisible into infinite the transforming, therefore

τοῦτο καὶ ἡ ἀλλοίωσις, ἀλλ' ἀθρόα γίγνεται πολλάκις, ὥσπερ ἡ

this and the alteration, but en masse occurs often, just as the

πῆξις. ἔτι ὅταν τι νοσήσῃ, ἀνάγκη χρόνον γενέσθαι ἐν

coagulation. further when something falls sick, necessity time to come to be in

ᾧ ὑγιασθήσεται, καὶ μὴ ἐν πέρατι χρόνου μεταβάλλειν· ἀνάγκη

which it will be healed, and not in beyond time to change; necessity

δὲ εἰς ὑγίειαν μεταβάλλειν καὶ μὴ εἰς ἄλλο μηθέν. ὥστε τὸ

and into health to change and not into anything else at all. so that the

φάναι συνεχῶς ἀλλοιοῦσθαι λίαν ἐστὶ τοῖς φανεροῖς ἀμφισβητεῖν.

saying continuously to be altered very is for those evident to dispute.

εἰς τοὐναντίον γὰρ ἡ ἀλλοίωσις· ὁ δὲ λίθος οὔτε σκληρότερος

into the opposite, for the alteration; the but stone neither harder

γίγνεται οὔτε μαλακώτερος. κατά τε τὸ φέρεσθαι θαυμαστὸν εἰ

becomes nor softer. according to and the to be brought wonderful if

λέληθεν ὁ λίθος κάτω φερόμενος ἢ μένων ἐπὶ τῆς γῆς. ἔτι

has escaped notice the stone down being carried or remaining upon the earth. yet

δ' ἡ γῆ καὶ τῶν ἄλλων ἕκαστον ἐξ ἀνάγκης μένουσι μὲν ἐν

but the earth and of the others each out of necessity remain indeed in

τοῖς οἰκείοις τόποις, κινοῦνται δὲ βιαίως ἐκ τούτων· εἴπερ οὖν
the familiar places, are moved but violently from these. if ever therefore
ἔνι' αὐτῶν ἐστιν ἐν τοῖς οἰκείοις τόποις, [254a] ἀνάγκη μηδὲ
there is of them is in the familiar places, [254a] necessity not even
κατὰ τόπον πάντα κινεῖσθαι. ὅτι μὲν οὖν ἀδύνατον ἢ ἀεὶ
according to place all to be moved. that indeed therefore impossible or always
πάντα κινεῖσθαι ἢ ἀεὶ πάντα ἠρεμεῖν, ἐκ τούτων καὶ ἄλλων
all to be moved or always all to be at rest, out of these and other
τοιούτων πιστεύσειεν ἄν τις.
such would believe if someone.

ἀλλὰ μὴν οὐδὲ τὰ μὲν ἀεὶ ἐνδέχεται ἠρεμεῖν, τὰ δ'
but now neither the on the one hand always is possible to be at rest, the but
ἀεὶ κινεῖσθαι, ποτὲ δ' ἠρεμεῖν καὶ ποτὲ κινεῖσθαι
always to be in motion, sometimes but to be at rest and sometimes to be in motion
μηδέν. λεκτέον δ' ὅτι ἀδύνατον, ὥσπερ ἐπὶ τῶν
nothing. it must be said but that it is impossible, just as in regard to the
εἰρημένων πρότερον, καὶ ἐπὶ τούτων (ὁρῶμεν γὰρ ἐπὶ
mentioned earlier, and in regard to these we see for in regard to
τῶν αὐτῶν γιγνομένας τὰς εἰρημένας μεταβολάς), καὶ πρὸς
the same occurring the mentioned changes), and in addition to
τούτοις ὅτι μάχεται τοῖς φανεροῖς ὁ ἀμφισβητῶν· οὔτε
these that it contends against the evident the debater; neither
γὰρ αὔξησις οὔθ' ἡ βίαιος ἔσται κίνησις, εἰ μὴ κινήσεται
for increase nor the violent will be motion, if not it does not move
παρὰ φύσιν ἠρεμοῦν πρότερον. γένεσιν οὖν ἀναιρεῖ καὶ
contrary to nature being at rest earlier. genesis therefore negates and

φθορὰν οὗτος ὁ λόγος. σχεδὸν δὲ καὶ τὸ κινεῖσθαι γίγνεσθαί

destruction this the argument. nearly but also the being in motion to become

τι καὶ φθείρεσθαι δοκεῖ πᾶσιν· εἰς ὃ μὲν γὰρ

something and to be destroyed seems to all; into what on the one hand for

μεταβάλλει, γίγνεται τοῦτο ἢ ἐν τούτῳ, ἐξ οὗ δὲ μεταβάλλει,

it changes, it becomes this or in this, out of which but it changes,

φθείρεται τοῦτο ἢ ἐντεῦθεν. ὥστε δῆλον ὅτι τὰ

it is destroyed this or from here. Therefore it is evident that the things

μὲν κινεῖται, τὰ δ' ἠρεμεῖ ἐνίοτε.

on one hand are in motion, the things but are at rest sometimes.

τὸ δὲ πάντα ἀξιοῦν ὁτὲ μὲν ἠρεμεῖν ὁτὲ δὲ κινεῖσθαι,

the and all are worthy whenever indeed to be at rest whenever and to be in motion,

τοῦτ' ἤδη συναπτέον πρὸς τοὺς πάλαι λόγους. ἀρχὴν δὲ

this already must be connected to the former arguments. a beginning and

πάλιν ποιητέον ἀπὸ τῶν νῦν διορισθέντων, τὴν αὐτὴν ἥνπερ

again must be made from the now defined, the same which

ἠρξάμεθα πρότερον. ἢ γάρ τοι πάντα ἠρεμεῖ, ἢ πάντα κινεῖται,

we began earlier. Or for indeed all is at rest, or all is in motion,

ἢ τὰ μὲν ἠρεμεῖ τὰ δὲ κινεῖται τῶν ὄντων. καὶ εἰ τὰ μὲν

or the indeed is at rest the and is in motion of the beings. And if the indeed

ἠρεμεῖ τὰ δὲ κινεῖται, ἀνάγκη ἤτοι πάντα ὁτὲ μὲν ἠρεμεῖν

is at rest the and is in motion, necessity either all whenever indeed to be at rest

ὁτὲ δὲ κινεῖσθαι, <ἢ τὰ μὲν ἀεὶ ἠρεμεῖν τὰ δὲ ἀεὶ

whenever and to be in motion, <Or the indeed always to be at rest the and always

κινεῖσθαι>, ἢ τὰ μὲν ἀεὶ ἠρεμεῖν τὰ δὲ ἀεὶ κινεῖσθαι

to be in motion>, or the indeed always to be at rest the and always to be moved

αὐτῶν, τὰ δ' ὁτὲ μὲν ἠρεμεῖν ὁτὲ δὲ

of them, the but sometimes on the one hand to be at rest sometimes on the other hand

κινεῖσθαι. ὅτι μὲν τοίνυν οὐχ οἷόν τε πάντ' ἠρεμεῖν,

to be moved. that on the one hand therefore not possible and all to be at rest,

εἴρηται μὲν καὶ πρότερον, εἴπωμεν δὲ καὶ νῦν. εἰ γὰρ

it has been said on the one hand and earlier, let us say but also now. if indeed

καὶ κατ' ἀλήθειαν οὕτως ἔχει καθάπερ φασί τινες, εἶναι τὸ

even according to truth thus it is as some say some, to be the

ὂν ἄπειρον καὶ ἀκίνητον, ἀλλ' οὔτι φαίνεταί γε κατὰ τὴν

being infinite and unmoving, but not at all it appears indeed in terms of the

αἴσθησιν, ἀλλὰ κινεῖσθαι πολλὰ τῶν ὄντων. εἴπερ οὖν ἔστιν

perception, but to be moved many of the beings. if indeed therefore there is

δόξα ψευδὴς ἢ ὅλως δόξα, καὶ κίνησις ἔστιν, κἂν εἰ φαντασία,

belief false or wholly belief, and movement exists, even if if imagination,

κἂν εἰ ὁτὲ μὲν οὕτως δοκεῖ ὁτὲ δ' ἑτέρως· ἡ

even if if sometimes on the one hand thus it seems sometimes but otherwise. the

γὰρ φαντασία καὶ ἡ δόξα κινήσεις τινὲς εἶναι δοκοῦσιν. ἀλλὰ τὸ

for imagination and the glory movements certain to be seem. but the

μὲν περὶ τούτου σκοπεῖν, καὶ ζητεῖν λόγον ὧν βέλτιον

on the one hand about this to examine, and to seek an account of which better

ἔχομεν ἢ λόγου δεῖσθαι, κακῶς κρίνειν ἐστὶν τὸ βέλτιον καὶ

we have or of a discourse to require, badly to judge is the better and

τὸ χεῖρον, καὶ τὸ πιστὸν καὶ τὸ μὴ πιστόν, καὶ ἀρχὴν καὶ

the worse, and the trustworthy and the not trustworthy, and a beginning and

μὴ ἀρχήν. ὁμοίως δὲ ἀδύνατον καὶ τὸ πάντα κινεῖσθαι, ἢ τὰ

not a beginning. likewise but impossible and the all to be moved, or the

μὲν ἀεὶ κινεῖσθαι τὰ δ' ἀεὶ ἠρεμεῖν. πρὸς

on the one hand always to be moved the on the other hand always to be at rest. concerning

ἅπαντα γὰρ ταῦτα ἱκανὴ μία πίστις· [254b] ὁρῶμεν γὰρ ἔνια

all for these things sufficient one belief· [254b] we see for some

ὁτὲ μὲν κινούμενα ὁτὲ δ' ἠρεμοῦντα.

at one time on the one hand moving at another time on the other hand at rest.

ὥστε φανερὸν ὅτι ἀδύνατον ὁμοίως τὸ πάντα ἠρεμεῖν καὶ τὸ

therefore evident that impossible likewise the all to be at rest and the

πάντα κινεῖσθαι συνεχῶς τῷ τὰ μὲν ἀεὶ κινεῖσθαι

all to be in motion continuously to the the on the one hand always to be in motion

τὰ δ' ἠρεμεῖν ἀεί. λοιπὸν οὖν θεωρῆσαι πότερον πάντα

the but to be at rest always therefore therefore to consider whether all

τοιαῦτα οἷα κινεῖσθαι καὶ ἠρεμεῖν, ἢ ἔνια μὲν οὕτως, ἔνια δ'

such as to be in motion and to be at rest or some indeed in this way some but

ἀεὶ ἠρεμεῖ, ἔνια δ' ἀεὶ κινεῖται· τοῦτο γὰρ δεικτέον ἡμῖν.

always is at rest some but always is in motion this for it must be shown to us.

Κεφάλαιο 4

Chapter 4

Τῶν δὴ κινούντων καὶ κινουμένων τὰ μὲν κατὰ συμβεβηκὸς

Of the indeed movers and moved the on one hand according to coincidental

κινεῖ καὶ κινεῖται, τὰ δὲ καθ' αὑτά, κατὰ

moves and is moved, the on the other hand according to themselves, according to

συμβεβηκὸς μὲν οἷον ὅσα τε τῷ ὑπάρχειν τοῖς κινοῦσιν

coincidental on one hand such as as many and to the existence of those moving

ἢ κινουμένοις καὶ τὰ κατὰ μόριον, τὰ δὲ καθ'

or being moved and the according to part, the on the other hand according to

αὑτά, ὅσα μὴ τῷ ὑπάρχειν τῷ κινοῦντι ἢ τῷ κινουμένῳ,

themselves, as many not to the existence of the mover or to the being moved,

μηδὲ τῷ μόριόν τι αὐτῶν κινεῖν ἢ κινεῖσθαι.

nor to the part any of them to move or to be moved.

τῶν δὲ καθ' αὐτὰ τὰ μὲν ὑφ' ἑαυτοῦ τὰ δ' ὑπ'

of the but according to themselves the on the one hand by itself the and by

ἄλλου, καὶ τὰ μὲν φύσει τὰ δὲ βίᾳ καὶ παρὰ

another, and the on the one hand by nature the but by force and contrary to

φύσιν. τό τε γὰρ αὐτὸ ὑφ' αὑτοῦ κινούμενον φύσει κινεῖται,

nature. the and for it by itself being moved by nature is moved,

οἷον ἕκαστον τῶν ζῴων (κινεῖται γὰρ τὸ ζῷον αὐτὸ ὑφ' αὑτοῦ,

such as each of the animals (is moved for the animal itself by itself,

ὅσων δ' ἡ ἀρχὴ ἐν αὐτοῖς τῆς κινήσεως, ταῦτα φύσει φαμὲν

of which and the principle in them of the motion, these by nature we say

κινεῖσθαι· διὸ τὸ μὲν ζῷον ὅλον φύσει αὐτὸ

to be moved. therefore the on the one hand animal as a whole by nature it

ἑαυτὸ κινεῖ, τὸ μέντοι σῶμα ἐνδέχεται καὶ φύσει καὶ παρὰ

itself moves, the however body may also by nature and contrary to

φύσιν κινεῖσθαι· διαφέρει γὰρ ὁποίαν τε ἂν κίνησιν κινούμενον

nature to be moved; differs for whatever and if motion being moved

τύχῃ καὶ ἐκ ποίου στοιχείου συνεστηκός), καὶ τῶν ὑπ' ἄλλου

it happens and out of which element being composed), and of the by another

κινουμένων τὰ μὲν φύσει κινεῖται τὰ δὲ παρὰ

moving the on the one hand naturally is moved the on the other hand contrary to

φύσιν, παρὰ φύσιν μὲν οἷον τὰ γεηρὰ ἄνω καὶ τὸ πῦρ

nature, contrary to nature on the one hand such as the heavy upward and the fire

κάτω, ἔτι δὲ τὰ μόρια τῶν ζῴων πολλάκις κινεῖται παρὰ

downward, still and the parts of the animals many times is moved contrary to

φύσιν, παρὰ τὰς θέσεις καὶ τοὺς τρόπους τῆς κινήσεως. καὶ
nature, contrary to the positions and the ways of the movement. and
μάλιστα τὸ ὑπό τινος κινεῖσθαι τὸ κινούμενον ἐν τοῖς παρὰ
especially the under someone to be moved the moved in those contrary to
φύσιν κινουμένοις ἐστὶ φανερὸν διὰ τὸ δῆλον εἶναι ὑπ' ἄλλου
nature moving is evident through the clear to be by another
κινούμενον. μετὰ δὲ τὰ παρὰ φύσιν τῶν κατὰ φύσιν τὰ
being moved. after and the contrary to nature of the according to nature the
αὐτὰ ὑφ' αὑτῶν, οἷον τὰ ζῷα· οὐ γὰρ τοῦτ' ἄδηλον, εἰ ὑπό
same under of them, like the animals. not for this is unclear, if by
τινος κινεῖται, ἀλλὰ πῶς δεῖ διαλαβεῖν αὐτοῦ τὸ κινοῦν
something is moved, but how it is necessary to understand it the mover
καὶ τὸ κινούμενον· ἔοικεν γὰρ ὥσπερ ἐν τοῖς πλοίοις καὶ τοῖς μὴ
and the moved; it seems for just as in the ships and the not
φύσει συνισταμένοις, οὕτω καὶ ἐν τοῖς ζῴοις εἶναι διῃρημένον τὸ
by nature constituted, so and in the animals to be divided the
κινοῦν καὶ τὸ κινούμενον, καὶ οὕτω τὸ ἅπαν αὐτὸ αὑτὸ κινεῖν.
mover and the moved, and so the whole itself itself to move.

μάλιστα δ' ἀπορεῖται τὸ λοιπὸν τῆς εἰρημένης τελευταίας διαιρέσεως·
indeed but is perplexed the remaining of the aforementioned last division.
τῶν γὰρ ὑπ' ἄλλου κινουμένων τὰ μὲν παρὰ φύσιν ἐθήκαμεν
of the for by another moving the on one hand contrary to nature, we have set
κινεῖσθαι, τὰ δὲ λείπεται ἀντιθεῖναι [255a] ὅτι φύσει. ταῦτα δ'
to move, the but it is left to oppose [255a] that by nature. these but
ἐστὶν ἃ τὴν ἀπορίαν παράσχοι ἂν ὑπὸ τίνος κινεῖται, οἷον τὰ
are what the perplexity might provide if by someone is moved, such as the

κοῦφα καὶ τὰ βαρέα. ταῦτα γὰρ εἰς μὲν τοὺς ἀντικειμένους

light and the heavy. these for to on one hand the opposing

τόπους βίᾳ κινεῖται, εἰς δὲ τοὺς οἰκείους, τὸ μὲν κοῦφον

places by force are moved, to but the own, the on one hand light

ἄνω τὸ δὲ βαρὺ κάτω, φύσει· τὸ δ' ὑπὸ τίνος οὐκέτι φανερόν,

upwards the but heavy downwards, by nature. the but by someone no longer clear,

ὥσπερ ὅταν κινῶνται παρὰ φύσιν. τό τε γὰρ αὐτὰ ὑφ'

just as when they are moved contrary to nature. the and for they under

αὑτῶν φάναι ἀδύνατον· ζωτικόν τε γὰρ τοῦτο καὶ τῶν ἐμψύχων

themselves to say impossible· vital and for this and of the living

ἴδιον, καὶ ἱστάναι ἂν ἐδύνατο αὐτὰ αὑτά (λέγω δ' οἷον, εἰ

peculiar, and to stand if it could them themselves (I say but such as, if

τοῦ βαδίζειν αἴτιον αὐτῷ, καὶ τοῦ μὴ βαδίζειν), ὥστ' εἰ ἐπ' αὐτῷ

the to walk cause to itself, and the not to walk), so that if upon it

τὸ ἄνω φέρεσθαι τῷ πυρί, δῆλον ὅτι ἐπ' αὐτῷ καὶ τὸ κάτω.

the above to be carried by the fire, it is clear that upon it and the below.

ἄλογον δὲ καὶ τὸ μίαν κίνησιν κινεῖσθαι μόνην ὑφ' αὑτῶν,

unreasonable but and the single motion to be moved alone under themselves,

εἴγε αὐτὰ ἑαυτὰ κινοῦσιν. ἔτι πῶς ἐνδέχεται συνεχές τι

if indeed they themselves are moving. still how is it possible continuous something

καὶ συμφυὲς αὐτὸ ἑαυτὸ κινεῖν; ᾗ γὰρ ἓν καὶ συνεχὲς μὴ

and of the nature it itself to move? In what way for one and continuous not

ἁφῇ, ταύτῃ ἀπαθές· ἀλλ' ᾗ κεχώρισται, ταύτῃ τὸ

touching, by this without feeling· but in which it has been separated, by this the

μὲν πέφυκε ποιεῖν τὸ δὲ πάσχειν. οὔτ' ἄρα τούτων οὐθὲν

indeed has come to be to make the but to suffer. neither then of these nothing

αὐτὸ ἑαυτὸ κινεῖ (συμφυῆ γάρ), οὔτ' ἄλλο συνεχὲς οὐδέν,

it itself moves (being composite for), neither another continuous nothing,

ἀλλ' ἀνάγκη διῃρῆσθαι τὸ κινοῦν ἐν ἑκάστῳ πρὸς τὸ κινούμενον,

but necessity must be divided the moving in each towards the moved,

οἷον ἐπὶ τῶν ἀψύχων ὁρῶμεν, ὅταν κινῇ τι τῶν ἐμψύχων.

such as upon the inanimate we observe, when it moves something of the living beings.

ἀλλὰ συμβαίνει καὶ ταῦτα ὑπό τινος ἀεὶ κινεῖσθαι· γένοιτο

but it happens also these things through someone always to be moved; let it be

δ' ἂν φανερὸν διαιροῦσι τὰς αἰτίας.

thus if clear they distinguish the causes.

ἔστιν δὲ καὶ ἐπὶ τῶν κινούντων λαβεῖν τὰ εἰρημένα· τὰ μὲν

is but and upon the moving to take the mentioned the on one hand

γὰρ παρὰ φύσιν αὐτῶν κινητικά ἐστιν, οἷον ὁ μοχλὸς οὐ

for contrary to nature of them kinetic is, such as the lever not

φύσει τοῦ βάρους κινητικός, τὰ δὲ φύσει, οἷον τὸ ἐνεργείᾳ

by nature of the weight kinetic, the but by nature, such as the in action

θερμὸν κινητικὸν τοῦ δυνάμει θερμοῦ. ὁμοίως δὲ καὶ ἐπὶ τῶν

hot kinetic of the power of heat. similarly but also upon the

ἄλλων τῶν τοιούτων.

other the such.

καὶ κινητὸν δ' ὡσαύτως φύσει τὸ δυνάμει ποιὸν ἢ ποσὸν ἢ

and moving but likewise by nature the potentially of what kind or of how much or

πού, ὅταν ἔχῃ τὴν ἀρχὴν τὴν τοιαύτην ἐν αὑτῷ καὶ μὴ κατὰ

where, whenever it has the principle the such in itself and not according to

συμβεβηκός (εἴη γὰρ ἂν τὸ αὐτὸ καὶ ποιὸν καὶ ποσόν,

accident (it would be for perhaps the same and of what kind and of how much,

ἀλλὰ θατέρῳ θάτερον συμβέβηκεν καὶ οὐ καθ' αὑτὸ ὑπάρχει).
but to another different it has happened and not according to itself exists).
τὸ δὴ πῦρ καὶ ἡ γῆ κινοῦνται ὑπό τινος βίᾳ μὲν ὅταν
the indeed fire and the earth are moved by something violently indeed whenever
παρὰ φύσιν, φύσει δ' ὅταν εἰς τὰς αὑτῶν ἐνεργείας δυνάμει
against nature, by nature but whenever into the their activities potentially
ὄντα. ἐπεὶ δὲ τὸ δυνάμει πλεοναχῶς λέγεται, τοῦτ' αἴτιον τοῦ
being. since but the potentially redundantly it is said, this is the cause of the
μὴ φανερὸν εἶναι ὑπὸ τίνος τὰ τοιαῦτα κινεῖται, οἷον τὸ πῦρ
not manifest to be by someone the such things is moved, such as the fire
ἄνω καὶ ἡ γῆ κάτω. ἔστι δὲ δυνάμει ἄλλως ὁ μανθάνων
upwards and the earth downwards. is but in power otherwise the learner
ἐπιστήμων καὶ ὁ ἔχων ἤδη καὶ μὴ ἐνεργῶν. ἀεὶ δ', ὅταν
expert and the having already and not acting. always but, whenever
ἅμα τὸ ποιητικὸν καὶ τὸ παθητικὸν ὦσιν, γίγνεται ἐνεργείᾳ
together the active and the passive are, it happens with activity
τὸ δυ [255b] νατόν, οἷον τὸ μανθάνον ἐκ δυνάμει ὄντος
the possible [255b] becoming, such as the learner from power being
ἕτερον γίγνεται δυνάμει (ὁ γὰρ ἔχων ἐπιστήμην μὴ θεωρῶν δὲ
another becomes by power (the for having knowledge not seeing but
δυνάμει ἐστὶν ἐπιστήμων πως, ἀλλ' οὐχ ὡς καὶ πρὶν μαθεῖν),
in power is an expert in a way, but not as also before learning),
ὅταν δ' οὕτως ἔχῃ, ἐάν τι μὴ κωλύῃ, ἐνεργεῖ καὶ θεωρεῖ,
whenever but thus has, if anything not hinders, acts and sees,
ἢ ἔσται ἐν τῇ ἀντιφάσει καὶ ἐν ἀγνοίᾳ. ὁμοίως δὲ ταῦτ' ἔχει
or will be in the in contradiction and in ignorance. Likewise but these have
καὶ ἐπὶ τῶν φυσικῶν· τὸ γὰρ ψυχρὸν δυνάμει θερμόν,
also concerning the natural things. For indeed the cold in potency hot,

ὅταν δὲ μεταβάλῃ, ἤδη πῦρ, καίει δέ, ἂν μή τι

when but it changes, it already becomes fire, it burns indeed, if not something

κωλύῃ καὶ ἐμποδίζῃ. ὁμοίως δ' ἔχει καὶ περὶ τὸ βαρὺ καὶ

hinders and obstructs. Likewise but it holds also concerning the heavy and

κοῦφον· τὸ γὰρ κοῦφον γίγνεται ἐκ βαρέος, οἷον ἐξ ὕδατος ἀήρ

light. For indeed the light becomes from the heavy, as from water air

(τοῦτο γὰρ δυνάμει πρῶτον), καὶ ἤδη κοῦφον, καὶ ἐνεργήσει γ'

(this indeed in potency first), and already light, and it will act indeed

εὐθύς, ἂν μή τι κωλύῃ. ἐνέργεια δὲ τοῦ κούφου τὸ ποὺ

immediately, if not something hinders. The activity but of the light the place

εἶναι καὶ ἄνω, κωλύεται δ', ὅταν ἐν τῷ ἐναντίῳ τόπῳ ᾖ. καὶ τοῦθ'

to be and upwards, is hindered but when in the opposite place. is. and this

ὁμοίως ἔχει καὶ ἐπὶ τοῦ ποσοῦ καὶ ἐπὶ τοῦ ποιοῦ. καίτοι τοῦτο

likewise has and upon the quantity and upon the quality. And yet this

ζητεῖται, διὰ τί ποτε κινεῖται εἰς τὸν αὐτῶν τόπον τὰ

is sought, because of what at any time is moved into the their place the

κοῦφα καὶ τὰ βαρέα. αἴτιον δ' ὅτι πέφυκέν οι, καὶ τοῦτ' ἔστιν

light and the heavy. The reason but that it has been created, and this is

τὸ κούφῳ καὶ βαρεῖ εἶναι, τὸ μὲν τῷ ἄνω τὸ δὲ

the light and heavy to be, the on the one hand the upper the on the other hand

τῷ κάτω διωρισμένον. δυνάμει δ' ἐστὶν κοῦφον καὶ βαρὺ πολλαχῶς,

the lower delineated. In power but it is light and heavy in many ways,

ὥσπερ εἴρηται· ὅταν τε γὰρ ᾖ ὕδωρ, δυνάμει γέ πώς

as has been stated; when both for is water, in power certainly somehow

ἐστι κοῦφον, καὶ ὅταν ἀήρ, ἔστιν ὡς ἔτι δυνάμει (ἐνδέχεται γὰρ

it is light, and when air, is as still in power (it may be for

ἐμποδιζόμενον μὴ ἄνω εἶναι)· ἀλλ' ἐὰν ἀφαιρεθῇ τὸ ἐμποδίζον,

resisting not upwards to be) but if it is removed the hindrance,

ἐνεργεῖ καὶ ἀεὶ ἀνωτέρω γίγνεται. ὁμοίως δὲ καὶ τὸ ποιὸν εἰς τὸ

it acts and always above it becomes. Similarly but and the quality into the

ἐνεργείᾳ εἶναι μεταβάλλει· εὐθὺς γὰρ θεωρεῖ τὸ ἐπιστῆμον,

activity to be it changes; immediately for it perceives the knowledge,

ἐὰν μή τι κωλύῃ· καὶ τὸ ποσὸν ἐκτείνεται, ἐὰν μή τι

if not anything hinders; and the quantity extends, if not anything

κωλύῃ. ὁ δὲ τὸ ὑφιστάμενον καὶ κωλῦον κινήσας ἔστιν ὡς κινεῖ

hinders. the but the underlying and hindering having moved it is as it moves

ἔστι δ' ὡς οὔ, οἷον ὁ τὸν κίονα ὑποσπάσας ἢ ὁ τὸν λίθον

it is but as not, like the the column having undermined or the the stone

ἀφελὼν ἀπὸ τοῦ ἀσκοῦ ἐν τῷ ὕδατι· κατὰ συμβεβηκὸς γὰρ

having removed from the bag in the water; according to chance for

κινεῖ, ὥσπερ καὶ ἡ ἀνακλασθεῖσα σφαῖρα οὐχ ὑπὸ τοῦ τοίχου

it moves, just as and the reflected sphere not under the wall

ἐκινήθη ἀλλ' ὑπὸ τοῦ βάλλοντος. ὅτι μὲν τοίνυν οὐδὲν

was moved but under the throwing. that on the one hand therefore nothing

τούτων αὐτὸ κινεῖ ἑαυτό, δῆλον· ἀλλὰ κινήσεως ἀρχὴν ἔχει,

of these it moves itself, it is clear; but of motion a principle it has,

οὐ τοῦ κινεῖν οὐδὲ τοῦ ποιεῖν, ἀλλὰ τοῦ πάσχειν.

not of the to move nor of the to make, but of the to suffer.

εἰ δὴ πάντα τὰ κινούμενα ἢ φύσει κινεῖται ἢ παρὰ φύσιν καὶ

if indeed all the moving or by nature is moved or contrary to nature and

βίᾳ, καὶ τά τε βίᾳ καὶ παρὰ φύσιν πάντα ὑπό τινος καὶ

by force, and the and by force and contrary to nature all under someone and

ὑπ' ἄλλου, τῶν δὲ φύσει πάλιν τά θ' ὑφ' αὑτῶν κινούμενα ὑπό

under another, of the but by nature again the and under themselves moving under

τινος κινεῖται καὶ τὰ μὴ ὑφ' αὑτῶν, οἷον τὰ κοῦφα καὶ τὰ βαρέα

someone is moved and the not under themselves, such as the light and the heavy

[256a] (ἢ γὰρ ὑπὸ τοῦ γεννήσαντος καὶ ποιήσαντος κοῦφον ἢ βαρύ,

[256a] (or for under the generator and creator light or heavy,

ἢ ὑπὸ τοῦ τὰ ἐμποδίζοντα καὶ κωλύοντα λύσαντος), ἅπαντα ἂν τὰ

or under the the obstructing and hindering loosening), all would the

κινούμενα ὑπό τινος κινοῖτο.

moving under someone would be moved.

Κεφάλαιο 5

Chapter 5

Τοῦτο δὲ διχῶς· ἢ γὰρ οὐ δι' αὐτὸ τὸ κινοῦν, ἀλλὰ δι' ἕτερον

This but in two ways: or for not by itself the moving, but by another

ὃ κινεῖ τὸ κινοῦν, ἢ δι' αὐτό, καὶ τοῦτο ἢ πρῶτον μετὰ τὸ

which moves the moving, or by it, and this or first after the

ἔσχατον ἢ διὰ πλειόνων, οἷον ἡ βακτηρία κινεῖ τὸν λίθον

last or through more, such as the rod moves the stone

καὶ κινεῖται ὑπὸ τῆς χειρὸς κινουμένης ὑπὸ τοῦ ἀνθρώπου, οὗτος

and is moved by the hand moving by the person, this

δ' οὐκέτι τῷ ὑπ' ἄλλου κινεῖσθαι. ἄμφω δὴ κινεῖν φαμέν,

but no longer to the by another to be moved. Both indeed to move we say,

καὶ τὸ τελευταῖον καὶ τὸ πρῶτον τῶν κινούντων, ἀλλὰ μᾶλλον

and the last and the first of the movers, but rather

τὸ πρῶτον· ἐκεῖνο γὰρ κινεῖ τὸ τελευ ταῖον, ἀλλ' οὐ τοῦτο

the first: that for moves the end, that is, but not this

τὸ πρῶτον, καὶ ἄνευ μὲν τοῦ πρώτου τὸ τελευταῖον οὐ

the first, and without on the one hand of the first the last not

κινήσει, ἐκεῖνο δ' ἄνευ τούτου, οἷον ἡ βακτηρία οὐ κινήσει μὴ
will move, that but without this, as if the bacterium not will move unless
κινοῦντος τοῦ ἀνθρώπου. εἰ δὴ ἀνάγκη πᾶν τὸ κινούμενον ὑπό
moving of the man. if indeed necessity all the moving by
τινός τε κινεῖσθαι, καὶ ἢ ὑπὸ κινουμένου ὑπ' ἄλλου ἢ μή,
something and to be moved, and either by something moving by another or not,
καὶ εἰ μὲν ὑπ' ἄλλου [κινουμένου], ἀνάγκη τι εἶναι
and if on the one hand by another [moving], necessity something to be
κινοῦν ὃ οὐχ ὑπ' ἄλλου πρῶτον, εἰ δὲ τοιοῦτο τὸ πρῶτον,
moving that which not by another first, if but such the first,
οὐκ ἀνάγκη θάτερον (ἀδύνατον γὰρ εἰς ἄπειρον ἰέναι τὸ κινοῦν
not necessity another (impossible for into infinity to go the moving
καὶ κινούμενον ὑπ' ἄλλου αὐτό· τῶν γὰρ ἀπείρων οὐκ ἔστιν
and moving by another itself; of the for infinite there is not is
οὐδὲν πρῶτον)—εἰ οὖν ἅπαν μὲν τὸ κινούμενον ὑπό
anything first)-if therefore all on the one hand the moving by
τινος κινεῖται, τὸ δὲ πρῶτον κινοῦν κινεῖται μέν, οὐχ ὑπ'
a certain is moved, the but first causing is moved on the one hand, not by
ἄλλου δέ, ἀνάγκη αὐτὸ ὑφ' αὑτοῦ κινεῖσθαι.
another on the other hand, necessity it by itself to be moved.

ἔτι δὲ καὶ ὧδε τὸν αὐτὸν τοῦτον λόγον ἔστιν ἐπελθεῖν. πᾶν
still and also here the same this argument it is to approach. everything
γὰρ τὸ κινοῦν τί τε κινεῖ καὶ τινί. ἢ γὰρ αὑτῷ κινεῖ τὸ
for the moving what and moves and to something. or for to itself moves the
κινοῦν ἢ ἄλλῳ, οἷον ἄνθρωπος ἢ αὐτὸς ἢ τῇ βακτηρίᾳ, καὶ
moving or to another, such as a human or itself or to the stick, and

ὁ ἄνεμος κατέβαλεν ἢ αὐτὸς ἢ ὁ λίθος ὃν ἔωσεν. ἀδύνατον
the wind has thrown down or itself or the stone which it has cast. impossible
δὲ κινεῖν ἄνευ τοῦ αὐτὸ αὑτῷ κινοῦντος τὸ ᾧ κινεῖ· ἀλλ'
and to move without the same to itself moving the to which it moves; but
εἰ μὲν αὐτὸ αὑτῷ κινεῖ, οὐκ ἀνάγκη ἄλλο εἶναι ᾧ κινεῖ,
if indeed it to itself moves, not necessary another to be to which it moves,
ἂν δὲ ᾖ ἕτερον τὸ ᾧ κινεῖ, ἔστιν τι ὃ κινήσει οὐ
if but it is another the to which it moves, it is something which will move not
τινὶ ἀλλ' αὑτῷ, ἢ εἰς ἄπειρον εἶσιν. εἰ οὖν κινούμενόν
to something but to itself, or into the infinite it goes. if therefore something moving
τι κινεῖ, ἀνάγκη στῆναι καὶ μὴ εἰς ἄπειρον ἰέναι· εἰ γὰρ
something moves, necessity to stand and not into the infinite to go; if for
ἡ βακτηρία κινεῖ τῷ κινεῖσθαι ὑπὸ τῆς χειρός, ἡ χεὶρ κινεῖ
the rod moves by the being moved by the hand, the hand moves
τὴν βακτηρίαν· εἰ δὲ καὶ ταύτῃ ἄλλο κινεῖ, καὶ ταύτην ἕτερόν
the rod; if but also in this another moves, and this other
τι τὸ κινοῦν. ὅταν δή τινι κινῇ ἀεὶ ἕτερον, ἀνάγκη
something the mover. whenever indeed to something moves always another, necessity
εἶναι πρότερον τὸ αὐτὸ αὑτῷ κινοῦν. εἰ οὖν κινεῖται μὲν
to be prior the same to itself moving. if therefore it is moved indeed
τοῦτο, μὴ ἄλλο δὲ τὸ κινοῦν αὐτό, ἀνάγκη αὐτὸ αὑτὸ [256b]
this, not another but the mover itself, necessity it itself [256b]
κινεῖν· ὥστε καὶ κατὰ τοῦτον τὸν λόγον ἤτοι εὐθὺς τὸ
to move; so that also according to this the account namely immediately the
κινούμενον ὑπὸ τοῦ αὐτὸ κινοῦντος κινεῖται, ἢ ἔρχεταί ποτε εἰς
moving by the self mover is moved, or comes at any time into
τὸ τοιοῦτον.
the such

πρὸς δὲ τοῖς εἰρημένοις καὶ ὧδε σκοποῦσι ταὐτὰ συμβήσεται ταῦτα.

to but the mentioned and here aiming the same will occur these.

εἰ γὰρ ὑπὸ κινουμένου κινεῖται τὸ κινούμενον πᾶν, ἤτοι τοῦτο

if for by the moving it is moved the moving everything, either this

ὑπάρχει τοῖς πράγμασιν κατὰ συμ

exists for the things according to (not translated, typically a part of longer term)

βεβηκός, ὥστε κινεῖν μὲν κινούμενον, οὐ μέντοι διὰ

having happened, so that to move on the one hand the moving body, not however through

τὸ κινεῖσθαι αὐτό, ἢ οὔ, ἀλλὰ καθ' αὑτό. πρῶτον μὲν

the being moved itself, or not, but according to itself. first on the one hand

οὖν εἰ κατὰ συμβεβηκός, οὐκ ἀνάγκη κινεῖσθαι τὸ κινοῦν.

therefore if according to contingent, not necessity to be moved the moving.

εἰ δὲ τοῦτο, δῆλον ὡς ἐνδέχεταί ποτε μηδὲν κινεῖσθαι

if but this, it is evident that it is possible at some time nothing to be moved

τῶν ὄντων· οὐ γὰρ ἀναγκαῖον τὸ συμβεβηκός, ἀλλ' ἐνδεχόμενον

of the beings. not for necessary the contingent, but it is possible

μὴ εἶναι. ἐὰν οὖν θῶμεν τὸ δυνατὸν εἶναι, οὐδὲν ἀδύνατον

not to be. if therefore we assume the possible to be, nothing impossible

συμβήσεται, ψεῦδος δ' ἴσως. ἀλλὰ τὸ κίνησιν μὴ εἶναι ἀδύνατον·

will occur, falsehood but perhaps. but the movement not to be impossible·

δέδεικται γὰρ πρότερον ὅτι ἀνάγκη κίνησιν ἀεὶ εἶναι.

it has been shown for earlier that necessity movement always to be.

καὶ εὐλόγως δὲ τοῦτο συμβέβηκεν. τρία γὰρ ἀνάγκη εἶναι, τό τε

and rightly but this has happened. three for necessity to be, the and

κινούμενον καὶ τὸ κινοῦν καὶ τὸ ᾧ κινεῖ. τὸ μὲν οὖν

moving and the moving and the which moves. the on one hand therefore

κινούμενον ἀνάγκη κινεῖσθαι, κινεῖν δ' οὐκ ἀνάγκη· τὸ δ' ᾧ

moving necessity to be moved, to move but not necessity. the but which

κινεῖ, καὶ κινεῖν καὶ κινεῖσθαι (συμμεταβάλλει γὰρ τοῦτο ἅμα

moves, and to move and to be moved (interchanges for this together

καὶ κατὰ τὸ αὐτὸ τῷ κινουμένῳ ὄν· δῆλον δ' ἐπὶ τῶν

and according to the same to the moving being. it is clear but upon those

κατὰ τόπον κινούντων· ἅπτεσθαι γὰρ ἀλλήλων ἀνάγκη μέχρι

according to place moving. to touch for one another necessity until

τινός)· τὸ δὲ κινοῦν οὕτως ὥστ' εἶναι μὴ ᾧ κινεῖ, ἀκίνητον.

something.) the but moving thus so that to be not which moves, immovable.

ἐπεὶ δ' ὁρῶμεν τὸ ἔσχατον, ὃ κινεῖσθαι μὲν δύναται,

since but we see the ultimate, which to be moved on one hand can,

κινήσεως δ' ἀρχὴν οὐκ ἔχει, καὶ ὃ κινεῖται μέν, οὐχ ὑπ' ἄλλου

of motion that beginning not has, and what is moved indeed, not by another

δὲ ἀλλ' ὑφ' αὑτοῦ, εὔλογον, ἵνα μὴ ἀναγκαῖον εἴπωμεν, καὶ

but rather under itself, reasonable, so that not necessary we may say, and

τὸ τρίτον εἶναι ὃ κινεῖ ἀκίνητον ὄν. διὸ καὶ Ἀναξαγόρας

the third to be that moves immovable being. therefore also Anaxagoras

ὀρθῶς λέγει, τὸν νοῦν ἀπαθῆ φάσκων καὶ ἀμιγῆ εἶναι, ἐπειδή γε

correctly says, the mind unmoved claiming and pure to be, since indeed

κινήσεως ἀρχὴν αὐτὸν εἶναι ποιεῖ· οὕτω γὰρ μόνως ἂν κινοίη

of motion the beginning of itself to be makes; thus for alone would move

ἀκίνητος ὢν καὶ κρατοίη ἀμιγὴς ὤν.

immovable being and would hold pure being.

ἀλλὰ μὴν εἰ μὴ κατὰ συμβεβηκὸς ἀλλ' ἐξ ἀνάγκης κινεῖται

but indeed if not according to coincidental but out of necessity it is moved

τὸ κινοῦν, εἰ δὲ μὴ κινοῖτο, οὐκ ἂν κινοίη, ἀνάγκη τὸ

the moving, if and not it were to be moved, not would it move, necessity the

κινοῦν, ᾗ κινεῖται, ἤτοι οὕτω κινεῖσθαι ὡς γε κατὰ τὸ

moving, by which it is moved, or thus to be moved as indeed according to the

αὐτὸ εἶδος τῆς κινήσεως, ἢ καθ' ἕτερον. λέγω δ' ἤτοι τὸ

same form of the movement, or according to another. I say but or the

θερμαῖνον καὶ αὐτὸ θερμαίνεσθαι καὶ τὸ ὑγιάζον ὑγιάζεσθαι

warming and itself to be warmed and the health-giving to be health-giving

καὶ τὸ φέρον φέρεσθαι, ἢ τὸ ὑγιάζον φέρεσθαι, τὸ δὲ φέρον

and the carrying to be carried, or the health-giving to be carried, the but carrying

αὐξάνεσθαι. ἀλλὰ φανερὸν ὅτι ἀδύνατον· δεῖ γὰρ μέχρι

to increase. but clear that it is impossible; it is necessary for until

[257a] τῶν ἀτόμων διαιροῦντα λέγειν, οἷον εἴ τι διδάσκει

[257a] of the atoms to divide to say, such as if anything teaches

γεωμετρεῖν, τοῦτο διδάσκεσθαι γεωμετρεῖν τὸ αὐτό, ἢ εἰ ῥιπτεῖ,

geometry, this to be taught to geometrize the same, or if it is thrown,

ῥιπτεῖσθαι τὸν αὐτὸν τρόπον τῆς ῥίψεως· ἢ οὕτως μὲν μή,

to be thrown the same manner of the throwing; or thus indeed not,

ἄλλο δ' ἐξ ἄλλου γένους, οἷον τὸ φέρον μὲν αὐξάνεσθαι, τὸ

another but from another kind, such as the bearing indeed to increase, the

δὲ τοῦτο αὔξον ἀλλοιοῦσθαι ὑπ' ἄλλου, τὸ δὲ τοῦτο ἀλλοιοῦν

but this increase to be altered by another, the but this altering

ἑτέραν τινὰ κινεῖσθαι κίνησιν. ἀλλ' ἀνάγκη στῆναι· πεπερασμέναι

another some to be moved motion. but necessity to stand; limited

γὰρ αἱ κινήσεις. τὸ δὲ πάλιν ἀνακάμπτειν καὶ τὸ ἀλλοιοῦν φάναι

for the motions. the but again to bend back and the altering to say

φέρεσθαι τὸ αὐτὸ ποιεῖν ἐστὶ κἂν εἰ εὐθὺς ἔφη τὸ φέρον

to be carried the same to make is even if if immediately it was said the bearing

φέρεσθαι καὶ διδάσκεσθαι τὸ διδάσκον (δῆλον γὰρ ὅτι κινεῖται
to be carried and to be taught the teacher (evident for that it is moved
καὶ ὑπὸ τοῦ ἀνωτέρω κινοῦντος τὸ κινούμενον πᾶν, καὶ μᾶλλον
and by the above of the mover the moved thing all, and rather
ὑπὸ τοῦ προτέρου τῶν κινούντων). ἀλλὰ μὴν τοῦτό γε
by the prior of the movers). But indeed this at least
ἀδύνατον· τὸ διδάσκον γὰρ συμβαίνει μανθάνειν, ὧν τὸ μὲν
is impossible; the teaching for it happens to learn, of which the one
μὴ ἔχειν τὸ δὲ ἔχειν ἐπιστήμην ἀναγκαῖον. ἔτι δὲ μᾶλλον
not to have the but to have knowledge is necessary. Moreover and rather
τούτων ἄλογον, ὅτι συμβαίνει πᾶν τὸ κινητικὸν κινητόν,
of these unreasonable, because it happens all the mobile thing,
εἴπερ ἅπαν ὑπὸ κινουμένου κινεῖται τὸ κινούμενον· ἔσται γὰρ
if indeed all by a mover is moved the moved thing; it will be for
κινητόν, ὥσπερ εἴ τις λέγοι πᾶν τὸ ὑγιαστικὸν [καὶ ὑγιάζον]
mobile, just as if someone might say all the healing [and health-giving]
ὑγιαστὸν εἶναι, καὶ τὸ οἰκοδομητικὸν οἰκοδομητόν, ἢ εὐθὺς ἢ
is healing, to be, and the constructive thing, or immediately or
διὰ πλειόνων· λέγω δ' οἷον εἰ κινητὸν μὲν ὑπ' ἄλλου
through more agents; I say but such as if something mobile indeed by another
πᾶν τὸ κινητικόν, ἀλλ' οὐ ταύτην τὴν κίνησιν κινητὸν ἣν κινεῖ τὸ
all the movable, but not this the motion movable which moves the
πλησίον, ἀλλ' ἑτέραν, οἷον τὸ ὑγιαστικὸν μαθητικόν, ἀλλὰ τοῦτο
nearby, but another, such as the healing learning, but this
ἐπαναβαῖνον ἥξει ποτὲ εἰς τὸ αὐτὸ εἶδος, ὥσπερ εἴπομεν
coming back will arrive at some time to the same form, just as we said
πρότερον. τὸ μὲν οὖν τούτων ἀδύνατον, τὸ δὲ
previously. the on one hand therefore of these impossible, the on the other hand

πλασματῶδες· ἄτοπον γὰρ τὸ ἐξ ἀνάγκης τὸ ἀλλοιωτικὸν

fictitious. inappropriate for the from necessity the altering

αὐξητὸν εἶναι. οὐκ ἄρα ἀνάγκη ἀεὶ κινεῖσθαι τὸ κινούμενον

increase to be. not therefore necessity always to be moved the moving

ὑπ' ἄλλου, καὶ τούτου κινουμένου· στήσεται ἄρα. ὥστε ἤτοι

by another, and this being moved; will be set. Therefore. so that either

ὑπὸ ἠρεμοῦντος κινήσεται τὸ κινούμενον πρῶτον, ἢ αὐτὸ ἑαυτὸ

by resting will move the moving first, or itself itself

κινήσει. ἀλλὰ μὴν καὶ εἴ γε δέοι σκοπεῖν πότερον

will move. but indeed also if at least it is necessary to consider either

αἴτιον κινήσεως καὶ ἀρχὴ τὸ αὐτὸ αὑτὸ κινοῦν ἢ τὸ ὑπ' ἄλλου

cause of movement and principle the same itself moving or the by another

κινούμενον, ἐκεῖνο πᾶς ἂν θείη· τὸ γὰρ αὐτὸ καθ' αὑτὸ

moved, that every if it would divine; the for itself according to itself

ὂν ἀεὶ πρότερον αἴτιον τοῦ καθ' ἕτερον καὶ αὐτοῦ ὄντος.

being always before cause of the according to another and of it being.

ὥστε τοῦτο σκεπτέον λαβοῦσιν ἄλλην ἀρχήν, εἴ τι

Therefore this must be considered having taken another principle, if anything

κινεῖ αὐτὸ αὑτό, πῶς κινεῖ καὶ τίνα τρόπον. ἀναγκαῖον δὴ

moves itself itself, how it moves and in what manner. necessary indeed

τὸ κινούμενον ἅπαν εἶναι διαιρετὸν εἰς ἀεὶ διαιρετά· τοῦτο γὰρ

the moving all to be divisible into always divisible; this for

δέδεικται πρότερον ἐν τοῖς [257b] καθόλου τοῖς περὶ φύσεως,

it has been shown before in the [257b] universally in the about nature,

ὅτι πᾶν τὸ καθ' αὑτὸ κινούμενον συνεχές. ἀδύνατον δὴ

that everything the according to itself moving is continuous. impossible indeed

τὸ αὐτὸ αὑτὸ κινοῦν πάντῃ κινεῖν αὐτὸ αὑτό· φέροιτο γὰρ ἂν

the same itself moving everywhere to move it itself. might bring for if

ὅλον καὶ φέροι τὴν αὐτὴν φοράν, ἓν ὂν καὶ ἄτομον τῷ

whole and might bring the same motion, one being and indivisible according to

εἴδει, καὶ ἀλλοιοῖτο καὶ ἀλλοιοῖ, ὥστε διδάσκοι ἂν καὶ μανθάνοι

form, and might be changed and changes, so that might teach if and might learn

ἅμα, καὶ ὑγιάζοι καὶ ὑγιάζοιτο τὴν αὐτὴν ὑγίειαν.

together, and might be healthy and might become healthy the same health.

ἔτι διώρισται ὅτι κινεῖται τὸ κινητόν· τοῦτο δ' ἐστὶν

further it has been defined that it is moved the mobile; this but is

δυνάμει κινούμενον, οὐκ ἐντελεχείᾳ, τὸ δὲ δυνάμει εἰς ἐντελέχειαν

potentially being moved, not in actuality, the but potentially into actuality

βαδίζει, ἔστιν δ' ἡ κίνησις ἐντελέχεια κινητοῦ ἀτελής. τὸ

it proceeds, it is but the motion actuality of the mobile imperfect. the

δὲ κινοῦν ἤδη ἐνεργείᾳ ἔστιν, οἷον θερμαίνει τὸ θερμὸν καὶ

but moving already in activity is, like heats the hot and

ὅλως γεννᾷ τὸ ἔχον τὸ εἶδος. ὥσθ' ἅμα τὸ αὐτὸ

altogether produces the having the form. so that at the same time the same

κατὰ τὸ αὐτὸ θερμὸν ἔσται καὶ οὐ θερμόν. ὁμοίως δὲ καὶ τῶν

according to the same warm will be and not warm. Similarly but also of the

ἄλλων ἕκαστον, ὅσων τὸ κινοῦν ἀνάγκη ἔχειν τὸ συνώνυμον. τὸ

others each, of those the moving necessity to have the synonym. the

μὲν ἄρα κινεῖ τὸ δὲ κινεῖται τοῦ αὐτὸ αὐτὸ κινοῦντος.

on the one hand therefore moves the but is moved by the same itself moving.

ὅτι δ' οὐκ ἔστιν αὐτὸ αὑτὸ κινοῦν οὕτως ὥσθ' ἑκάτερον

that but not is it itself moving in such a way so that each one

ὑφ' ἑκατέρου κινεῖσθαι, ἐκ τῶνδε φανερόν. οὔτε γὰρ ἔσται

under either to be moved, from these it is apparent. neither for will be

πρῶτον κινοῦν οὐδέν, εἴ γε αὐτὸ ἑαυτὸ κινήσει ἑκάτερον (τὸ

first moving nothing, if indeed it itself will move each one (the

γὰρ πρότερον αἰτιώτερον τοῦ κινεῖσθαι τοῦ ἐχομένου καὶ κινήσει
for previous more causative of the to be moved of the possessing and will move
μᾶλλον· διχῶς γὰρ κινεῖν ἦν, τὸ μὲν τὸ ὑπ' ἄλλου κινούμενον
rather; in two ways for to move was, the on one hand the by another being moved
αὐτό, τὸ δ' αὑτῷ· ἐγγύτερον δὲ τὸ πορρώτερον τοῦ κινουμένου
itself, the but to itself. closer and the farther of the moving
τῆς ἀρχῆς ἢ τὸ μεταξύ)· ἔτι οὐκ ἀνάγκη τὸ κινοῦν κινεῖσθαι εἰ
of the source or the between)· still not necessity the mover to be moved if
μὴ ὑφ' αὑτοῦ· κατὰ συμβεβηκὸς ἄρα ἀντικινεῖ θάτερον.
not by itself· according to incidental therefore moves against. another.
ἔλαβον τοίνυν ἐνδέχεσθαι μὴ κινεῖν· ἔστιν ἄρα τὸ μὲν
I received therefore to be possible not to move· it is therefore the on one hand
κινούμενον τὸ δὲ κινοῦν ἀκίνητον. ἔτι οὐκ ἀνάγκη τὸ κινοῦν
being moved the but mover immobile. still not necessity the mover
ἀντικινεῖσθαι, ἀλλ' ἢ ἀκίνητόν γέ τι κινεῖν ἀνάγκη ἢ αὐτὸ
to be moved against, but or immovable indeed something to move necessity or it
ὑφ' αὑτοῦ κινούμενον, εἴπερ ἀνάγκη ἀεὶ κίνησιν εἶναι. ἔτι
by itself being moved, if indeed necessity always motion to exist. still
ἦν κινεῖ κίνησιν, κινοῖτ' ἄν, ὥστε τὸ θερμαῖνον θερμαίνεται.
which moves motion, let it move. if, thus the heating is heated.
ἀλλὰ μὴν οὐδὲ τοῦ πρώτως αὐτὸ αὑτὸ κινοῦντος οὔτε ἓν μόριον
but indeed nor of the first itself itself moving nor one part
οὔτε πλείω κινήσει αὐτὸ αὑτὸ ἕκαστον. τὸ γὰρ ὅλον εἰ κινεῖται
nor more will move itself itself each. the for whole if it is moved
αὐτὸ ὑφ' αὑτοῦ, ἤτοι ὑπὸ τῶν αὐτοῦ τινὸς κινήσεται ἢ ἢ ὅλον
itself by itself, namely by the its some will move or or whole
ὑφ' ὅλου. εἰ μὲν οὖν τῷ κινεῖσθαί τι μόριον αὐτὸ ὑφ' αὑτοῦ,
by the whole. if indeed then to the being moved some part itself by itself,

τοῦτ' ἂν εἴη τὸ πρῶτον αὐτὸ αὑτὸ κινοῦν (χωρισθὲν γὰρ τοῦτο
this would be the first itself itself moving (separated for this
μὲν κινήσει αὐτὸ αὑτό, τὸ δὲ ὅλον οὐκέτι)· εἰ δὲ ὅλον ὑφ'
indeed will move itself itself, the but whole no longer). if but whole by
ὅλου κινεῖται, κατὰ συμβεβηκὸς ἂν ταῦτα κινοῖ αὐτὰ
the whole it is moved, according to what happens would these things it moves those
ἑαυτά. ὥστε εἰ μὴ ἀναγκαῖον, εἰλήφθω μὴ κινούμενα [258a]
themselves. so that if not necessary, let it be taken not moving [258a]
ὑφ' αὑτῶν. τῆς ὅλης ἄρα τὸ μὲν κινήσει ἀκίνητον ὂν
by themselves. the whole thus the on one hand it will move the unmoving being
τὸ δὲ κινηθήσεται· μόνως γὰρ οὕτως οἷόν τέ
the on the other hand it will be moved. only for thus it is possible and
τι αὐτοκίνητον εἶναι. ἔτι εἴπερ ἡ ὅλη αὐτὴ αὑτὴν κινεῖ,
something self-moving to be. further if indeed the whole it itself itself moves,
τὸ μὲν κινήσει αὐτῆς, τὸ δὲ κινήσεται. ἡ ἄρα
the on one hand will move of it, the on the other hand will be moved. the thus
AB ὑφ' αὑτῆς τε κινηθήσεται καὶ ὑπὸ τῆς A. ἐπεὶ δὲ κινεῖ τὸ
AB by itself also will be moved and by the A. when but it moves the
μὲν κινούμενον ὑπ' ἄλλου τὸ δ' ἀκίνη τον ὄν,
on one hand the moving by another the on the other hand unmoving being being,
καὶ κινεῖται τὸ μὲν κινοῦν τὸ δὲ οὐδὲν κινοῦν,
and it is moved the on one hand moving the on the other hand nothing moves,
τὸ αὐτὸ αὑτὸ κινοῦν ἀνάγκη ἐξ ἀκινήτου εἶναι κινοῦντος δέ, καὶ
the it this moving necessity from immovable to be mover but, and
ἔτι ἐκ κινουμένου μὴ κινοῦντος δ' ἐξ ἀνάγκης, ἀλλ' ὁπότερ'
still out of moving not mover but from necessity, but whichever
ἔτυχεν. ἔστω γὰρ τὸ A κινοῦν μὲν ἀκίνητον δέ, τὸ δὲ
it happens. let it be for the A moving on the one hand immovable but, the but

Β κινούμενόν τε ὑπὸ τοῦ Α καὶ κινοῦν τὸ ἐφ' ᾧ Γ, τοῦτο δὲ

B moving and by the A and moving the on which C, this but

κινούμενον μὲν ὑπὸ τοῦ Β, μὴ κινοῦν δὲ μηδέν· εἴπερ

being moved on the one hand by the B, not moving but anything; if

γὰρ καὶ διὰ πλειόνων ἥξει ποτὲ εἰς τὸ Γ, ἔστω δι' ἑνὸς

for also through more will come at some time to the C, let it be by one

μόνου. τὸ δὴ ἅπαν ΑΒΓ αὐτὸ ἑαυτὸ κινεῖ. ἀλλ' ἐὰν ἀφέλω τὸ Γ,

alone. the indeed all ABC it itself moves. but if I remove the C,

τὸ μὲν ΑΒ κινήσει αὐτὸ ἑαυτό, τὸ μὲν Α κινοῦν

the on the one hand AB will move it itself, the on the one hand A moving

τὸ δὲ Β κινούμενον, τὸ δὲ Γ οὐ κινήσει αὐτὸ

the on the other hand B being moved, the on the other hand C not will move it

ἑαυτό, οὐδ' ὅλως κινήσεται. ἀλλὰ μὴν οὐδ' ἡ ΒΓ κινήσει αὐτὴ

itself, nor at all will be moved. but indeed nor the BC will move it

ἑαυτὴν ἄνευ τοῦ Α· τὸ γὰρ Β κινεῖ τῷ κινεῖσθαι ὑπ' ἄλλου, οὐ

itself without the A· the for B moves the being moved by another, not

τῷ ὑφ' αὑτοῦ τινὸς μέρους. τὸ ἄρα ΑΒ μόνον αὐτὸ ἑαυτὸ κινεῖ.

the by itself some part. the therefore AB only it itself moves.

ἀνάγκη ἄρα τὸ αὐτὸ ἑαυτὸ κινοῦν ἔχειν τὸ κινοῦν ἀκίνητον

necessity therefore the itself itself moving to have the moving unmoving

δέ, καὶ τὸ κινούμενον μηδὲν δὲ κινοῦν ἐξ ἀνάγκης, ἁπτόμενα

indeed, and the being moved nothing but moving out of necessity, touching

ἤτοι ἄμφω ἀλλήλων ἢ θατέρου θάτερον. εἰ μὲν οὖν συνεχές

either both of each other or of the other other. if indeed therefore continuous

ἐστι τὸ κινοῦν (τὸ μὲν γὰρ κινούμενον ἀναγκαῖον εἶναι συνεχές),

is the moving (the indeed for moved necessary to be continuous),

ἅψεται ἑκάτερον ἑκατέρου. δῆλον δὴ ὅτι τὸ πᾶν αὐτὸ

it will be connected each of each. it is clear indeed that the whole it

ἑαυτὸ κινεῖ οὐ τῷ αὐτοῦ τι εἶναι τοιοῦτον οἷον αὐτὸ αὐτὸ

itself moves not by the same thing to be such as it itself

κινεῖν, ἀλλ' ὅλον κινεῖ αὐτὸ ἑαυτό, κινούμενόν τε καὶ κινοῦν τῷ

to move, but whole moves it itself, being moved and and moving by the

αὐτοῦ τι εἶναι τὸ κινοῦν καὶ τὸ κινούμενον. οὐ γὰρ ὅλον κι νεῖ

same thing to be the moving and the moved. not for whole and moves

οὐδ' ὅλον κινεῖται, ἀλλὰ κινεῖ μὲν ἡ τὸ A, κινεῖται δὲ ἡ τὸ B

nor whole is moved, but it moves indeed the the A, is moved but the the B

μόνον [τὸ δὲ Γ ὑπὸ τοῦ A οὐκέτι· ἀδύνατον γάρ].

only [the but C by the A no longer; impossible for].

ἀπορίαν δ' ἔχει, ἐὰν ἀφέλῃ τις ἢ τῆς A, εἰ συνεχὲς τὸ

perplexity but has, if someone removes anyone or of the A, if continuous the

κινοῦν μὲν ἀκίνητον δέ, ἢ τῆς B τῆς κινουμένης· ἡ λοιπὴ

moving indeed unmoving but, or of the B of the moving. the remaining

ἆρα κινήσει τῆς A ἢ τῆς B κινηθήσεται; εἰ γὰρ τοῦτο, οὐκ ἂν

therefore will move the A or the B will be moved? if for this, not would

εἴη πρώ τως κινουμένη ὑφ' αὑτῆς ἡ τὸ AB· ἀφαιρεθείσης γὰρ

be first at all moving by itself the the AB. Having been removed for

ἀπὸ τῆς AB, ἔτι κινήσει αὐτὴν ἡ λοιπὴ AB. ἢ δυνάμει μὲν

from the AB, still will move it the remaining AB. Or in power indeed

[258b] ἑκάτερον οὐδὲν κωλύει ἢ θάτερον, τὸ κινούμενον, διαιρετὸν

[258b] either nothing hinders or the other, the moving, divisible

εἶναι, ἐντελεχείᾳ δ' ἀδιαίρετον· ἐὰν δὲ διαιρεθῇ, μηκέτι εἶναι

to be, in actuality but indivisible. If but it is divided, no longer to be

ἔχον τὴν αὐτὴν φύσιν· ὥστ' οὐδὲν κωλύει ἐν διαιρετοῖς δυνάμει

having the same nature. so that nothing prevents in divisible power

πρώτως ἐνεῖναι.

first to be.

φανερὸν τοίνυν ἐκ τούτων ὅτι ἔστιν τὸ πρώτως κινοῦν ἀκίνητον·

evident therefore from these that there is the first mover unmoved;

εἴτε γὰρ εὐθὺς ἵσταται τὸ κινούμενον, ὑπό τι νος δὲ

whether for immediately stands the moved, by something of it and

κινούμενον, εἰς ἀκίνητον τὸ πρῶτον, εἴτε εἰς κινούμενον μέν,

moving, into unmoved the first, whether into moving on the one hand,

αὐτὸ δ' αὑτὸ κινοῦν καὶ ἱστάν, ἀμφοτέρως συμβαίνει τὸ πρώτως

it but itself moving and standing, in both cases it happens the first

κινοῦν ἅπασιν εἶναι τοῖς κινουμένοις ἀκίνητον.

mover to all to be for those being moved unmoved.

Κεφάλαιο 6

Chapter 6

Ἐπεὶ δὲ δεῖ κίνησιν ἀεὶ εἶναι καὶ μὴ διαλείπειν, ἀνάγκη

Since but it is necessary movement always to be and not to cease, necessity

εἶναί τι ἀΐδιον ὃ πρῶτον κινεῖ, εἴτε ἓν εἴτε πλείω· καὶ

to be something eternal which first moves, whether one or more; and

τὸ πρῶτον κινοῦν ἀκίνητον. ἕκαστον μὲν οὖν ἀΐδιον εἶναι

the first moving unchanging. Each indeed therefore eternal to be

τῶν ἀκινήτων μὲν κινούντων δὲ οὐδὲν πρὸς τὸν νῦν

of the unchanging indeed moving but nothing in relation to the current

λόγον· ὅτι δ' ἀναγκαῖον εἶναί τι τὸ ἀκίνητον μὲν αὐτὸ

argument; since but it is necessary to be something the unchanging indeed itself

πάσης ἐκτὸς μεταβολῆς, καὶ ἁπλῶς καὶ κατὰ συμβεβηκός,

of all external to change, and simply and according to accidental,

κινητικὸν δ' ἑτέρου, δῆλον ὧδε σκοποῦσιν. ἔστω δή, εἴ

moving but of another, it is clear thus they aim. Let it be indeed, if

τις βούλεται, ἐπί τινων ἐνδεχόμενον ὥστε εἶναί ποτε

someone wishes, concerning some possible so that to be at some time

καὶ μὴ εἶναι ἄνευ γενέσεως καὶ φθορᾶς (τάχα γὰρ ἀναγκαῖον, εἴ

and not to be without generation and decay. perhaps for necessary, if

τι ἀμερὲς ὁτὲ μὲν ἔστιν ὁτὲ δὲ μὴ ἔστιν,

something imperfect at one time on the one hand it exists at one time but not it exists,

ἄνευ τοῦ μεταβάλλειν ὁτὲ μὲν εἶναι ὁτὲ δὲ μὴ

without of the changing at one time on the one hand to be at one time but not

εἶναι πᾶν τὸ τοιοῦτον). καὶ τῶν ἀρχῶν τῶν ἀκινήτων μὲν

to be all the such). and of the beginnings of the unmoved on the one hand

κινητικῶν δ' ἐνίας ὁτὲ μὲν εἶναι ὁτὲ δὲ μὴ εἶναι,

moved and of some at one time on the one hand to be at one time but not to be,

ἐνδεχέσθω καὶ τοῦτο. ἀλλ' οὔ τί γε πάσας δυνατόν·

let it be inferred and this. but not anything at all all possible·

δῆλον γὰρ ὡς αἴτιον τοῖς αὐτὰ ἑαυτὰ κινοῦσίν ἐστί τι

clear for that cause to the same themselves movements is something

τοῦ ὁτὲ μὲν εἶναι ὁτὲ δὲ μή. τὸ μὲν γὰρ

of the at one time on the one hand to be at one time but not. the on the one hand for

αὐτὸ ἑαυτὸ κινοῦν ἅπαν ἔχειν ἀνάγκη μέγεθος, εἰ μηδὲν κινεῖται

it itself moving all to have necessity, greatness, if nothing is moved

ἀμερές, τὸ δὲ κινοῦν οὐδεμία ἀνάγκη ἐκ τῶν εἰρημένων. τοῦ

indivisible, the and moving no necessity from the aforementioned. of the

δὴ τὰ μὲν γίγνεσθαι τὰ δὲ φθείρεσθαι, καὶ

indeed the on the one hand to become the on the other hand to be destroyed, and

τοῦτ' εἶναι συνεχῶς, οὐδὲν αἴτιον τῶν ἀκινήτων μὲν μὴ ἀεὶ δ'

this to be constantly, nothing cause of the unmovable for not always but

ὄντων, οὐδ' αὖ τωνδὶ μὲν ταδί [κινούντων], τούτων δ' ἕτερα. τοῦ

existing, nor again of those for such [moving], these and others. for

γὰρ ἀεὶ καὶ συνεχοῦς οὔτε ἕκαστον αὐτῶν οὔτε πάντα αἴτια·

indeed always and continuous neither each of them nor all causes.

τὸ μὲν γὰρ οὕτως ἔχειν ἀΐδιον καὶ ἐξ ἀνάγκης, τὰ δὲ

the on the one hand for thus to have eternal and from necessity, the but

πάντα ἄπειρα, καὶ οὐχ ἅμα πάντα ὄντα. δῆλον τοίνυν

all infinite, and not at the same time all beings. it is clear therefore

ὅτι, εἰ καὶ μυριάκις ἕνια [ἀρχαὶ] [259a] τῶν ἀκινήτων μὲν

that, if also countless some [principles] [259a] of the unmovable on the one hand

κινούντων δέ, καὶ πολλὰ τῶν αὐτὰ ἑαυτὰ κινούντων, φθείρεται,

of moving but, and many of the same themselves moving, deteriorates,

τὰ δ' ἐπιγίγνεται, καὶ τόδε μὲν ἀκίνητον ὂν τόδε κινεῖ,

the but occurs, and this on the one hand immobile being this moves,

ἕτερον δὲ τοδί, ἀλλ' οὐδὲν ἧττον ἔστιν τι ὃ περιέχει, καὶ

another but this, yet nothing less exists something which contains, and

τοῦτο παρ' ἕκαστον, ὅ ἐστιν αἴτιον τοῦ τὰ μὲν εἶναι

this by each, which is the cause of the on the one hand to be

τὰ δὲ μὴ καὶ τῆς συνεχοῦς μεταβολῆς· καὶ τοῦτο μὲν

the but not and of the continuous change. And this on the one hand

τούτοις, ταῦτα δὲ τοῖς ἄλλοις αἴτια κινήσεως. εἴπερ οὖν ἀΐδιος

to these, these but to the others causes of movement. If indeed then eternal

ἡ κίνησις, ἀΐδιον καὶ τὸ κινοῦν ἔσται πρῶτον, εἰ ἕν· εἰ δὲ

the movement, eternal and the moving will be first, if one; if but

πλείω, πλείω τὰ ἀΐδια. ἓν δὲ μᾶλλον ἢ πολλά, καὶ πεπερασμένα

more, more the eternal. One but rather or many, and finite

ἢ ἄπειρα, δεῖ νομίζειν. τῶν αὐτῶν γὰρ συμβαινόντων αἰεὶ

or infinite, it is necessary to think. of the same for occurrences always

τὰ πεπερασμένα μᾶλλον ληπτέον· ἐν γὰρ τοῖς φύσει δεῖ

the finite rather must be taken; in for the nature it is necessary

τὸ πεπερασμένον καὶ τὸ βέλτιον, ἂν ἐνδέχηται, ὑπάρχειν μᾶλλον.

the finite and the better, if it should happen, to exist rather.

ἱκανὸν δὲ καὶ ἕν, ὃ πρῶτον τῶν ἀκινήτων ἀΐδιον ὂν ἔσται

Sufficient and also one, which first of the immovable eternal being will be

ἀρχὴ τοῖς ἄλλοις κινήσεως.

the beginning for the others of motion.

φανερὸν δὲ καὶ ἐκ τοῦδε ὅτι ἀνάγκη εἶναί τι ἓν καὶ ἀΐδιον τὸ

apparent but and from this that necessity to be something one and eternal the

πρῶτον κινοῦν. δέδεικται γὰρ ὅτι ἀνάγκη ἀεὶ κίνησιν εἶναι. εἰ

first moving has been shown for that necessity always motion to be if

δὲ ἀεί, ἀνάγκη συνεχῆ εἶναι· καὶ γὰρ τὸ ἀεὶ συνεχές, τὸ δ'

but always, necessity continuous to be. and for the always continuous, the but

ἐφεξῆς οὐ συνεχές. ἀλλὰ μὴν εἴ γε συνεχής, μία. μία δ' ἡ

subsequently not continuous. but indeed if at least continuous, one. one but the

ὑφ' ἑνός τε τοῦ κινοῦντος καὶ ἑνὸς τοῦ κινουμένου· εἰ γὰρ ἄλλο

by one and the moving and one the moved. if for another

καὶ ἄλλο κινήσει, οὐ συνεχὴς ἡ ὅλη κίνησις, ἀλλ' ἐφεξῆς.

and another will move, not continuous the whole motion, but subsequently.

ἔκ τε δὴ τούτων πιστεύσειεν ἄν τις εἶναί τι πρῶτον

from and indeed these he would believe would anyone to be something first

ἀκίνητον, καὶ πάλιν ἐπιβλέψας ἐπὶ τὰς ἀρχάς [τῶν κινούντων]. τὸ
unchanging, and again having looked upon the principles of the movers]. the
μὲν δὴ εἶναι ἄττα τῶν ὄντων ἃ ὁτὲ μὲν κινεῖται
on one hand indeed to be something of the beings which sometimes indeed is in motion
ὁτὲ δὲ ἠρεμεῖ φανερόν. καὶ διὰ τούτου γέγονε δῆλον
sometimes but is at rest is evident. And through this it has become evident
ὅτι οὔτε πάντα κινεῖται οὔτε πάντα ἠρεμεῖ οὔτε τὰ μὲν
that neither all is in motion nor all is at rest nor the on one hand
ἀεὶ ἠρεμεῖ τὰ δὲ ἀεὶ κινεῖται· τὰ γὰρ ἐπαμφοτερίζοντα καὶ
always is at rest the but always is in motion. the for mutually living and
δύναμιν ἔχοντα τοῦ κινεῖσθαι καὶ ἠρεμεῖν δείκνυσιν περὶ αὐτῶν.
power having of being moved and being at rest shows about them.
ἐπεὶ δὲ τὰ μὲν τοιαῦτα δῆλα πᾶσι, βουλόμεθα δὲ
Since and the on one hand such things are evident to all, we wish and
δεῖξαι καὶ τοῖν δυοῖν ἑκατέρου τὴν φύσιν, ὅτι ἔστιν τὰ
to demonstrate and the of the two of each the nature, that it is the
μὲν ἀεὶ ἀκίνητα τὰ δὲ ἀεὶ κινούμενα, προϊόντες δ' ἐπὶ
on one hand always unmoved the but always moved, producing and upon
τοῦτο καὶ θέντες ἅπαν τὸ κινούμενον ὑπό τινος κινεῖσθαι,
this and having placed all the moving by someone moves, to
καὶ τοῦτ' εἶναι ἢ ἀκίνητον ἢ κινούμενον, καὶ κινούμενον ἢ
and this to be either unmoved or moving, and moving or
ὑφ' αὑτοῦ ἢ ὑπ' ἄλλου ἀεί, προήλθομεν ἐπὶ τὸ λαβεῖν ὅτι τῶν
by itself or by another always, we have come to the take that of the
κινουμένων ἐστὶν ἀρχὴ κινουμένων μὲν ὃ αὐτὸ ἑαυτὸ
moving things there is a source of moving things on one hand which itself itself
[259b] κινεῖ, πάντων δὲ τὸ ἀκίνητον, ὁρῶμεν δὲ καὶ φανερῶς ὄντα
[259b] moves, of all but the unmoved, we see but and clearly existing

τοιαῦτα ἃ κινεῖ αὐτὰ ἑαυτά, οἷον τὸ τῶν ἐμψύχων καὶ τὸ τῶν
such as moves these themselves, such as the of the living beings and the of the
ζῴων γένος, ταῦτα δὲ καὶ δόξαν παρεῖχε μή ποτε ἐνδέχεται
of living beings kind, these but and the glory provided not ever is it possible
κίνησιν ἐγγίγνεσθαι μὴ οὖσαν ὅλως, διὰ τὸ ἐν τούτοις ὁρᾶν
motion to emerge not being at all, through the in these to see
ἡμᾶς τοῦτο συμβαῖνον (ἀκίνητα γάρ ποτε ὄντα κινεῖ ται πάλιν, ὡς
us this happening (motionless for ever being moves they again, as
δοκεῖ), τοῦτο δὴ δεῖ λαβεῖν, ὅτι μίαν κίνησιν αὐτὰ κινεῖ, καὶ ὅτι
it seems), this indeed must be taken, that one motion itself moves, and that
ταύτην οὐ κυρίως· οὐ γὰρ ἐξ αὐτοῦ τὸ αἴτιον, ἀλλ' ἔνεισιν ἄλλαι
this not essentially; not for from it the cause, but belongs to other
κινήσεις φυσικαὶ τοῖς ζῴοις, ἃς οὐ κινοῦνται δι' αὐτῶν,
movements natural to the living beings, which not are moved by themselves,
οἷον αὔξησις φθίσις ἀναπνοή, ἃς κινεῖται τῶν ζῴων ἕκαστον
such as growth decay, respiration, which is moved of the living beings each
ἠρεμοῦν καὶ οὐ κινούμενον τὴν ὑφ' αὐτοῦ κίνησιν. τούτου δ' αἴτιον
calming and not moved the by itself motion. of this but cause
τὸ περιέχον καὶ πολλὰ τῶν εἰσιόντων, οἷον ἐνίων ἡ τροφή·
the encompassing and many of the enterers, such as some the nourishment;
πεττομένης μὲν γὰρ καθεύδουσιν, διακρινομένης δ'
when being fallen, on the one hand for they are sleeping, when being distinguished but
ἐγείρονται καὶ κινοῦσιν ἑαυτούς, τῆς πρώτης ἀρχῆς ἔξωθεν
they arise and move themselves, of the first principle from outside
οὔσης, διὸ οὐκ ἀεὶ κινοῦνται συνεχῶς ὑφ' αὐτῶν· ἄλλο
being, therefore not always they are moved continuously by themselves. Another
γὰρ τὸ κινοῦν, αὐτὸ κινούμενον καὶ μεταβάλλον πρὸς ἕκαστον
for the mover, it being moved and changing towards each

τῶν κινούντων ἑαυτά. ἐν πᾶσι δὲ τούτοις κινεῖται τὸ κινοῦν
of the movers, themselves. In all but these it is moved the mover
πρῶτον καὶ τὸ αἴτιον τοῦ αὐτὸ ἑαυτὸ κινεῖν ὑφ' αὐτοῦ, κατὰ
first and the cause of the itself itself to move by itself, according to
συμβεβηκὸς μέντοι· μεταβάλλει γὰρ τὸν τόπον τὸ σῶμα, ὥστε
the concomitant however. It changes for the place the body, so that
καὶ τὸ ἐν τῷ σώματι ὂν καὶ τῇ μοχλείᾳ κινοῦν ἑαυτό. ἐξ
also the in the body being and the lever to move itself. out of
ὧν ἔστιν πιστεῦσαι ὅτι εἴ τί ἐστι τῶν ἀκινήτων μὲν
which is to believe that if anything is of the unmoving on the one hand
κινούντων δὲ καὶ αὐτὰ κατὰ συμβεβηκός, ἀδύνατον συνεχῆ
moving but and themselves according to accidental, impossible continuous
κίνησιν κινεῖν. ὥστ' εἴπερ ἀνάγκη συνεχῶς εἶναι κίνησιν, εἶναί
motion to move. so that if ever necessity to be continuous to be motion, to be
τι δεῖ τὸ πρῶτον κινοῦν ἀκίνητον καὶ κατὰ συμβεβηκός, εἰ
something must the first moving unmoving and according to accidental, if
μέλλει, καθάπερ εἴπομεν, ἔσεσθαι ἐν τοῖς οὖσιν ἄπαυστός τις
it will be, just as we said, it will be in the beings ceaseless there is
καὶ ἀθάνατος κίνησις, καὶ μενεῖν τὸ ὂν αὐτὸ ἐν αὑτῷ καὶ ἐν τῷ
and immortal motion, and to remain the being itself in itself and in the
αὐτῷ· τῆς γὰρ ἀρχῆς μενούσης ἀνάγκη καὶ τὸ πᾶν μένειν
same; of the for of the principle remaining necessity and the all to remain
συνεχὲς ὂν πρὸς τὴν ἀρχήν. οὐκ ἔστιν δὲ τὸ αὐτὸ τὸ κινεῖσθαι
continuously being towards the beginning. not is but the same the to be moved
κατὰ συμβεβηκὸς ὑφ' αὐτοῦ καὶ ὑφ' ἑτέρου· τὸ μὲν γὰρ
according to contingent by itself and by another; the on one hand for
ὑφ' ἑτέρου ὑπάρχει καὶ τῶν ἐν τῷ οὐρανῷ ἐνίαις ἀρχαῖς, ὅσα
by another exists and of the in the heaven certain principles, as much as

πλείους φέρεται φοράς, θάτερον δὲ τοῖς φθαρτοῖς μόνον. ἀλλὰ

more is carried in cycles, the other but to the perishable only. But

μὴν εἴ γε ἔστιν τι ἀεὶ τοιοῦτον, κινοῦν μέν τι

indeed if indeed it is something always such, moving indeed something

ἀκίνητον δὲ αὐτὸ καὶ ἀΐδιον, ἀνάγκη καὶ τὸ πρῶτον ὑπὸ τούτου

unmovable but itself and eternal, necessity and the first by this

[260a] κινούμενον ἀΐδιον εἶναι. ἔστιν δὲ τοῦτο δῆλον μὲν καὶ ἐκ

[260a] being moved eternally to be. It is but this clear indeed and from

τοῦ μὴ ἂν ἄλλως εἶναι γένεσιν καὶ φθορὰν καὶ μεταβολὴν τοῖς

the not if otherwise to be a generation and decay and change to the

ἄλλοις, εἰ μή τι κινήσει κινούμενον· τὸ μὲν γὰρ ἀκίνητον

others, if not something it will move moving. the indeed for immovable

[τὴν αὐτὴν] ἀεὶ τὸν αὐτὸν κινήσει τρόπον καὶ μίαν κίνησιν, ἅτε

the same] always the same will move way and one motion, since

οὐδὲν αὐτὸ μεταβάλλον πρὸς τὸ κινούμενον. τὸ δὲ κινούμενον

nothing it changing towards the moving. the but moving

ὑπὸ τοῦ κινουμένου μέν, ὑπὸ τοῦ ἀκινήτου δὲ κινουμένου ἤδη,

by the moving indeed, by the immovable but moving, already,

διὰ τὸ ἄλλως καὶ ἄλλως ἔχειν πρὸς τὰ πράγματα, οὐ τῆς

because of the otherwise and differently to have towards the things, not of the

αὐτῆς ἔσται κινήσεως αἴτιον, ἀλλὰ διὰ τὸ ἐν ἐναντίοις εἶναι

same will be of motion cause, but through the in opposites being

τόποις ἢ εἴδεσιν ἐναντίως παρέξεται κινούμενον ἕκαστον τῶν

places or forms oppositely it will differ moving each of the

ἄλλων, καὶ ὁτὲ μὲν ἠρεμοῦν ὁτὲ δὲ κινούμενον.

others, and sometimes indeed at rest sometimes but moving.

φανερὸν δὴ γέγονεν ἐκ τῶν εἰρημένων καὶ ὃ κατ' ἀρχὰς

evident indeed has become from the stated and what according to beginnings

ἠποροῦμεν, τί δή ποτε οὐ πάντα ἢ κινεῖται ἢ ἠρεμεῖ,

we were puzzled, what indeed at any time not all or is moved or is at rest,

ἢ τὰ μὲν κινεῖται ἀεὶ τὰ δ' ἀεὶ ἠρεμεῖ, ἀλλ' ἔνια ὁτὲ

or the indeed is moved always the but always is at rest, but some sometimes

μὲν ὁτὲ δ' οὔ. τούτου γὰρ τὸ αἴτιον δῆλόν

on the one hand sometimes on the other hand no. of this for the cause is clear

ἐστι νῦν, ὅτι τὰ μὲν ὑπὸ ἀκινήτου κινεῖται ἀϊδίου, διὸ

it is now, that the indeed by the unchanging is moved of the eternal, therefore

ἀεὶ κινεῖται, τὰ δ' ὑπὸ κινουμένου καὶ μεταβάλλοντος, ὥστε

always is moved, the but by the moving and changing, so that

καὶ αὐτὰ ἀναγκαῖον μεταβάλλειν. τὸ δ' ἀκίνητον, ὥσπερ

also they themselves must change. the but the unchanging, as

εἴρηται, ἅτε ἁπλῶς καὶ ὡσαύτως καὶ ἐν τῷ αὐτῷ διαμένον, μίαν

has been stated, since simply and likewise and in the same remaining, one

καὶ ἁπλῆν κινήσει κίνησιν.

and simple motion movement.

Κεφάλαιο 7

Chapter 7

Οὐ μὴν ἀλλὰ καὶ ἄλλην ποιησαμένοις ἀρχὴν μᾶλλον ἔσται

Not indeed but also another those who will have produced principle rather will be

περὶ τούτων φανερόν. σκεπτέον γὰρ πότερον ἐνδέχεταί

regarding these clear. It must be considered for whether it is possible

τινα κίνησιν εἶναι συνεχῆ ἢ οὔ, καὶ εἰ ἐνδέχεται, τίς αὕτη, καὶ

any motion to be continuous or not, and if it is possible, what this, and

τίς πρώτη τῶν κινήσεων· δῆλον γὰρ ὡς εἴπερ ἀναγκαῖον

what first of the movements. it is evident for that if indeed necessary

μὲν ἀεὶ κίνησιν εἶναι, πρώτη δὲ ἥδε καὶ συνεχής, ὅτι τὸ
on the one hand always motion to be, first but this and continuous, because the
πρῶτον κινοῦν κινεῖ ταύτην τὴν κίνησιν, ἣν ἀναγκαῖον μίαν καὶ
first moving moves this the motion, which necessary one and
τὴν αὐτὴν εἶναι καὶ συνεχῆ καὶ πρώτην.
the same to be and continuous and first.

τριῶν δ' οὐσῶν κινήσεων, τῆς τε κατὰ μέγεθος καὶ τῆς
of three and existing movements, the and according to size and the
κατὰ πάθος καὶ τῆς κατὰ τόπον, ἣν καλοῦμεν φοράν,
according to quality and the according to place, which we call change,
ταύτην ἀναγκαῖον εἶναι πρώτην. ἀδύνατον γὰρ αὔξησιν εἶναι ἀλλοιώσεως
this necessary to be first. impossible for increase to be of a change
μὴ προϋπαρχούσης· τὸ γὰρ αὐξανόμενον ἔστιν μὲν ὡς
not pre-existing; the for increasing is on one hand as
ὁμοίῳ αὐξάνεται, ἔστιν δ' ὡς ἀνομοίῳ· τροφὴ γὰρ λέγεται
similar, it grows, it is but as dissimilar; nourishment for is referred to
τῷ ἐναντίῳ τὸ ἐναντίον. προσγίγνεται δὲ πᾶν γιγνόμενον
the opposite the contrary. it is added and everything that is becoming
ὅμοιον ὁμοίῳ. ἀνάγκη οὖν ἀλλοίωσιν εἶναι τὴν εἰς τἀναντία
similar to the same. necessity therefore change to be the into the opposites
μεταβολήν. [260b] ἀλλὰ μὴν εἴ γε ἀλλοιοῦται, δεῖ
transformation. [260b] but indeed if at least it is changed, it is necessary
τι εἶναι τὸ ἀλλοιοῦν καὶ ποιοῦν ἐκ τοῦ δυνάμει θερμοῦ
something to be the changing and producing from the potential of heat
ἐνεργείᾳ θερμόν. δῆλον οὖν ὅτι τὸ κινοῦν οὐχ ὁμοίως ἔχει,
in actuality warm. evident therefore that the moving not similarly has,

ἀλλ’ ὁτὲ μὲν ἐγγύτερον ὁτὲ δὲ πορρώτερον τοῦ

but at one time on one hand nearer at another time but farther of the

ἀλλοιουμένου ἐστίν. ταῦτα δ’ ἄνευ φορᾶς οὐκ ἐνδέχεται ὑπάρχειν.

changing it is. these and without motion not it is possible to exist.

εἰ ἄρα ἀνάγκη ἀεὶ κίνησιν εἶναι, ἀνάγκη καὶ φορὰν ἀεὶ εἶναι

if therefore necessity always motion to be, necessity also motion always to be

πρώτην τῶν κινήσεων, καὶ φορᾶς, εἰ ἔστιν ἡ μὲν πρώτη ἡ

first of the motions, and motions, if there is the indeed first the

δ’ ὑστέρα, τὴν πρώτην. ἔτι δὲ πάντων τῶν παθημάτων ἀρχὴ

but latter, the first. further and of all the phenomena origin

πύκνωσις καὶ μάνωσις· καὶ γὰρ βαρὺ καὶ κοῦφον καὶ μαλακὸν

density and rarefaction; and for heavy and light and soft

καὶ σκληρὸν καὶ θερμὸν καὶ ψυχρὸν πυκνότητες δοκοῦσιν καὶ

and hard and hot and cold densities seem and

ἀραιότητες εἶναί τινες. πύκνωσις δὲ καὶ μάνωσις σύγκρισις καὶ

rarefactions to be some. thickening but and increasing comparison and

διάκρισις, καθ’ ἃς γένεσις καὶ φθορὰ λέγεται τῶν οὐσιῶν.

distinction according to which generation and decay is said of the substances.

συγκρινόμενα δὲ καὶ διακρινόμενα ἀνάγκη κατὰ τόπον

compared but and distinguished necessity in accordance with place

μεταβάλλειν. ἀλλὰ μὴν καὶ τοῦ αὐξανομένου καὶ φθίνοντος

to change. but truly and of the increasing and decreasing

μεταβάλλει κατὰ τόπον τὸ μέγεθος.

changes in accordance with place the size.

ἔτι καὶ ἐντεῦθεν ἐπισκοποῦσιν ἔσται φανερὸν ὅτι ἡ φορὰ

yet and from here they observe it will be evident that the movement

πρώτη. τὸ γὰρ πρῶτον, ὥσπερ ἐφ' ἑτέρων, οὕτω καὶ ἐπὶ κινήσεως
is first. the for first, just as on others, so and upon movement
ἂν λέγοιτο πλεοναχῶς. λέγεται δὲ πρότερον οὗ τε μὴ ὄντος
may be said redundantly. it is said but earlier of which and not being
οὐκ ἔσται τἆλλα, ἐκεῖνο δὲ ἄνευ τῶν ἄλλων, καὶ τὸ τῷ
not there will be other things, that one but without the others, and the to the
χρόνῳ, καὶ τὸ κατ' οὐσίαν. ὥστ' ἐπεὶ κίνησιν μὲν ἀναγκαῖον
time, and the according to substance. thus since movement indeed necessary
εἶναι συνεχῶς, εἴη δ' ἂν συνεχῶς ἢ συνεχὴς οὖσα ἢ
to be continuously, it would be but if continuous or being continuous being or
ἐφεξῆς, μᾶλλον δ' ἡ συνεχής, καὶ βέλτιον συνεχῆ ἢ ἐφεξῆς
successive, rather but the continuous, and better continuous than successive
εἶναι, τὸ δὲ βέλτιον ἀεὶ ὑπολαμβάνομεν ἐν τῇ φύσει ὑπάρχειν, ἂν
to be, the but better always we assume in the nature to exist, if
ᾖ δυνατόν, δυνατὸν δὲ συνεχῆ εἶναι (δειχθήσεται δ' ὕστερον·
it is possible, possible but continuous to be (it will be shown that later;
νῦν δὲ τοῦτο ὑποκείσθω), καὶ ταύτην οὐδεμίαν ἄλλην οἷόν τε
now but this let it be subject), and this none other kind and
εἶναι ἀλλ' ἢ φοράν, ἀνάγκη τὴν φορὰν εἶναι πρώτην. οὐδεμία γὰρ
to be other than a form, necessity the form to be first. none for
ἀνάγκη οὔτε αὔξεσθαι οὔτε ἀλλοιοῦσθαι τὸ φερόμενον,
necessity neither to be increased nor to be altered the that which is brought,
οὐδὲ δὴ γίγνεσθαι ἢ φθείρεσθαι· τούτων δὲ οὐδεμίαν ἐνδέχεται
nor indeed to become or to be destroyed; of these but none is possible
τῆς συνεχοῦς μὴ οὔσης, ἣν κινεῖ τὸ πρῶτον κινοῦν.
of the continuous not being, which moves the first mover.

ἔτι χρόνῳ πρώτην· τοῖς γὰρ ἀϊδίοις μόνον ἐνδέχεται κινεῖσθαι
still time first to the for eternal only is possible to be moved
ταύτην. ἀλλ' ἐφ' ἑνὸς μὲν ὁτουοῦν τῶν ἐχόντων γένεσιν
this. but in regard to one indeed whatever of those having a generation
τὴν φορὰν ἀναγκαῖον ὑστάτην εἶναι τῶν κινήσεων· μετὰ γὰρ
the movement necessary last to be of the movements; after for
τὸ γενέσθαι πρῶτον ἀλλοίωσις καὶ αὔξησις, φορὰ δ' ἤδη
the having come to be first alteration and increase, movement but already
τετελειωμένων κίνησίς ἐστιν. [261a] ἀλλ' ἕτερον ἀνάγκη κινούμενον
of those completed the movement it is. [261a] but another necessity to be moved
εἶναι κατὰ φορὰν πρότερον, ὃ καὶ τῆς γενέσεως αἴτιον
to be according to movement earlier, which also of the generation cause
ἔσται τοῖς γιγνομένοις, οὐ γιγνόμενον, οἷον τὸ γεννῆσαν τοῦ
will be to the becoming, not becoming, such as the begetting of the
γεννηθέντος, ἐπεὶ δόξειέ γ' ἂν ἡ γένεσις εἶναι πρώτη τῶν
born, since it will seem indeed if the generation to be first of the
κινήσεων διὰ τοῦτο, ὅτι γενέσθαι δεῖ τὸ πρᾶγμα πρῶτον.
movements through this, that it must come to be necessary the thing first.
τὸ δ' ἐφ' ἑνὸς μὲν ὁτουοῦν τῶν γιγνομένων οὕτως ἔχει, ἀλλ'
the and on one indeed whatever of the beings thus it is, but
ἕτερον ἀναγκαῖον πρότερόν τι κινεῖσθαι τῶν γιγνομένων
another necessary prior something to be moved of the beings
ὂν αὐτὸ καὶ μὴ γιγνόμενον, καὶ τούτου ἕτερον πρότερον. ἐπεὶ δὲ
being it and not becoming, and of this another prior. since but
γένεσιν ἀδύνατον εἶναι πρώτην (πάντα γὰρ ἂν εἴη τὰ κινούμενα
genesis impossible to be first (for all indeed if it were the moving
φθαρτά), δῆλον ὡς οὐδὲ τῶν ἐφεξῆς κινήσεων οὐδεμία
perishables), it is clear that nor of the subsequent movements no

προτέρα· λέγω δ' ἐφεξῆς αὔξησιν, εἶτ' ἀλλοίωσιν καὶ φθί σιν

prior. I say and subsequently increase, then alteration and decay -

καὶ φθοράν· πᾶσαι γὰρ ὕστεραι γενέσεως, ὥστ' εἰ μηδὲ γένεσις

and destruction. all indeed later of genesis, so that if nor genesis

προτέρα φορᾶς, οὐδὲ τῶν ἄλλων οὐδεμία μεταβολῶν.

prior of movement, nor of the others no changes.

ὅλως τε φαίνεται τὸ γιγνόμενον ἀτελὲς καὶ ἐπ' ἀρχὴν

altogether and appears the occurring incomplete and to the beginning

ἰόν, ὥστε τὸ τῇ γενέσει ὕστερον τῇ φύσει πρότερον εἶναι.

going, so that the to the generation later to the nature earlier to be.

τελευταῖον δὲ φορὰ πᾶσιν ὑπάρχει τοῖς ἐν γενέσει. διὸ τὰ

last but movement to all exists for the in generation. therefore the

μὲν ὅλως ἀκίνητα τῶν ζώντων δι' ἔνδειαν [τοῦ ὀργάνου],

on one hand altogether unmoving of the living through lack of the instrument],

οἷον τὰ φυτὰ καὶ πολλὰ γένη τῶν ζῴων, τοῖς δὲ τελειουμένοις

such as the plants and many kinds of the animals, for the but being perfected

ὑπάρχει. ὥστ' εἰ μᾶλλον ὑπάρχει φορὰ τοῖς μᾶλλον ἀπειληφόσιν

exists. so that if rather exists movement for the rather threatened

τὴν φύσιν, καὶ ἡ κίνησις αὕτη πρώτη τῶν ἄλλων ἂν εἴη

the nature, and the motion this first of the others if it would be

κατ' οὐσίαν, διά τε ταῦτα καὶ διότι ἥκιστα τῆς οὐσίας

according to essence, through and these and because least of the essence

ἐξίσταται τὸ κινούμενον τῶν κινήσεων ἐν τῷ φέρεσθαι· κατὰ

exists the moving of the movements in the being carried; according to

μόνην γὰρ οὐδὲν μεταβάλλει τοῦ εἶναι, ὥσπερ ἀλλοιουμένου

the only for nothing changes of the being, just as that which is changing

μὲν τὸ ποιόν, αὐξανομένου δὲ καὶ φθίνοντος τὸ

on the one hand the quality, on the other hand increasing and also diminishing the

ποσόν. μάλιστα δὲ δῆλον ὅτι τὸ κινοῦν αὐτὸ αὑτὸ μάλιστα

quantity. especially and it is clear that the moving itself itself especially

ταύτην κινεῖ κυρίως, τὴν κατὰ τόπον· καίτοι φαμὲν τοῦτο

this moves primarily, the according to place; and yet we say this

εἶναι τῶν κινουμένων καὶ κινούντων ἀρχὴν καὶ πρῶτον τοῖς

to be of the things in motion and movers the principle and first to the

κινουμένοις, τὸ αὐτὸ αὑτὸ κινοῦν.

things being moved, the same itself moving.

ὅτι μὲν τοίνυν τῶν κινήσεων ἡ φορὰ πρώτη, φανερὸν ἐκ

that on the one hand therefore of the movements the rate first, evident from

τούτων· τίς δὲ φορὰ πρώτη, νῦν δεικτέον. ἅμα

these. who on the other hand rate first, now must be shown. at the same time

δὲ καὶ τὸ νῦν καὶ πρότερον ὑποτεθέν, ὅτι ἐνδέχεταί τινα κίνησιν

and also the now and before proposed, that it may happen some movement

εἶναι συνεχῆ καὶ ἀΐδιον, φανερὸν ἔσται τῇ αὐτῇ μεθόδῳ. ὅτι

to be continuous and eternal, evident it will be in the same method. that

μὲν οὖν τῶν ἄλλων κινήσεων οὐδεμίαν ἐνδέχεται

on the one hand therefore of the other movements no may occur

συνεχῆ εἶναι, ἐκ τῶνδε φανερόν. ἅπασαι γὰρ ἐξ ἀντικειμένων

continuous to be, from these evident. all for from opposing

εἰς ἀντικείμενά εἰσιν αἱ κινήσεις καὶ μεταβολαί, οἷον γενέσει

into opposing things are the movements and changes, such as generation

μὲν καὶ φθορᾷ τὸ ὂν καὶ τὸ μὴ ὂν ὅροι, ἀλλοιώσει

on the one hand and decay the being and the non-being being definitions, transformation

δὲ τὰ ἐναντία πάθη, αὐξήσει δὲ καὶ φθίσει ἢ

on the other hand the contrary passions, will increase and and it will perish or

μέγεθος καὶ μικρότης ἢ τελειότης μεγέθους καὶ ἀτέλεια· ἐναντίαι

greatness and smallness or perfection of greatness and imperfection. Opposites

δ' αἱ [261b] εἰς τὰ ἐναντία. τὸ δὲ μὴ αἰεὶ κινούμενον τήνδε τὴν

but the [261b] into the opposites. the but not always moving this the

κίνησιν, ὂν δὲ πρότερον, ἀνάγκη πρότερον ἠρεμεῖν. φανερὸν

motion, being and previously, necessity previously to be at rest. It is clear

οὖν ὅτι ἠρεμήσει ἐν τῷ ἐναντίῳ τὸ μεταβάλλον. ὁμοίως δὲ

therefore that it will be at rest in the opposite the changing. Likewise and

καὶ ἐπὶ τῶν μεταβολῶν· ἀντίκειται γὰρ φθορὰ καὶ γένεσις

also concerning the changes; it is opposed for destruction and generation

ἁπλῶς καὶ ἡ καθ' ἕκαστον τῇ καθ' ἕκαστον. ὥστ' εἰ

simply and the according to each the according to each. Therefore, if

ἀδύνατον ἅμα μεταβάλλειν τὰς ἀντικειμένας, οὐκ

it is impossible at the same time to change the opposites, there will not

ἔσται συνεχὴς ἡ μεταβολή, ἀλλὰ μεταξὺ ἔσται αὐτῶν χρόνος.

be continuous the change, but between there will be of them time.

οὐδὲν γὰρ διαφέρει ἐναντίας ἢ μὴ ἐναντίας εἶναι τὰς κατ'

Nothing for differs contrary or not contrary to be the according to

ἀντίφασιν μεταβολάς, εἰ μόνον ἀδύνατον ἅμα τῷ αὐτῷ

contradiction changes, if only impossible at the same time with that

παρεῖναι (τοῦτο γὰρ τῷ λόγῳ οὐδὲν χρήσιμον), οὐδ' εἰ μὴ ἀνάγκη

to be present (this for the reason nothing useful), nor if not necessity

ἠρεμῆσαι ἐν τῇ ἀντιφάσει, μηδ' ἐστὶν μεταβολὴ ἠρεμίᾳ ἐναντίον

to rest in the opposition, nor is a change in rest contrary

(οὐ γὰρ ἴσως ἠρεμεῖ τὸ μὴ ὄν, ἡ δὲ φθορὰ εἰς τὸ μὴ ὄν),

(not for perhaps it will rest the not being, the but decay into the not being),

ἀλλ’ εἰ μόνον μεταξὺ γίγνεται χρόνος· οὕτω γὰρ οὐκ ἔστιν ἡ

but if only in between does happen time· thus for not is the

μεταβολὴ συνεχής· οὐδὲ γὰρ ἐν τοῖς πρότερον ἡ ἐναντίωσις

change continuous· nor for in the former the opposition

χρήσιμον, ἀλλὰ τὸ μὴ ἐνδέχεσθαι ἅμα ὑπάρχειν. οὐ δεῖ δὲ

useful, but the not to be possible at the same time to exist. not should but

ταράττεσθαι ὅτι τὸ αὐτὸ πλείοσιν ἔσται ἐναντίον, οἷον ἡ κίνησις

be disturbed that the same to more will be opposed, as the motion

καὶ στάσει καὶ κινήσει τῇ εἰς τοὐναντίον, ἀλλὰ μόνον τοῦτο

and state and movement to the into the opposite, but only this

λαμβάνειν, ὅτι ἀντίκειταί πως καὶ τῇ κινήσει καὶ τῇ ἠρεμίᾳ ἡ

to take, that it is opposed somehow and to the motion and to the rest the

κίνησις ἡ ἐναντία, καθάπερ τὸ ἴσον καὶ τὸ μέτριον τῷ ὑπερέχοντι

motion the opposite, just as the equal and the moderate to the surpassing

καὶ τῷ ὑπερεχομένῳ, καὶ ὅτι οὐκ ἐνδέχεται ἅμα τὰς

and to the surpassing, and that not is possible at the same time the

ἀντικειμένας οὔτε κινήσεις οὔτε μεταβολὰς ὑπάρχειν. ἔτι δ’ ἐπί

opposing nor motions nor changes to exist. moreover but upon

τε τῆς γενέσεως καὶ τῆς φθορᾶς καὶ παντελῶς ἄτοπον ἂν εἶναι

and the generation and the decay and entirely absurd if to be

δόξειεν, εἰ γενόμενον εὐθὺς ἀνάγκη φθαρῆναι καὶ μηδένα

it would seem, if having become immediately necessity to perish and no

χρόνον διαμεῖναι. ὥστε ἐκ τούτων ἂν ἡ πίστις γένοιτο

time to remain. so that from these (modal particle) the faith might become

ταῖς ἄλλαις· φυσικὸν γὰρ τὸ ὁμοίως ἔχειν ἐν ἁπάσαις.

the others natural for the likewise to have in all

Κεφάλαιο 8

Chapter 8

Ὅτι δ' ἐνδέχεται εἶναί τινα ἄπειρον, μίαν οὖσαν καὶ συνεχῆ, καὶ

That but it is possible to be some infinite, one being and continuous, and

αὕτη ἐστὶν ἡ κύκλῳ, λέγωμεν νῦν. πᾶν μὲν γὰρ κινεῖται τὸ

this is the circle, let us say now. Everything indeed for moves the

φερόμενον ἢ κύκλῳ ἢ εὐθεῖαν ἢ μικτήν, ὥστ' εἰ μηδ' ἐκείνων

moving or circle or straight line or mixed, so that if not even those

ἡ ἑτέρα συνεχής, οὐδὲ τὴν ἐξ ἀμφοῖν οἷόν τ' εἶναι συγκειμένην·

the other continuous, nor the from both capable of being composed.

ὅτι δὲ τὸ φερόμενον τὴν εὐθεῖαν καὶ πεπερασμένην οὐ φέρεται

That but the moving the straight line and finite not is carried

συνεχῶς, δῆλον· ἀνακάμπτει γάρ, τὸ δ' ἀνακάμπτον τὴν εὐθεῖαν

continuously, evident: it bends for, the but bending the straight line

τὰς ἐναντίας κινεῖται κινήσεις· ἐναντία γὰρ κατὰ τόπον ἡ

the opposite moves; movements: opposite for according to place the

ἄνω τῇ κάτω καὶ ἡ εἰς τὸ πρόσθεν τῇ εἰς τοὔπισθεν καὶ ἡ

upward the downward and the towards the front the into the back and the

εἰς ἀριστερὰ τῇ εἰς δεξιά· τόπου γὰρ ἐναντιώσεις αὗται. τίς δ'

into left to the into right. of place for oppositions these. who but

ἐστὶν ἡ μία καὶ [262a] συνεχὴς κίνησις, διώρισται πρότερον,

is the one and [262a] continuous motion, has been defined previously,

ὅτι ἡ τοῦ ἑνὸς καὶ ἐν ἑνὶ χρόνῳ καὶ ἐν ἀδιαφόρῳ κατ'

that the of the one and in one time and in indifferent according to

εἶδος (τρία γὰρ ἦν, τό τε κινούμενον, οἷον ἄνθρωπος ἢ

kind (three for there was, the and moving, such as a human or

θεός, καὶ ὅτε, οἷον χρόνος, καὶ τρίτον τὸ ἐν ᾧ· τοῦτο δ'

a god, and when, such as time, and a third the in which; . this but

ἐστὶν τόπος ἢ πάθος ἢ εἶδος ἢ μέγεθος). τὰ δ' ἐναντία διαφέρει

is place or state or kind or magnitude). the but opposites differ

τῷ εἴδει, καὶ οὐχ ἕν· τόπου δ' αἱ εἰρημέναι διαφοραί. σημεῖον

to the kind, and not one; of place but the mentioned differences. sign

δ' ὅτι ἐναντία ἡ κίνησις ἡ ἀπὸ τοῦ Α πρὸς τὸ Β τῇ ἀπὸ τοῦ

but that opposite the motion the from the A towards the B the from the

Β πρὸς τὸ Α, ὅτι ἱστᾶσιν καὶ παύουσιν ἀλλήλας, ἐὰν ἅμα

B towards the A, that they settle and cease each other, if together

γίγνωνται. καὶ ἐπὶ κύκλου ὡσαύτως, οἷον ἡ ἀπὸ τοῦ Α ἐπὶ τὸ

they come to be. and on circle likewise, such as the from the A to the

Β τῇ ἀπὸ τοῦ Α ἐπὶ τὸ Γ (ἱστᾶσι γάρ, κἂν συνεχεῖς ὦσιν καὶ

B the from the A to the C they settle for, and if continuous they be and

μὴ γίγνηται ἀνάκαμψις, διὰ τὸ τἀναντία φθείρειν καὶ κωλύειν

not it comes to be a rebound, through the opposites to destroy and to hinder

ἄλληλα)· ἀλλ' οὐχ ἡ εἰς τὸ πλάγιον τῇ ἄνω. μάλιστα δὲ

each other). but not the towards the oblique the upward. especially but

φανερὸν ὅτι ἀδύνατον εἶναι συνεχῆ τὴν ἐπὶ τῆς εὐθείας κίνησιν,

evident that impossible to be continuous the on of the straight motion,

ὅτι ἀνακάμπτον ἀναγκαῖον στῆναι, οὐ μόνον ἐπ' εὐθείας, ἀλλὰ

because turning back necessary to stand, not only in straight, but

κἂν κύκλον φέρηται. οὐ γὰρ ταὐτὸν κύκλῳ φέρεσθαι καὶ

even if a circle is carried. not for the same with a circle to be carried and

κύκλον· ἔστιν γὰρ ὁτὲ μὲν συνείρειν κινούμενον, ὁτὲ

a circle. There is for sometimes indeed to converge moving, sometimes

δ' ἐπὶ τὸ αὐτὸ ἐλθὸν ὅθεν ὡρμήθη ἀνακάμψαι πάλιν. ὅτι δ'

but at the same going, whence it was urged to turn back again. because but

ἀνάγκη ἵστασθαι, ἡ πίστις οὐ μόνον ἐπὶ τῆς αἰσθήσεως ἀλλὰ

necessity to stand, the belief not only in the perception but also

καὶ ἐπὶ τοῦ λόγου. ἀρχὴ δὲ ἥδε. τριῶν γὰρ ὄντων, ἀρχῆς
and in the reason. Beginning but this is. of three for beings, of beginning
μέσου τελευτῆς, τὸ μέσον πρὸς ἑκάτερον ἄμφω ἐστίν, καὶ τῷ
middle and end, the middle towards each both is, and to the
μὲν ἀριθμῷ ἕν, τῷ λόγῳ δὲ δύο. ἔτι δὲ ἄλλο ἐστὶν τὸ δυνάμει
one count one, to the reason but two. still but other is the by power
καὶ τὸ ἐνεργείᾳ, ὥστε τῆς εὐθείας τῶν ἐντὸς τῶν ἄκρων ὁτιοῦν
and the by activity, so that of the straight of the within the extremes of any
σημεῖον δυνάμει μέν ἐστι μέσον, ἐνεργείᾳ δ' οὐκ ἔστιν, ἐὰν
sign by power on the one hand is a middle, by activity but not is, if
μὴ διέλῃ ταύτῃ καὶ ἑπιστὰν πάλιν ἄρξηται κινεῖσθαι· οὕτω
not it passes through this and having stood again may begin to move; thus
δὲ τὸ μέσον ἀρχὴ γίγνεται καὶ τελευτή, ἀρχὴ μὲν τῆς
and the middle is a beginning becomes and an end, beginning indeed of the
ὕστερον, τελευτὴ δὲ τῆς πρώτης (λέγω δ' οἷον ἐὰν φερόμενον τὸ
later, end but of the first (I say but such as if being carried the
A στῇ ἐπὶ τοῦ B καὶ πάλιν φέρηται ἐπὶ τὸ Γ). ὅταν δὲ συνεχῶς
A stands upon the B and again is carried upon the C). when but continuously
φέρηται, οὔτε γεγονέναι οὔτε ἀπογεγονέναι οἷόν τε τὸ A
it is carried, neither has become nor has ceased to be such as also the A
κατὰ τὸ B σημεῖον, ἀλλὰ μόνον εἶναι ἐν τῷ νῦν, ἐν
in accordance with the B sign, but only to be in the present, in
χρόνῳ δ' οὐδενὶ πλὴν οὗ τὸ νῦν ἐστιν διαί ρεσις, ἐν τῷ
time of no one except of whom the present is division operations, in the
ὅλῳ [τῷ ΑΒΓ]. (εἰ δὲ γεγονέναι τις θήσει καὶ ἀπογεγονέναι,
whole [the ABG]. (if but having occurred someone will place and having been removed,
ἀεὶ στήσεται τὸ A φερόμενον· ἀδύνατον [262b] γὰρ τὸ A
always will stand the A being brought; impossible [262b] for the A

ἅμα γεγονέναι τε ἐπὶ τοῦ Β καὶ ἀπογεγονέναι. ἐν ἄλλῳ
simultaneously having occurred and upon the B and having been removed. in another
ἄρα καὶ ἄλλῳ σημείῳ χρόνου. χρόνος ἄρα ἔσται ὁ ἐν μέσῳ.
thus and another sign of time. Time thus will be the in the middle.
ὥστε ἠρεμήσει τὸ Α ἐπὶ τοῦ Β. ὁμοίως δὲ καὶ ἐπὶ τῶν ἄλλων
so that it will calm the A upon the B. Similarly but and upon the other
σημείων· ὁ γὰρ αὐτὸς λόγος ἐπὶ πάντων. ὅταν δὴ χρήσηται
signs; the for itself word upon all. whenever indeed it is used
τὸ φερόμενον Α τῷ Β μέσῳ καὶ τελευτῇ καὶ ἀρχῇ, ἀνάγκη
the carried A to the B midst and end and beginning, necessity
στῆναι διὰ τὸ δύο ποιεῖν, ὥσπερ ἂν εἰ καὶ νοήσειεν.)
to stand through the two to make, just as if it were also it would understand.)
ἀλλ' ἀπὸ μὲν τοῦ Α σημείου ἀπογέγονε τῆς ἀρχῆς, ἐπὶ
but from on one hand the A sign, it has descended from the beginning, upon
δὲ τοῦ Γ γέγονεν, ὅταν τελευτήσῃ καὶ στῇ. διὸ καὶ πρὸς
but the G it has occurred, whenever it finishes and stands. therefore and towards
τὴν ἀπορίαν τοῦτο λεκτέον· ἔχει γὰρ ἀπορίαν τήνδε. εἰ γὰρ εἴη
the difficulty this must be said: has for a difficulty this. if for it were
ἡ τὸ Ε τῇ Ζ ἴση καὶ τὸ Α φέροιτο συνεχῶς ἀπὸ τοῦ ἄκρου
the the E to the Z equal and the A would carry constantly from the extreme
πρὸς τὸ Γ, ἅμα δ' εἴη τὸ Α ἐπὶ τῷ Β σημείῳ, καὶ
towards the G, at the same time but it would be the A upon the B point, and
τὸ Δ φέροιτο ἀπὸ τῆς Ζ ἄκρας πρὸς τὸ Η ὁμαλῶς καὶ τῷ αὐτῷ
the D might bring from the Z peak towards the H smoothly and the same
τάχει τῷ Α, τὸ Δ ἔμπροσθεν ἥξει ἐπὶ τὸ Η ἢ τὸ Α ἐπὶ τὸ
speed the A, the D before will arrive upon the H or the A upon the
Γ· τὸ γὰρ πρότερον ὁρμῆσαν καὶ ἀπελθὸν πρότερον ἐλθεῖν
G· the for previously they rushed and having gone away previously to come

ἀνάγκη. οὐ γὰρ ἅμα γέγονε τὸ A ἐπὶ τῷ B καὶ ἀπογέγονεν

necessity. not for together has taken place the A upon the B and has descended

ἀπ' αὐτοῦ, διὸ ὑστερίζει. εἰ γὰρ ἅμα, οὐχ ὑστεριεῖ,

from it, therefore it lags behind. if for together, not it will lag behind,

ἀλλ' ἀνάγκη ἔσται ἵστασθαι. οὐκ ἄρα θετέον, ὅτε τὸ A

but necessity will be to stand. not therefore it must be set, when the A

ἐγένετο κατὰ τὸ B, τὸ Δ ἅμα κινεῖσθαι ἀπὸ τοῦ Z ἄκρου

has come to be according to the B, the D together to be moved from the Z apex

(εἰ γὰρ ἔσται γεγονὸς τὸ A ἐπὶ τοῦ B, ἔσται καὶ τὸ

if for it will be having come to be the A upon the B, it will be and the

ἀπογενέσθαι, καὶ οὐχ ἅμα), ἀλλ' ἦν ἐν τομῇ χρόνου καὶ οὐκ

arising, and not together), but it was in part of time and not

ἐν χρόνῳ. ἐνταῦθα μὲν οὖν ἀδύνατον οὕτως λέγειν ἐπὶ

in time. Here on the one hand therefore impossible thus to speak upon

τῆς συνεχοῦς· ἐπὶ δὲ τοῦ ἀνακάμπτοντος ἀνάγκη λέγειν οὕτως.

of the continuous; upon but the returning necessity to speak thus.

εἰ γὰρ ἡ τὸ H φέροιτο πρὸς τὸ Δ καὶ πάλιν ἀνακάμψασα κάτω

if for the the H it would bear towards the D and again having returned down

φέροιτο, τῷ ἄκρῳ ἐφ' οὗ Δ τελευτῇ καὶ ἀρχῇ κέχρηται,

it would bear, to the apex upon which of which D it ends and begins it is used,

τῷ ἑνὶ σημείῳ ὡς δύο· διὸ στῆναι ἀνάγκη· καὶ οὐχ ἅμα

to the one point as two; therefore to stand necessity; and not at once

γέγονεν ἐπὶ τῷ Δ καὶ ἀπελήλυθεν ἀπὸ τοῦ Δ· ἐκεῖ γὰρ ἂν

has come to be upon the D and has departed from the D. there for should

ἅμα εἴη καὶ οὐκ εἴη ἐν τῷ αὐτῷ νῦν. ἀλλὰ μὴν τήν γε πάλαι

at once be and not be in the same now. but indeed the at least long ago

λύσιν οὐ λεκτέον· οὐ γὰρ ἐνδέχεται λέγειν ὅτι ἐστὶν κατὰ

release not to be spoken. not for it is likely to say that it is according to

τὸ Δ ἢ τὸ Η ἐν τομῇ, οὐ γέγονε δὲ οὐδ' ἀπογέγονεν.

the D the the H in a division, not has come to be but not even has departed.

ἀνάγκη γὰρ ἐπὶ τέλος ἐλθεῖν τὸ ἐνεργείᾳ ὄν, μὴ δυνάμει. τὰ

necessity for to an end to arrive the in actuality being, not in potential. the

μὲν οὖν ἐν μέσῳ δυνάμει ἔστι, τοῦτο δ' ἐνεργείᾳ, καὶ

on one hand therefore in the middle potentiality is, this but in actuality, and

τελευτὴ μὲν κάτωθεν, [263a] ἀρχὴ δὲ ἄνωθεν·

the end on one hand, from below, [263a] the beginning on the other hand from above.

καὶ τῶν κινήσεων ἄρα ὡσαύτως. ἀνάγκη ἄρα στῆναι τὸ

and of the movements therefore likewise. necessity therefore to stand the

ἀνακάμπτον ἐπὶ τῆς εὐθείας. οὐκ ἄρα ἐνδέχεται συνεχῆ

turning toward the straight line. not therefore it is possible continuous

κίνησιν εἶναι ἐπὶ τῆς εὐθείας ἀΐδιον.

motion to be toward the straight line everlasting.

τὸν αὐτὸν δὲ τρόπον ἀπαντητέον καὶ πρὸς τοὺς ἐρωτῶντας τὸν

the same but manner must be answered and towards the questioners the

Ζήνωνος λόγον, [καὶ ἀξιοῦντας,] εἰ ἀεὶ τὸ ἥμισυ διιέναι δεῖ,

of Zeno argument, and demanding, if always the half to pass is necessary,

ταῦτα δ' ἄπειρα, τὰ δ' ἄπειρα ἀδύνατον διεξελθεῖν, ἢ ὡς τὸν

these things but infinite, the but infinite impossible to traverse, or as the

αὐτὸν τοῦτον λόγον τινὲς ἄλλως ἐρωτῶσιν, ἀξιοῦντες ἅμα

same this argument some otherwise question, demanding together

τῷ κινεῖσθαι τὴν ἡμίσειαν πρότερον ἀριθμεῖν καθ' ἕκαστον

with moving the half beforehand to count according to each

γιγνόμενον τὸ ἥμισυ, ὥστε διελθόντος τὴν ὅλην ἄπειρον συμβαίνει

occurrence the half, so that having passed the whole infinite it occurs

ἠριθμηκέναι ἀριθμόν· τοῦτο δ' ὁμολογουμένως ἐστὶν ἀδύνατον.

to be counted number. this but admittedly is impossible.

ἐν μὲν οὖν τοῖς πρώτοις λόγοις τοῖς περὶ κινήσεως

in on the one hand therefore the first arguments the regarding motion

ἐλύομεν διὰ τοῦ τὸν χρόνον ἄπειρα ἔχειν ἐν αὑτῷ· οὐδὲν γὰρ

we resolved through the the time infinite to have in itself. nothing for

ἄτοπον εἰ ἐν ἀπείρῳ χρόνῳ ἄπειρα διέρχεταί τις· ὁμοίως δὲ

strange if in infinite time infinite passes someone; similarly and

τὸ ἄπειρον ἔν τε τῷ μήκει ὑπάρχει καὶ ἐν τῷ χρόνῳ. ἀλλ' αὕτη

the infinite in and the length exists and in the time. but this

ἡ λύσις πρὸς μὲν τὸν ἐρωτῶντα ἱκανῶς ἔχει (ἠρωτᾶτο γὰρ

the solution to indeed the questioner sufficiently has (was asking for

εἰ ἐν πεπερασμένῳ ἄπειρα ἐνδέχεται διεξελθεῖν ἢ ἀριθμῆσαι),

if in finite infinite it is possible to go through or to count),

πρὸς δὲ τὸ πρᾶγμα καὶ τὴν ἀλήθειαν οὐχ ἱκανῶς· ἂν γάρ τις

to but the matter and the truth not sufficiently; if for someone

ἀφέμενος τοῦ μήκους καὶ τοῦ ἐρωτᾶν εἰ ἐν πεπερασμένῳ

having relinquished the length and the questioning if in finite

χρόνῳ ἐνδέχεται ἄπειρα διεξελθεῖν, πυνθάνηται ἐπ' αὐτοῦ τοῦ

time it is possible infinite to go through, he/she may learn about that the

χρόνου ταῦτα (ἔχει γὰρ ὁ χρόνος ἀπείρους διαιρέσεις), οὐκέτι

time these (has for the time infinite divisions), no longer

ἱκανὴ ἔσται αὕτη ἡ λύσις, ἀλλὰ τὸ ἀληθὲς λεκτέον, ὅπερ

sufficient will be this the solution, but the truth must be said, that which

εἴπομεν ἐν τοῖς ἄρτι λόγοις. ἐὰν γάρ τις τὴν συνεχῆ διαιρῇ εἰς

we said in the recent words. if for anyone the continuous divides into

δύο ἡμίση, οὗτος τῷ ἑνὶ σημείῳ ὡς δυσὶ χρῆται· ποιεῖ γὰρ αὐτὸ

two halves, this the one point as two he uses; he makes for it

ἀρχὴν καὶ τελευτήν. οὕτω δὲ ποιεῖ ὅ τε ἀριθμῶν καὶ ὁ εἰς
a beginning and an end. Thus and he makes that both of numbers and the into
τὰ ἡμίση διαιρῶν. οὕτω δὲ διαιροῦντος οὐκ ἔσται συνεχὴς οὔθ' ἡ
the halves divides. Thus and of one dividing not will be continuous neither the
γραμμὴ οὔθ' ἡ κίνησις· ἡ γὰρ συνεχὴς κίνησις συνεχοῦς ἐστιν,
line nor the motion; the for continuous motion of continuity is,
ἐν δὲ τῷ συνεχεῖ ἔνεστι μὲν ἄπειρα ἡμίση, ἀλλ' οὐκ ἐντελεχείᾳ
in but the continuous exists indeed infinitely halves, but not in completeness
ἀλλὰ δυνάμει. ἂν δὲ ποιῇ ἐντελεχείᾳ, οὐ ποιήσει
but by power. if but he/she/it does in actuality, not he/she/it will do
συνεχῆ, ἀλλὰ στήσει, ὅπερ ἐπὶ τοῦ ἀριθμοῦντος τὰ ἡμίσεα
continuously, but will establish, which upon the numbering the halves
φανερόν ἐστιν ὅτι συμβαίνει· τὸ γὰρ ἓν σημεῖον ἀνάγκη [263b]
is clear is that that it happens; the for one sign necessity [263b]
αὐτῷ ἀριθμεῖν δύο· τοῦ μὲν γὰρ ἑτέρου τελευτὴ ἡμίσεος
to him to number two; the on the one hand for of another the end of half
τοῦ δ' ἑτέρου ἀρχὴ ἔσται, ἂν μὴ μίαν ἀριθμῇ τὴν συνεχῆ, ἀλλὰ
the but of another beginning will be, if not one in number the continuous, but
δύο ἡμισείας. ὥστε λεκτέον πρὸς τὸν ἐρωτῶντα εἰ ἐνδέχεται
two halves. thus it must be said to the questioner if it is possible
ἄπειρα διεξελθεῖν ἢ ἐν χρόνῳ ἢ ἐν μήκει, ὅτι ἔστιν ὡς, ἔστιν
infinitely to proceed or in time or in length, as it is, as, it is
δ' ὡς οὔ. ἐντελεχείᾳ μὲν γὰρ ὄντα οὐκ ἐνδέχεται, δυνάμει
but as not. In actuality on one hand for being not possible, by power
δὲ ἐνδέχεται· ὁ γὰρ συνεχῶς κινούμενος κατὰ συμβεβηκὸς
but it is possible; the for constantly moving according to incident
ἄπειρα διελήλυθεν, ἁπλῶς δ' οὔ· συμβέβηκε γὰρ τῇ γραμμῇ
infinite has passed through, simply but not; it has happened for to the line

ἄπειρα ἡμίσεα εἶναι, ἡ δ' οὐσία ἐστὶν ἑτέρα καὶ τὸ εἶναι. δῆλον
infinite halves to be, the but substance is another and the being. clear
δὲ καὶ ὅτι ἐὰν μή τις ποιῇ τοῦ χρόνου τὸ διαιροῦν σημεῖον τὸ
and also that if not someone does of the time the dividing sign the
πρότερον καὶ ὕστερον ἀεὶ τοῦ ὑστέρου τῷ πράγματι, ἔσται
earlier and later always of the later with the matter, it will be
ἅμα τὸ αὐτὸ ὂν καὶ οὐκ ὄν, καὶ ὅτε γέγονεν οὐκ ὄν. τὸ
at once the same being and not being, and when it has become not being. the
σημεῖον μὲν οὖν ἀμφοῖν κοινόν, καὶ τοῦ προτέρου καὶ τοῦ
sign indeed therefore of both common, and of the former and of the
ὑστέρου, καὶ ταὐτὸν καὶ ἓν ἀριθμῷ, λόγῳ δ' οὐ ταὐτόν (τοῦ
latter, and the same and one number, in speech but not the same (of the
μὲν γὰρ τελευτή, τοῦ δ' ἀρχή)· τῷ δὲ πράγματι ἀεὶ
on the one hand for end, of the but beginning). to the and fact always
τοῦ ὑστέρου πάθους ἐστίν. χρόνος ἐφ' ᾧ ΑΓΒ, πρᾶγμα
of the later condition is. time on which which ABC, thing
ἐφ' ᾧ Δ. τοῦτο ἐν μὲν τῷ Α χρόνῳ λευκόν, ἐν
on which which D. this in on the one hand the A time white, in
δὲ τῷ Β οὐ λευκόν· ἐν τῷ ἄρα Γ λευκὸν καὶ οὐ λευκόν. ἐν
but the B not white. in the thus C white and not white. in
ὁτῳοῦν γὰρ τοῦ Α λευκὸν ἀληθὲς εἰπεῖν, εἰ πάντα τὸν χρόνον
whatever for of the A white true to say, if all the time
τοῦτον ἦν λευκόν, καὶ ἐν τῷ Β οὐ λευκόν· τὸ δὲ Γ ἐν ἀμφοῖν.
this was white, and in the B not white. the and C in both.
οὐκ ἄρα δοτέον ἐν παντί, ἀλλὰ πλὴν τοῦ τελευταίου νῦν
not therefore to be given in every but except the of the last now
ἐφ' οὗ τὸ Γ· τοῦτο δ' ἤδη τοῦ ὑστέρου. καὶ εἰ ἐγίγνετο
upon which the G. this and already of the later. And if it became

οὐ λευκὸν καὶ ἐφθείρετο <τὸ> λευκὸν ἐν τῷ Α παντί,

not white and it was destroyed the white in the A altogether,

γέγονεν ἢ ἔφθαρται ἐν τῷ Γ. ὥστε λευκὸν ἢ μὴ λευκὸν

it has become or it has been corrupted in the G. so that white or not white

ἐν ἐκείνῳ πρῶτον ἀληθὲς εἰπεῖν, ἢ ὅτε γέγονεν οὐκ ἔσται,

in that first true to say, or when it has become not it will be,

καὶ ὅτε ἔφθαρται ἔσται, ἢ ἅμα λευκὸν καὶ οὐ λευκὸν

and when it is corrupted it will be, or at the same time white and not white

καὶ ὅλως ὂν καὶ μὴ ὂν ἀνάγκη εἶναι. εἰ δ' ὃ ἂν ᾖ

and altogether being and not being necessity to be. If however whatever may be

πρότερον μὴ ὄν, ἀνάγκη γίγνεσθαι ὄν, καὶ ὅτε γίγνεται μὴ ἔστιν,

before not being, necessity to become being, and when it becomes not it is,

οὐχ οἷόν τε εἰς ἀτόμους χρόνους διαιρεῖσθαι τὸν χρόνον. εἰ γὰρ ἐν

not such and into atoms times to be divided the time. if for in

τῷ Α τὸ Δ ἐγίγνετο λευκόν, γέγονε δ' ἅμα καὶ ἔστιν

the A the D was generated white, it has become but at the same time and is

ἐν ἑτέρῳ ἀτόμῳ χρόνῳ ἐχομένῳ δ', ἐν τῷ Β-εἰ ἐν τῷ Α ἐγίγνετο,

in another atom time having but, in the B-if in the A was generated,

οὐκ ἦν, ἐν δὲ τῷ Β ἐστί-, γένεσιν δεῖ τινὰ εἶναι μεταξύ, ὥστε

not was, in but the B it is-, generation must some to be between, so that

καὶ [264a] χρόνον ἐν ᾧ ἐγίγνετο. οὐ γὰρ ὁ αὐτὸς ἔσται

also [264a] time in which it was generated. not for the same will be

λόγος καὶ τοῖς μὴ ἄτομα λέγουσιν, ἀλλ' αὐτοῦ τοῦ χρόνου,

word and to those not atoms say, but of itself the time,

ἐν ᾧ ἐγίγνετο, γέγονε καὶ ἔστιν ἐν τῷ ἐσχάτῳ σημείῳ,

in which it was generated, it has become and is in the last sign,

οὗ οὐδὲν ἐχόμενόν ἐστιν οὐδ' ἐφεξῆς· οἱ δὲ ἄτομοι χρόνοι

of which nothing holding is nor subsequently the but indivisible times

ἐφεξῆς. φανερὸν δ' ὅτι εἰ ἐν τῷ Α ὅλῳ χρόνῳ ἐγίγνετο, οὐκ

subsequently. evident but that if in the A whole time it came to be, not

ἔστιν πλείων χρόνος ἐν ᾧ γέγονεν καὶ ἐγίγνετο ἢ ἐν ᾧ

is more time in which it has come to be and it came to be or in which

ἐγίγνετο μόνον παντί.

it came to be only universally.

οἷς μὲν οὖν ἄν τις ὡς οἰκείοις πιστεύσειε

to whom on the one hand therefore if anyone as those belonging might believe

λόγοις, οὗτοι καὶ τοιοῦτοί τινές εἰσιν· λογικῶς δ' ἐπισκοποῦσι

in arguments, these and such some are. Logically, but they observe

κἂν ἐκ τῶνδε δόξειέ τῳ ταὐτὸ τοῦτο συμβαίνειν. ἅπαν γὰρ

even if from these it seems to him the same this to happen. All for

τὸ κινούμενον συνεχῶς, ἂν ὑπὸ μηδενὸς ἐκκρούηται, εἰς ὅπερ

the moving continuously, if by no is obstructed, into that which

ἦλθεν κατὰ τὴν φοράν, εἰς τοῦτο καὶ ἐφέρετο πρότερον, οἷον

has come according to the process, into this and was carried previously, just as

εἰ ἐπὶ τὸ Β ἦλθε, καὶ ἐφέρετο ἐπὶ τὸ Β, καὶ οὐχ ὅτε πλησίον

if to the B has come, and was carried to the B, and not when nearby

ἦν, ἀλλ' εὐθὺς ὡς ἤρξατο κινεῖσθαι· τί γὰρ μᾶλλον νῦν ἢ

he was, but immediately as he began to move. What for more now than

πρότερον; ὁμοίως δὲ καὶ ἐπὶ τῶν ἄλλων. τὸ δὴ ἀπὸ τοῦ Α [ἐπὶ

previously? Similarly but also for the others. The indeed from of the A upon

τὸ Γ] φερόμενον, ὅταν ἐπὶ τὸ Γ ἔλθῃ, πάλιν ἥξει ἐπὶ τὸ Α

the G being carried, when upon the G comes, again it will arrive upon the A

συν εχῶς κινούμενον. ὅτε ἄρα ἀπὸ τοῦ Α φέρεται πρὸς τὸ

with having moved. when therefore from the A it is brought towards the

Γ, τότε καὶ εἰς τὸ Α φέρεται τὴν ἀπὸ τοῦ Γ κίνησιν, ὥσθ'

G, then and into the A it is brought the from the the G motion, so that

ἅμα τὰς ἐναντίας· ἐναντίαι γὰρ αἱ κατ' εὐθεῖαν. ἅμα

together the oppositions. Oppositions for the according to straight. together

δὲ καὶ ἐκ τούτου μεταβάλλει ἐν ᾧ οὐκ ἔστιν. εἰ οὖν τοῦτ'

but also from this it changes in which not is. if therefore this

ἀδύνατον, ἀνάγκη ἵστασθαι ἐπὶ τοῦ Γ. οὐκ ἄρα μία ἡ κίνησις·

impossible, necessity to stand upon the G. not therefore one the motion;

ἡ γὰρ διαλαμ βανομένη στάσει οὐ μία.

the for lasting it is shining in state not one.

ἔτι καὶ ἐκ τῶνδε φανερὸν καθόλου μᾶλλον περὶ πάσης κινήσεως.

still and out of these evident in general rather concerning all motion.

εἰ γὰρ ἅπαν τὸ κινούμενον τῶν εἰρημένων τινὰ κινεῖται κινήσεων

if for all the moving of those mentioned some is moved by motions

καὶ ἠρεμεῖ τῶν ἀντικειμένων ἠρεμιῶν (οὐ γὰρ ἦν ἄλλη παρὰ

and rests of the objects at rest (not for it was another than

ταύτας), τὸ δὲ μὴ αἰεὶ κινούμενον τήνδε τὴν κίνησιν (λέγω δ'

these), the but not always moving this the motion (I say but

ὅσαι ἕτεραι τῷ εἴδει, καὶ μὴ εἴ τι μόριόν ἐστιν τῆς ὅλης)

as many other to the kind, and not if anything part it is of the whole)

ἀνάγκη πρότερον ἠρεμεῖν τὴν ἀντικειμένην ἠρεμίαν (ἡ γὰρ ἠρεμία

necessity before to rest the opposing rest (the for rest

στέρησις κινήσεως)· εἰ οὖν ἐναντίαι μὲν κινήσεις αἱ

is the absence of motion); if therefore opposing on the one hand movements the

κατ' εὐθεῖαν, ἅμα δὲ μὴ ἐνδέχεται κινεῖσθαι τὰς ἐναντίας,

in straight, at the same time but not can be moved the opposites,

τὸ ἀπὸ τοῦ Α πρὸς τὸ Γ φερόμενον οὐκ ἂν φέροιτο ἅμα καὶ
the from of the A towards the G being brought not would be brought together and
ἀπὸ τοῦ Γ πρὸς τὸ Α· ἐπεὶ δ' οὐχ ἅμα φέρεται, κινήσεται
from the G towards the A. since but not together is brought, it will move
δὲ ταύτην τὴν κίνησιν, ἀνάγκη πρότερον ἠρεμῆσαι πρὸς τῷ
and this the motion, necessity beforehand to be at rest towards the
Γ· αὕτη γὰρ ἦν ἡ ἀντικειμένη ἠρεμία τῇ ἀπὸ τοῦ Γ κινήσει.
G. this for was the opposing calmness to the from the G movement.
δῆλον τοίνυν [264b] ἐκ τῶν εἰρημένων ὅτι οὐκ ἔσται συνεχὴς
it is clear therefore [264b] from the aforementioned that not will be continuous
ἡ κίνησις.
the movement.

ἔτι δὲ καὶ ὅδε ὁ λόγος μᾶλλον οἰκεῖος τῶν εἰρημένων. ἅμα
still and also this the word rather appropriate of the said. at the same time
γὰρ ἔφθαρται τὸ οὐ λευκὸν καὶ γέγονε λευκόν. εἰ οὖν
for it is corrupted the not white and has become white. if therefore
συνεχὴς ἡ ἀλλοίωσις εἰς λευκὸν καὶ ἐκ λευκοῦ καὶ μὴ μένει
continuous the change into white and from the white and not remains
τινὰ χρόνον, ἅμα ἔφθαρται τὸ οὐ λευκὸν καὶ γέγονε
any time, at the same time it is corrupted the not white and has become
λευκὸν καὶ γέγονεν οὐ λευκόν· τριῶν γὰρ ἔσται ὁ αὐτὸς
white and has become not white; of three for there will be the same
χρόνος.
time.

ἔτι οὐκ εἰ συνεχὴς ὁ χρόνος, καὶ ἡ κίνησις, ἀλλ' ἐφεξῆς. πῶς
still not if continuous the time, and the movement, but subsequently. how

δ' ἂν εἴη τὸ ἔσχατον τὸ αὐτὸ τῶν ἐναντίων, οἷον λευκότητος

then would it be the last the same of the opposites, such as whiteness

καὶ μελανίας;

and blackness?

ἡ δ' ἐπὶ τῆς περιφεροῦς ἔσται μία καὶ συνεχής· οὐθὲν γὰρ ἀδύνατον

the but upon the revolving will be one and continuous· nothing for impossible

συμβαίνει· τὸ γὰρ ἐκ τοῦ A κινούμενον ἅμα κινήσεται εἰς

happens· the for from the A moving at the same time will move into

τὸ A κατὰ τὴν αὐτὴν πρόθεσιν (εἰς ὃ γὰρ ἥξει, καὶ

the A according to the same intention (to which for it will come, and

κινεῖται εἰς τοῦτο), ἀλλ' οὐχ ἅμα κινήσεται τὰς ἐναντίας

it is moving into this), but not at the same time will move the opposites

οὐδὲ τὰς ἀντικειμένας· οὐ γὰρ ἅπασα ἡ εἰς τοῦτο τῇ ἐκ τούτου

nor the contradicting· not for all the into this the from this

ἐναντία οὐδ' ἀντικειμένη, ἀλλ' ἐναντία μὲν ἡ κατ' εὐθεῖαν

opposite nor contradicting, but opposite on one hand the on direct

(ταύτῃ γὰρ ἔστιν ἐναντία κατὰ τόπον, οἷον τὰ κατὰ διάμε

(in this way for it is opposite by place, such as the according to diameter

τρον· ἀπέχει γὰρ πλεῖστον), ἀντικειμένη δὲ ἡ κατὰ τὸ

through· it is distant for the greatest), contradicting but the according to the

αὐτὸ μῆκος. ὥστ' οὐδὲν κωλύει συνεχῶς κινεῖσθαι καὶ μηδένα

same length. so nothing prevents continuously to be moved and no one

χρόνον διαλείπειν· ἡ μὲν γὰρ κύκλῳ κίνησίς ἐστιν ἀφ' αὑτοῦ εἰς

time to cease; the one for circular motion is from itself to

αὑτό, ἡ δὲ κατ' εὐθεῖαν ἀφ' αὑτοῦ εἰς ἄλλο· καὶ ἡ μὲν ἐν τῷ

itself, the other along the straight from itself to another; and the one in the

κύκλῳ οὐδέποτε ἐν τοῖς αὐτοῖς, ἡ δὲ κατ' εὐθεῖαν πολλάκις ἐν
circle never in the same, the other along the straight often in
τοῖς αὐτοῖς. τὴν μὲν οὖν ἀεὶ ἐν ἄλλῳ καὶ ἄλλῳ γιγνομένην
the same. the one therefore always in another and another being generated
ἐνδέχεται κινεῖσθαι συνεχῶς, τὴν δ' ἐν τοῖς αὐτοῖς πολλάκις
it is possible to be moved continuously, the other in the same often
οὐκ ἐνδέχεται· ἀνάγκη γὰρ ἅμα κινεῖσθαι τὰς ἀντικειμένας. ὥστ'
not it is possible; necessity for at once to be moved the opposites. so
οὐδ' ἐν τῷ ἡμικυκλίῳ οὐδ' ἐν ἄλλῃ περιφερείᾳ οὐδεμιᾷ ἐνδέχεται
neither in the semicircle neither in another circumstance none is possible
συνεχῶς κινεῖσθαι· πολλάκις γὰρ ἀνάγκη ταὐτὰ κινεῖσθαι καὶ τὰς
continuously to be moved often for necessity the same to be moved and the
ἐναντίας μεταβάλλειν μεταβολάς· οὐ γὰρ συνάπτει τῇ ἀρχῇ τὸ
opposite to change changes not for connects to the beginning the
πέρας. ἡ δὲ τοῦ κύκλου συνάπτει, καὶ ἔστι μόνη τέλειος.
end. the but of the circle connects, and is only perfect.

φανερὸν δὲ ἐκ ταύτης τῆς διαιρέσεως ὅτι οὐδὲ τὰς ἄλλας ἐνδέχεται
evident but from this the division that nor the others is possible
κινήσεις εἶναι συνεχεῖς· ἐν ἁπάσαις γὰρ ταὐτὰ συμβαίνει κινεῖσθαι
movements to be continuous. in all for the same occurs to be moved
πολλάκις, οἷον ἐν ἀλλοιώσει τὰ μεταξύ, καὶ ἐν τῇ τοῦ ποσοῦ
often, such as in alteration the between, and in the of the quantity
τὰ ἀνὰ μέσον μεγέθη, καὶ ἐν γενέσει καὶ φθορᾷ ὡσαύτως·
the by the means sizes, and in generation and decay likewise.
οὐδὲν γὰρ διαφέρει ὀλίγα ἢ πολλὰ ποιῆσαι, ἐν [265a] οἷς ἐστὶν
nothing for differs a little or much to produce, in [265a] which is

ἡ μεταβολή, οὐδὲ μεταξὺ θεῖναί τι ἢ ἀφελεῖν· ἀμφοτέρως
the change, nor between to place something or to remove. both ways
γὰρ συμβαίνει ταὐτὰ κινεῖσθαι πολλάκις. δῆλον οὖν ἐκ τούτων
for occurs the same to be moved often. clear therefore from these
ὅτι οὐδ' οἱ φυσιολόγοι καλῶς λέγουσιν οἱ πάντα τὰ αἰσθητὰ
that neither the physiologists rightly say the all the perceptible
κινεῖσθαι φάσκοντες ἀεί· κινεῖσθαι γὰρ ἀνάγκη τούτων τινὰ τῶν
to be moved claiming always· to be moved for necessity of these some of the
κινήσεων, καὶ μάλιστα κατ' ἐκείνους [ἐστὶν] ἀλλοιοῦσθαι·
movements, and particularly according to those [it is] to be altered·
ῥεῖν γάρ φασιν ἀεὶ καὶ φθίνειν, ἔτι δὲ καὶ τὴν γένεσιν
to flow for they say always and to diminish, furthermore but and the generation
καὶ τὴν φθορὰν ἀλλοίωσιν λέγουσιν. ὁ δὲ λόγος νῦν
and the destruction they say alteration. they say. the but discourse now
εἴρηκε καθόλου περὶ πάσης κινήσεως ὅτι κατ' οὐδεμίαν
has stated in general concerning all movement that according to no
κίνησιν ἐνδέχεται κινεῖσθαι συνεχῶς ἔξω τῆς κύκλῳ, ὥστε οὔτε
movement is possible to be moved continuously outside of the circle, so that neither
κατ' ἀλλοίωσιν οὔτε κατ' αὔξησιν. ὅτι μὲν οὖν
according to alteration nor according to increase. that on the one hand therefore
οὔτ' ἄπειρός ἐστι μεταβολὴ οὐδεμία οὔτε συνεχὴς ἔξω τῆς κύκλῳ
nor infinite it is a change no nor continuous outside of the circle
φορᾶς ἔστω τοσαῦθ' ἡμῖν εἰρημένα.
of movement let it be thus much to us said.

Κεφάλαιο 9

Chapter 9

Ὅτι δὲ τῶν φορῶν ἡ κυκλοφορία πρώτη, δῆλον. πᾶσα γὰρ φορά,

That and of the bearers the circulation is first, clear. Every for bearing,

ὥσπερ καὶ πρότερον εἴπομεν, ἢ κύκλῳ ἢ ἐπ' εὐθείας ἢ μικτή.

just as and before we said, either circular or upon straight or mixed.

ταύτης δὲ ἀνάγκη προτέρας εἶναι ἐκείνας· ἐξ ἐκείνων γὰρ συνέστηκεν.

Of this and necessity prior to be of those; from those for it has been constitute

τῆς δ' εὐθείας ἡ κύκλῳ· ἁπλῆ γὰρ καὶ τέλειος μᾶλλον. ἄπειρον

The and straight the circular; simple for and perfect rather. Infinite

μὲν γὰρ οὐκ ἔστιν εὐθεῖαν φέρεσθαι (τὸ γὰρ οὕτως

indeed for not is to bear a straight to be carried (for for thus

ἄπειρον οὐκ ἔστιν· ἅμα δ' οὐδ' εἰ ἦν, ἐκινεῖτ'

infinite not is; at the same time but not if it was, it would move

ἂν οὐδέν· οὐ γὰρ γίγνεται τὸ ἀδύνατον, διελθεῖν δὲ τὴν

indeed nothing; not for it becomes the impossible, to pass through but the

ἄπειρον ἀδύνατον)· ἡ δ' ἐπὶ τῆς πεπερασμένης ἀνακάμπτουσα

infinite impossible); the and upon the finite bending

μὲν συνθετὴ καὶ δύο κινήσεις, μὴ ἀνακάμπτουσα δὲ ἀτελὴς καὶ

indeed compositional and two movements, not retreating but incomplete and

φθαρτή. πρότερον δὲ καὶ φύσει καὶ λόγῳ καὶ χρόνῳ τὸ τέλειον

perishable. Previously but also by nature and by reason and by time the perfect

μὲν τοῦ ἀτελοῦς, τοῦ φθαρτοῦ δὲ τὸ ἄφθαρτον. ἔτι προτέρα

indeed of the incomplete, of the perishable but the imperishable. Moreover earlier

ἦν ἐνδέχεται ἀΐδιον εἶναι τῆς μὴ ἐνδεχομένης· τὴν μὲν

which it is possible to be eternal to be of the not possible; that indeed

οὖν κύκλῳ ἐνδέχεται ἀΐδιον εἶναι, τῶν δὲ ἄλλων οὔτε

therefore in a circle it is possible to be eternal to be, of the but other neither

φορὰν οὔτε ἄλλην οὐδεμίαν· στάσιν γὰρ δεῖ γενέσθαι, εἰ

movement nor another any; a state for it is necessary to happen, if

δὲ στάσις, ἔφθαρται ἡ κίνησις. εὐλόγως δὲ συμβέβηκε τὸ τὴν
but a state, is perishable the movement. Reasonably but it has happened that the
κύκλῳ μίαν εἶναι καὶ συνεχῆ, καὶ μὴ τὴν ἐπ' εὐθείας· τῆς
circular one to be and continuous, and not the on the straight line; of the
μὲν γὰρ ἐπ' εὐθείας ὥρισται καὶ ἀρχὴ καὶ τέλος καὶ
indeed for on of straightness has been defined and beginning and end and
μέσον, καὶ πάντ' ἔχει ἐν αὑτῇ, ὥστ' ἔστιν ὅθεν ἄρξεται τὸ
middle, and all has in itself, so that it is from where it will begin the
κινούμενον καὶ οὗ τελευτήσει (πρὸς γὰρ τοῖς πέρασιν ἠρεμεῖ
moving and where it will end (for for the boundaries it is at rest
πᾶν, ἢ ὅθεν ἢ οὗ), τῆς δὲ περιφεροῦς ἀόριστα· τί γὰρ
all, or from where or where), the but of the revolving indefinitely. what for
μᾶλλον ὁποιονοῦν πέρας τῶν ἐπὶ τῆς γραμμῆς; ὁμοίως γὰρ
rather any boundary of the upon the line? similarly for
ἕκαστον καὶ ἀρχὴ καὶ μέσον καὶ τέλος, ὥστ' ἀεί τε [265b]
each and beginning and middle and end, so that always and [265b]
εἶναι ἐν ἀρχῇ καὶ ἐν τέλει καὶ μηδέποτε. διὸ κινεῖταί τε καὶ
to be in beginning and in end and never. therefore it is moved and it also
ἠρεμεῖ πως ἡ σφαῖρα· τὸν αὐτὸν γὰρ κατέχει τόπον. αἴτιον
is at rest somehow the sphere; the same for it holds a place. cause
δ' ὅτι πάντα συμβέβηκε ταῦτα τῷ κέντρῳ· καὶ γὰρ ἀρχὴ καὶ
but that all has happened these to the center; and for beginning and
μέσον τοῦ μεγέθους καὶ τέλος ἐστίν, ὥστε διὰ τὸ ἔξω εἶναι
middle of the size and end is, so that through the outside to be
τοῦτο τῆς περιφερείας οὐκ ἔστιν ὅπου τὸ φερόμενον ἠρεμήσει ὡς
this of the circumference not is where the carried will rest as
διεληλυθός (ἀεὶ γὰρ φέρεται περὶ τὸ μέσον, ἀλλ' οὐ πρὸς
having been passed (always for it is carried around the middle, but not towards

τὸ ἔσχατον), διὰ δὲ τὸ τοῦτο μένειν ἀεί τε ἠρεμεῖ πως
the end), through but the this remaining always and will be at rest somehow
τὸ ὅλον καὶ κινεῖται συνεχῶς. συμβαίνει δ' ἀντιστρόφως· καὶ
the whole and moves continuously. it happens but conversely; and
γὰρ ὅτι μέτρον τῶν κινήσεων ἡ περιφορά, πρώτην ἀναγκαῖον
for that a measure of the movements the revolution, first necessary
αὐτὴν εἶναι (ἅπαντα γὰρ μετρεῖται τῷ πρώτῳ), καὶ διότι πρώτη,
it to be (all for it is measured by the first), and because first,
μέτρον ἐστὶν τῶν ἄλλων. ἔτι δὲ καὶ ὁμαλῆ ἐνδέχεται εἶναι τὴν
measure is of the others. still but also smooth may be to be the
κύκλῳ μόνην· τὰ γὰρ ἐπ' εὐθείας ἀνωμαλῶς ἀπὸ τῆς ἀρχῆς
circular only· for the for on straight irregularly from the beginning
φέρεται καὶ πρὸς τὸ τέλος· πάντα γὰρ ὅσῳπερ ἂν ἀφίστηται
is carried and towards the end· all for as much as may be separated
[πλεῖον] τοῦ ἠρεμοῦντος, φέρεται θᾶττον· τῆς δὲ κύκλῳ μόνης
[more] of the resting, is carried faster· the but circular only
οὔτ' ἀρχὴ οὔτε τέλος ἐν αὐτῇ πέφυκεν, ἀλλ' ἐκτός. ὅτι δ'
neither beginning nor end in it has been but outside. because but
ἡ κατὰ τόπον φορὰ πρώτη τῶν κινήσεων, μαρτυροῦσι πάντες
the by place motion first of the motions, testify all
ὅσοι περὶ κινήσεως πεποίηνται μνείαν· τὰς γὰρ ἀρχὰς αὐτῆς
who regarding motion have accepted mention· the for beginnings of it
ἀποδιδόασιν τοῖς κινοῦσι τοιαύτην κίνησιν. διάκρισις γὰρ καὶ
attribute to those moving such motion. distinction for and
σύγκρισις κινήσεις κατὰ τόπον εἰσίν, οὕτω δὲ κινοῦσιν
comparison motions according to place they are, thus and they move
ἡ φιλία καὶ τὸ νεῖκος· τὸ μὲν γὰρ διακρίνει, τὸ δὲ
the friendship and the strife. the on one hand for it distinguishes, the but

συγκρίνει αὐτῶν. καὶ τὸν νοῦν δέ φησιν Ἀναξαγόρας διακρίνειν

it unites them. And the mind also says Anaxagoras to distinguish

τὸν κινήσαντα πρῶτον. ὁμοίως δὲ καὶ ὅσοι τοιαύτην μὲν

the mover first. Likewise and also all those such on one hand

οὐδεμίαν αἰτίαν λέγουσιν, διὰ δὲ τὸ κενὸν κινεῖσθαί φασιν·

no cause they say, on account of but the void to be moved they say.

καὶ γὰρ οὗτοι τὴν κατὰ τόπον κίνησιν κινεῖσθαι τὴν φύσιν

And for these the according to place motion to be moved the nature

λέγουσιν (ἡ γὰρ διὰ τὸ κενὸν κίνησις φορά ἐστιν καὶ ὡς ἐν

they say (the for by the void motion movement is and as in

τόπῳ), τῶν δ' ἄλλων οὐδεμίαν ὑπάρχειν τοῖς πρώτοις ἀλλὰ

place), of the but others no to exist for the first ones but

τοῖς ἐκ τούτων οἴονται· αὐξάνεσθαι γὰρ καὶ φθίνειν καὶ

for those from these they believe. to increase for and to perish and

ἀλλοιοῦσθαι συγκρινομένων καὶ διακρινομένων τῶν

to be altered of things being compared and of things being distinguished of the

ἀτόμων σωμάτων φασίν. τὸν αὐτὸν δὲ τρόπον καὶ ὅσοι διὰ

atoms bodies they say. the same but manner and as many as through

πυκνότητα ἢ μανότητα κατασκευάζουσι γένεσιν καὶ φθοράν·

density or rarity they construct generation and decay.

συγκρίσει γὰρ καὶ διακρίσει ταῦτα διακοσμοῦσιν. ἔτι δὲ παρὰ

for comparison for and distinction these they embellish. yet but from

τούτους οἱ τὴν ψυχὴν αἰτίαν ποιοῦντες κινήσεως· τὸ γὰρ

these the the soul cause producing of motion. for the for

αὐτὸ αὑτὸ κινοῦν ἀρχὴν εἶναί φασιν τῶν κινουμένων, κινεῖ

same itself moving principle to be they say of the moving, it moves

δὲ τὸ ζῷον καὶ πᾶν τὸ ἔμ [266a] ψυχον τὴν κατὰ τόπον

and the animal and all the living [266a] soul the according to place

αὐτὸ κίνησιν. καὶ κυρίως δὲ κινεῖσθαί φαμεν μόνον τὸ κινούμενον

itself motion. and primarily but to be moved we say only the moving

[τὴν] κατὰ τόπον [κίνησιν]· ἂν δ' ἠρεμῇ μὲν ἐν τῷ

[the] according to place [motion]. if and it will be at rest indeed in the

αὐτῷ, αὐξάνηται δ' ἢ φθίνῃ ἢ ἀλλοιούμενον τυγχάνῃ, πῇ

same, it increases and or it diminishes or it is altered it happens, whither

κινεῖσθαι, ἁπλῶς δὲ κινεῖσθαι οὔ φαμεν. ὅτι μὲν οὖν ἀεί

to be moved, simply and to be moved not let us say. that indeed therefore always

τε κίνησις ἦν καὶ ἔσται τὸν ἅπαντα χρόνον, καὶ τίς ἀρχὴ

and motion it was and it will be the all time, and what beginning

τῆς ἀϊδίου κινήσεως, ἔτι δὲ τίς πρώτη κίνησις, καὶ τίνα κίνησιν

of the eternal motion, moreover and what first motion, and what motion

ἄϊδιον ἐνδέχεται μόνην εἶναι, καὶ τὸ κινοῦν πρῶτον ὅτι ἀκίνητον,

eternal it is possible alone to be, and the mover first since it is unmoved,

εἴρηται.

it has been stated.

Κεφάλαιο 10

Chapter 10

Ὅτι δὲ τοῦτ' ἀμερὲς ἀναγκαῖον εἶναι καὶ μηδὲν ἔχειν μέγεθος, νῦν

That and this part necessary to be and nothing to have size, now

λέγωμεν, πρῶτον περὶ τῶν προτέρων αὐτοῦ διορίσαντες.

let us say, first about the previous of it having defined.

τούτων δ' ἓν μέν ἐστιν ὅτι οὐχ οἷόν τε οὐδὲν πεπερασμένον

Of these but one indeed is that not possible and nothing finite

κινεῖν ἄπειρον χρόνον. τρία γὰρ ἔστιν, τὸ κινοῦν, τὸ κινούμενον,

to move infinite time. Three for there is, the mover, the moved,

τὸ ἐν ᾧ τρίτον, ὁ χρόνος. ταῦτα δὲ ἢ πάντα ἄπειρα ἢ πάντα
the in which third, the time. These but or all infinite or all
πεπερασμένα ἢ ἔνια, οἷον τὰ δύο ἢ τὸ ἕν. ἔστω δὴ τὸ A
finite or some, such as the two or the one. Let it be indeed the A
τὸ κινοῦν, τὸ δὲ κινούμενον B, χρόνος ἄπειρος ἐφ' οὗ
the mover, the and moved B, time infinite in which of which
Γ. τὸ δὴ Δ τῆς B κινείτω τι μέρος, τὸ ἐφ' οὗ E. οὐ
C. the indeed D of the B may move something part, the on which E. not
δὴ ἐν ἴσῳ τῷ Γ· ἐν πλείονι γὰρ τὸ μεῖζον. ὥστ' οὐκ ἄπειρος
indeed in equal to the G. in more for the greater. thus not infinite
ὁ χρόνος ὁ τὸ Z. οὕτω δὴ τῇ Δ προστιθεὶς καταναλώσω
the time the the Z. thus indeed to the D adding I will consume
τὸ A καὶ τῇ E τὸ B· τὸν δὲ χρόνον οὐ καταναλώσω ἀεὶ
the A and to the E the B. the but time not I will consume always
ἀφαιρῶν ἴσον· ἄπειρος γάρ· ὥστε ἡ πᾶσα A τὴν ὅλην B
taking away equal; infinite for; so that the whole A the entire B
κινήσει ἐν πεπερασμένῳ χρόνῳ τοῦ Γ. οὐκ ἄρα οἷόν τε ὑπὸ
will move in finite time of the G. not therefore possible and by
πεπερασμένου κινεῖσθαι οὐδὲν ἄπειρον κίνησιν.
finite to be moved nothing infinite motion.

ὅτι μὲν οὖν οὐκ ἐνδέχεται τὸ πεπερασμένον ἄπειρον
that on the one hand therefore not it is possible the finite infinite
κινεῖν χρόνον, φανερόν· ὅτι δ' ὅλως οὐκ ἐνδέχεται ἐν πεπερασμένῳ
to move time, clearly; that but altogether not it is possible in finite
μεγέθει ἄπειρον εἶναι δύναμιν, ἐκ τῶνδε δῆλον. ἔστω γὰρ ἡ
magnitude to be infinite to be a power, from these evident. Let it be for the

πλείων δύναμις ἀεὶ ἡ τὸ ἴσον ἐν ἐλάττονι χρόνῳ ποιοῦσα, οἷον
greater power always the the equal in lesser time producing, such as
θερμαίνουσα ἢ γλυκαίνουσα ἢ ῥιπτοῦσα καὶ ὅλως κινοῦσα.
warming or sweetening or tossing and altogether moving.
ἀνάγκη ἄρα καὶ ὑπὸ τοῦ πεπερασμένου μὲν ἄπειρον δ'
Necessity therefore and by the finite on one hand infinite but
ἔχοντος δύναμιν πάσχειν τι τὸ πάσχον, καὶ πλεῖον ἢ ὑπ'
having power to suffer something the suffering, and more than under
ἄλλου· πλείων γὰρ ἡ ἄπειρος. ἀλλὰ μὴν χρόνον γε οὐκ
another; greater for the infinite. But indeed time at least not
ἐνδέχεται εἶναι οὐδένα. εἰ γάρ ἐστιν ὁ ἐφ' οὗ A χρόνος ἐν ᾧ
it is possible to be any. if for is the on which A time in which
ἡ ἄπειρος ἰσχὺς ἐθέρμανεν ἢ ἔωσεν, ἐν τῷ δὲ AB πεπερασμένη
the infinite power has warmed or has shone, in the but AB finite
τις, πρὸς ταύτην [266b] μείζω λαμβάνων ἀεὶ πεπερασμένην
someone, towards this [266b] greater taking always finite
ἥξω ποτὲ εἰς τὸ ἐν τῷ A χρόνῳ κεκινηκέναι· πρὸς
I will arrive at some point into the in the A time to have moved; towards
πεπερασμένον γὰρ ἀεὶ προστιθεὶς ὑπερβαλῶ παντὸς ὡρισμένου,
finite for always adding I will surpass of all defined,
καὶ ἀφαιρῶν ἐλλείψω ὡσαύτως. ἐν ἴσῳ ἄρα χρόνῳ κινήσει τῇ
and removing I will leave out likewise. in equal thus time it will move the
ἀπείρῳ ἡ πεπερασμένη. τοῦτο δὲ ἀδύνατον· οὐδὲν ἄρα πεπερασμένον
infinite the finite. this but impossible; nothing thus finite
ἐνδέχε ται ἄπειρον δύναμιν ἔχειν. οὐ τοίνυν οὐδ' ἐν ἀπείρῳ πεπε(6)
may have it infinite power to have. not therefore nor in infinite finite
ρασμένην· καίτοι ἐνδέχεται ἐν ἐλάττονι μεγέθει πλείω δύναμιν
finite; and yet it may happen in lesser magnitude more power

εἶναι· ἀλλ' ἔτι μᾶλλον ἐν μείζονι πλείω. ἔστω δὴ τὸ ἐφ'

to be. But still more in greater more. Let it be indeed the on

οὗ AB ἄπειρον. τὸ δὴ BΓ ἔχει δύναμίν τινα, ἣ ἔν τινι

which AB infinite. The indeed BG has power some, which in some

χρόνῳ ἐκίνησεν τὴν Δ, ἐν τῷ χρόνῳ ἐφ' οὗ EZ. ἂν δὴ τῆς

time has moved the D, in the time on which EZ. If indeed of the

BΓ διπλασίαν λαμβάνω, ἐν ἡμίσει χρόνῳ τοῦ EZ (ἔστω γὰρ

BG double I take, in half time of the EZ (let it be for

αὕτη ἡ ἀναλογία), ὥστε ἐν τῷ ZΘ κινήσει. οὐκοῦν οὕτω

this the analogy), so that in the ZΘ it will move. Therefore in this way

λαμβάνων ἀεὶ τὴν μὲν AB οὐδέποτε διέξειμι, τοῦ χρόνου

taking always the indeed AB never will I pass, of the time

δὲ τοῦ δοθέντος αἰεὶ ἐλάττω λήψομαι. ἄπειρος ἄρα ἡ

but of the given always I will take less. I will receive. Infinite thus the

δύναμις ἔσται· πάσης γὰρ πεπερασμένης ὑπερβάλλει δυνάμεως, εἴ

power will be; of all for finite surpasses of power, if

γε πάσης πεπερασμένης δυνάμεως ἀνάγκη πεπερασμένον εἶναι καὶ

indeed of all finite power necessity finite to be and

τὸν χρόνον (εἰ γὰρ ἔν τινι ἡ τοσηδί, ἡ μείζων ἐν ἐλάττονι

the time (if for in a certain the such the greater in lesser

μὲν ὡρισμένῳ δὲ χρόνῳ κινήσει, κατὰ τὴν ἀντιστροφὴν

indeed definite but time motion, according to the converse

τῆς ἀναλογίας)· ἄπειρος δὲ πᾶσα δύναμις, ὥσπερ καὶ πλῆθος

of the proportion)· infinite but every power, just as and multitude

καὶ μέγεθος τὸ ὑπερβάλλον παντὸς ὡρισμένου. ἔστιν δὲ καὶ ὧδε

and magnitude the exceeding of all definite. it is but also thus

δεῖξαι τοῦτο· ληψόμεθα γάρ τινα δύναμιν τὴν αὐτὴν τῷ

to demonstrate this· we shall take for some power the same to the

γένει τῇ ἐν τῷ ἀπείρῳ μεγέθει, ἐν πεπερασμένῳ μεγέθει οὖσαν, ἢ

kind the in the infinite magnitude, in finite magnitude being, which

καταμετρήσει τὴν ἐν τῷ ἀπείρῳ πεπερασμένην δύναμιν.

will measure the in the infinite finite power.

ὅτι μὲν οὖν οὐκ ἐνδέχεται ἄπειρον εἶναι δύναμιν ἐν

that on the one hand therefore not it is possible infinite to be a power in

πεπερασμένῳ μεγέθει, οὐδ' ἐν ἀπείρῳ πεπερασμένην, ἐκ τούτων

finite magnitude, nor in infinite finite, from these

δῆλον. περὶ δὲ τῶν φερομένων ἔχει καλῶς διαπορῆσαί

it is evident. Regarding but the carried it has well to have a difficulty

τινα ἀπορίαν πρῶτον. εἰ γὰρ πᾶν τὸ κινούμενον κινεῖται ὑπὸ

some perplexity first. If for everything the moved is moved by

τινός, ὅσα μὴ αὐτὰ ἑαυτὰ κινεῖ, πῶς κινεῖται ἔνια συνεχῶς

something, whatever not itself itself moves, how is it moved in some continuously

μὴ ἁπτομένου τοῦ κινήσαντος, οἷον τὰ ῥιπτούμενα; εἰ δ'

without being touched by the mover, such as the thrown? If and

ἅμα κινεῖ καὶ ἄλλο τι ὁ κινήσας, οἷον τὸν ἀέρα, ὃς

at the same time moves and another something the mover, such as the air, which

κινούμενος κινεῖ, ὁμοίως ἀδύνατον τοῦ πρώτου μὴ ἁπτομένου μηδὲ

being moved moves, similarly impossible of the first not being touched nor

κινοῦντος κινεῖσθαι, ἀλλ' ἅμα πάντα <καὶ> κινεῖσθαι

of the mover to be moved, but at the same time everything and to be moved

καὶ [267a] πεπαῦσθαι ὅταν τὸ πρῶτον κινοῦν παύσηται, καὶ εἰ

and [267a] to have ceased whenever the first moving should stop, and if

ποιεῖ, ὥσπερ ἡ λίθος, οἷόν τε κινεῖν ὃ ἐκίνησεν. ἀνάγκη δὴ

it does, just as the stone, such as and to move that it moved. necessity indeed

τοῦτο μὲν λέγειν, ὅτι τὸ πρῶτον κινῆσαν ποιεῖ οἷόν τε κινεῖν
this on one hand to say, that the first have moved it does such as and to move
ἢ τὸν ἀέρα [τοιοῦτον] ἢ τὸ ὕδωρ ἤ τι ἄλλο τοιοῦτον ὃ
or the air [such] or the water or something else such that
πέφυκε κινεῖν καὶ κινεῖσθαι· ἀλλ' οὐχ ἅμα παύεται
has a nature to move and to be moved; but not at the same time does it cease
κινοῦν καὶ κινούμενον, ἀλλὰ κινούμενον μὲν ἅμα ὅταν
from moving and being moved, but being moved indeed at the same time when
ὁ κινῶν παύσηται κινῶν, κινοῦν δὲ ἔτι ἐστίν. διὸ καὶ κινεῖ
the mover ceases moving, while moving but still it is. therefore also it moves
τι ἄλλο ἐχόμενον· καὶ ἐπὶ τούτου ὁ αὐτὸς λόγος. παύεται
something else that is held; and on this matter the same reasoning. ceases
δέ, ὅταν ἀεὶ ἐλάττων ἡ δύναμις τοῦ κινεῖν ἐγγίγνηται τῷ
but, whenever always less the power of the to move comes to be to the
ἐχομένῳ. τέλος δὲ παύεται, ὅταν μηκέτι ποιήσῃ τὸ πρότερον
existing. finally but ceases, whenever no longer it does the prior
κινοῦν, ἀλλὰ κινούμενον μόνον. ταῦτα δ' ἀνάγκη ἅμα παύεσθαι,
moving, but being moved only. these and necessity together to cease,
τὸ μὲν κινοῦν τὸ δὲ κινούμενον, καὶ τὴν ὅλην
the on one hand moving the on the other hand being moved, and the whole
κίνησιν. αὕτη μὲν οὖν ἐν τοῖς ἐνδεχομένοις ὁτὲ μὲν
motion. this indeed therefore in those conceivable sometimes indeed
κινεῖσθαι ὁτὲ δ' ἠρεμεῖν ἐγγίγνεται ἡ κίνησις, καὶ οὐ
to be moved and sometimes but to be at rest comes to be the motion, and not
συνεχής, ἀλλὰ φαίνεται· ἢ γὰρ ἐφεξῆς ὄντων ἢ ἁπτομένων
continuous, but it appears; or for successively existing or touching
ἐστίν· οὐ γὰρ ἓν τὸ κινοῦν, ἀλλ' ἐχόμενα ἀλλήλων. διὸ
it is: not for one the moving, but held to each other. therefore

ἐν ἀέρι καὶ ὕδατι γίγνεται ἡ τοιαύτη κίνησις, ἣν λέγουσί τινες
in air and water it becomes the such motion, which they say some
ἀντιπερίστασιν εἶναι. ἀδύνατον δὲ ἄλλως τὰ ἀπορηθέντα λύειν,
resistance to be. impossible but otherwise the perplexing to resolve,
εἰ μὴ τὸν εἰρημένον τρόπον. ἡ δ' ἀντιπερίστασις ἅμα πάντα
if not the mentioned way. the and resistance together all
κινεῖσθαι ποιεῖ καὶ κινεῖν, ὥστε καὶ παύεσθαι· νῦν δὲ φαίνεταί
to move causes and to move, so that also to cease; now but it appears
τι ἓν κινούμενον συνεχῶς· ὑπὸ τίνος οὖν; οὐ γὰρ ὑπὸ τοῦ
something one moving continuously; by whom then? not for by the
αὐτοῦ. ἐπεὶ δ' ἐν τοῖς οὖσιν ἀνάγκη κίνησιν εἶναι συνεχῆ, αὕτη
same. since and in the beings necessity motion to be continuous, this
δὲ μία ἐστίν, ἀνάγκη δὲ τὴν μίαν μεγέθους τέ τινος εἶναι (οὐ
and one is, necessity but the one of magnitude and of some to be (not
γὰρ κινεῖται τὸ ἀμέγεθες) καὶ ἑνὸς καὶ ὑφ' ἑνός (οὐ γὰρ ἔσται
for is moved the unmeasured) and of one and by one (not for will be
συνεχής, ἀλλ' ἐχομένη ἑτέρα ἑτέρας καὶ διῃρημένη), τὸ δὴ
continuous, but having another of another and divided), the indeed
κινοῦν εἰ ἕν, ἢ κινούμενον κινεῖ ἢ ἀκίνητον ὄν. εἰ μὲν
moving if one, or being moved moves or immovable being. if on the one hand
δὴ κινού μενον, συνακολουθεῖν δεήσει καὶ μεταβάλλειν
indeed moving being, it will need to follow it will require and to change
αὐτό, ἅμα δὲ [267b] κινεῖσθαι ὑπό τινος, ὥστε στήσεται
itself, at the same time but [267b] to be moved under something, so that it will set
καὶ ἥξει εἰς τὸ κινεῖσθαι ὑπὸ ἀκινήτου. τοῦτο γὰρ οὐκ ἀνάγκη
and will arrive at the moving under the immovable. this for not necessity
συμμεταβάλλειν, ἀλλ' ἀεί τε δυνήσεται κινεῖν (ἄπονον γὰρ τὸ
to change, but always and will be able to move (without effort for the

οὕτω κινεῖν) καὶ ὁμαλὴς αὕτη ἡ κίνησις ἢ μόνη ἢ μάλιστα· οὐ
thus to move) and smooth this the motion or alone or especially; not
γὰρ ἔχει μεταβολὴν τὸ κινοῦν οὐδεμίαν. δεῖ δὲ οὐδὲ τὸ
for has change the moving none. it is necessary but not even the
κινούμενον πρὸς ἐκεῖνο ἔχειν μεταβολήν, ἵνα ὁμοία ᾖ ἡ
being moved towards that to have change, in order that similar may be the
κίνησις. ἀνάγκη δὴ ἢ ἐν μέσῳ ἢ ἐν κύκλῳ εἶναι· αὗται γὰρ
motion. necessity indeed or in the midst or in circle to be; these for
αἱ ἀρχαί. ἀλλὰ τάχιστα κινεῖται τὰ ἐγγύτατα τοῦ κινοῦντος.
the principles. but most swiftly it moves the nearest of the mover.
τοιαύτη δ' ἡ τοῦ κύκλου κίνησις· ἐκεῖ ἄρα τὸ κινοῦν. ἔχει δ'
such and the of the circle motion; there therefore the mover. has and
ἀπορίαν εἰ ἐνδέχεταί τι κινούμενον κινεῖν συνεχῶς, ἀλλὰ μὴ
uncertainty if it is possible something being moved to move continuously, but not
ὥσπερ τὸ ὠθοῦν πάλιν καὶ πάλιν, τῷ ἐφεξῆς εἶναι συνεχῶς· ἢ
as the pushing back and again, by in succession to be continuously; or
γὰρ αὐτὸ δεῖ ἀεὶ ὠθεῖν ἢ ἕλκειν ἢ ἄμφω, ἢ ἕτερόν τι
for it necessitates always to push or to pull or both, or another something
ἐκδεχόμενον ἄλλο παρ' ἄλλου, ὥσπερ πάλαι ἐλέχθη ἐπὶ τῶν
that is received another from another, as in the past was said about of the
ῥιπτουμένων, εἰ διαιρετὸς ὢν ὁ ἀὴρ [ἢ τὸ ὕδωρ] κινεῖ ἄλλος ἀεὶ
falling, if divisible being the air or the water] moves another always
κινούμενος. ἀμφοτέρως δ' οὐχ οἷόν τε μίαν εἶναι, ἀλλ' ἐχομένην.
moving. in both ways but not possible and one to be, but having.
μόνη ἄρα συνεχὴς ἦν κινεῖ τὸ ἀκίνητον· ἀεὶ γὰρ ὁμοίως ἔχον
only thus continuous which moves the unmoving; always for similarly having
καὶ πρὸς τὸ κινούμενον ὁμοίως ἕξει καὶ συνεχῶς.
and towards the moving similarly will have and continuously.

διωρισμένων δὲ τούτων φανερὸν ὅτι ἀδύνατον τὸ πρῶτον κινοῦν

defined but of these it is clear that impossible the first to move

καὶ ἀκίνητον ἔχειν τι μέγεθος. εἰ γὰρ μέγεθος ἔχει, ἀνάγκη ἤτοι

and unmoved to have any size. if for size has, necessity either

πεπερασμένον αὐτὸ εἶναι ἢ ἄπειρον. ἄπειρον μὲν οὖν

finite it to be or infinite. infinite on the one hand therefore

ὅτι οὐκ ἐνδέχεται μέγεθος εἶναι, δέδεικται πρότερον ἐν τοῖς

that not it is possible size to be, it has been shown before in the

φυσικοῖς· ὅτι δὲ τὸ πεπερασμένον ἀδύνατον ἔχειν δύναμιν ἄπειρον,

physical. that but the finite impossible to have strength infinite,

καὶ ὅτι ἀδύνατον ὑπὸ πεπερασμένου κινεῖσθαί τι ἄπειρον

and that impossible by finite to be moved anything infinite

χρόνον, δέδεικται νῦν. τὸ δέ γε πρῶτον κινοῦν ἀΐδιον κινεῖ

time, it has been shown now. the but indeed first to move eternal moves

κίνησιν καὶ ἄπειρον χρόνον. φανερὸν τοίνυν ὅτι ἀδιαίρετόν ἐστι καὶ

motion and infinite time. it is clear therefore that indivisible is and

ἀμερὲς καὶ οὐδὲν ἔχον μέγεθος.

unmeasurable and nothing having size.